



THRIVE IN EVERY HARVEST!

GRADE 9 AGRICULTURE AND NUTRITION

**TOPICAL REVISION BOOK
CBE KJSEA EDITION**



IGNITING PASSION FOR SEEDS AND HEALTHY FEEDS



IGNITING PASSION FOR SEEDS AND HEALTHY FEEDS

STRAND 1.0: CONSERVATION OF RESOURCES

SUB-STRAND 1.1: CONSERVING ANIMAL FEED (HAY)

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

Instructions: Choose the best answer from the given alternatives.

1. A farmer in Kericho cuts grass during the dry season for his cows.



What is this method of conserving animal feed called?

- A. Silage
 - B. Hay making
 - C. Grazing
 - D. Mulching
2. Hay should be harvested when grasses are:
- A. Overgrown and dry
 - B. Young and leafy
 - C. Flowering and full of seeds
 - D. Rotten and brown
3. Which type of grass is most suitable for making hay?
- A. Napier grass at mature stage
 - B. Maize stalks after harvest
 - C. Natural pasture grasses
 - D. Sugarcane tops
4. The main purpose of making hay is to:
- A. Increase soil fertility
 - B. Feed animals during dry season
 - C. Protect crops from pests
 - D. Produce bedding material
5. The best time to cut grass for hay is when it is:
- A. Very tall and brown
 - B. Still young and green
 - C. At flowering stage
 - D. At seeding stage



KITCHEN AND BACKYARD GARDENING



COOKING

6. Which of the following ensures high quality hay?
 - A. Storing in the open field
 - B. Drying under the sun until brittle
 - C. Harvesting at late maturity
 - D. Mixing with weeds
7. A farmer notices mold on stored hay. The most likely cause is:
 - A. Overdrying
 - B. Proper storage
 - C. Excess moisture
 - D. Good ventilation
8. The structure below is used for storing hay.



What is it called?

- A. Granary
 - B. Hay barn
 - C. Silo
 - D. Storehouse
9. Which of the following animals benefit most from hay?
 - A. Poultry
 - B. Cows, goats, and sheep
 - C. Fish
 - D. Pigs
 10. The difference between hay and silage is that:
 - A. Hay is dried, silage is stored wet
 - B. Hay is stored underground, silage is stored on top
 - C. Hay is green, silage is brown
 - D. Hay is for pigs, silage is for cows
 11. The ideal moisture content of hay for safe storage is:
 - A. 15–20%
 - B. 25–30%
 - C. 35–40%
 - D. 45–50%



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COOKING

12. A farmer makes hay during the rainy season and stores it when wet. What problem is likely to occur?
A. Increased nutrition
B. Fire outbreak
C. Fermentation and rotting
D. Better palatability
13. Which of the following is NOT a characteristic of good quality hay?
A. Green in color
B. Sweet smell
C. High in leaves
D. Moldy appearance
14. The most suitable tool for cutting grass for hay is:
A. Hoe
B. Panga
C. Sickle
D. Rake
15. Why should hay be stored in bales?
A. For easy transport and storage
B. To prevent animals from eating it
C. To increase soil fertility
D. To reduce labor in cutting
16. A farmer in Narok notices that his hay turns brown and loses leaves after storage. What is the main cause?



- A. Proper storage
- B. Overdrying during preparation
- C. Correct baling
- D. Adequate ventilation



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COOKING

17. The following are advantages of hay over fresh pasture EXCEPT:
- A. Can be stored for long
 - B. Easy to transport
 - C. Retains high moisture
 - D. Available during dry season

18. Which of the following practices will produce poor quality hay?

- A. Cutting grass at early flowering stage
- B. Drying grass under the sun
- C. Mixing hay with weeds
- D. Storing hay in bales

19. Which weather condition is most suitable for hay drying?

- A. Cloudy days
- B. Rainy days
- C. Sunny and windy days

20. A farmer observes smoke and heat coming from stored hay.



What is the most likely cause?

- A. Firewood near the store
- B. Spontaneous combustion due to wet hay
- C. Properly dried hay
- D. Good ventilation

21. Which method of feeding hay reduces wastage?

- A. Throwing it on the ground
- B. Using hay racks
- C. Leaving animals to pick freely
- D. Mixing it with soil

22. The main disadvantage of feeding animals on poor quality hay is:

- A. It increases milk production
- B. It lowers animal health and productivity
- C. It reduces cost of feeding
- D. It prevents soil erosion



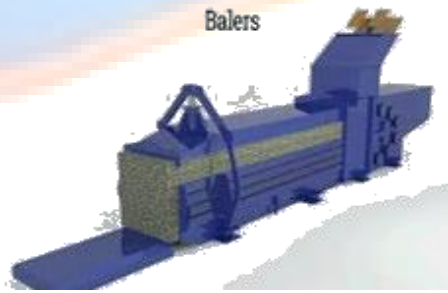
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COOKING

23. Which of the following is NOT a tool used in hay making?

A. Sickle



B. Rake



C. Shovel



D. Baler



24. Which type of farmers benefit most from hay making?

A. Fishermen

B. Dairy and beef farmers

C. Poultry farmers

D. Pig farmers

25. Which practice ensures hay maintains nutrients during storage?

A. Storing in damp places

B. Baling when slightly green

C. Harvesting overgrown grasses

D. Leaving in the field uncovered



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26. If hay is harvested too early before flowering, it may have:
- A. Low yield
 - B. High fiber content
 - C. More seeds
 - D. Moldy smell
27. The main source of energy in hay is from:
- A. Starch in leaves and stems
 - B. Protein in flowers
 - C. Fat in seeds
 - D. Minerals in roots
28. The practice of tying dried hay into compact blocks is called:
- A. Chopping
 - B. Baling
 - C. Bundling
 - D. Raking
29. Which factor affects the nutritive value of hay the most?
- A. The time of harvesting
 - B. The type of soil
 - C. The type of baler used
 - D. The color of barn roof
30. Hay stored in open fields without cover is easily destroyed by:
- A. Sunshine only
 - B. Rain and pests
 - C. Good ventilation
31. The best grass stage for hay harvesting is when:
- A. Leaves have fallen
 - B. The grass is flowering
 - C. Grass is dry and withered
 - D. Grass is overgrown and tall
32. Which statement about hay is TRUE?
- A. Hay is fed only to poultry
 - B. Hay is rich in water content
 - C. Hay is dried grass stored for feeding animals
 - D. Hay is a type of fertilizer
33. A farmer notices that his hay is dark brown and smells sour. What does this show?
- A. High quality hay
 - B. Properly dried hay
 - C. Spoiled hay due to poor storage
 - D. Sweet-smelling hay



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34. Why is hay making an important practice in semi-arid regions of Kenya?

- A. Because grass is always plenty
- B. It provides feed during droughts
- C. It makes land fertile
- D. It reduces farm labor

35. Which of the following reduces the labor of hay making?

- A. Cutting with panga
- B. Manual drying
- C. Use of a hay baler
- D. Storing in heaps

36. The best color that shows good quality hay is:

- A. Black
- B. Greenish
- C. Brownish
- D. Grey

37. A farmer prepares hay but forgets to remove weeds. What effect will this have?

- A. Increases animal nutrition
- B. Reduces palatability and quality
- C. Makes hay smell better
- D. Increases hay nutrients

38. The main reason why hay should be turned during drying is to:

- A. Increase labor costs
- B. Allow even drying
- C. Make hay heavier
- D. Reduce leaf content

39. Which structure is commonly used to store hay in Kenya?

- A. Zero grazing unit
- B. Hay barn
- C. Pit silo
- D. Chicken coop

40. Feeding hay to animals prevents:

- A. Soil erosion
- B. Drought in the farm
- C. Starvation during dry season
- D. Loss of soil fertility

41. Which of the following is a disadvantage of hay compared to silage?

- A. Requires dry weather for preparation
- B. Easier to store
- C. Lasts longer
- D. Retains more nutrients



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42. Which animal would least benefit from hay?
- Dairy cows
 - Sheep
 - Poultry
 - Goats
43. Which of the following shows that hay is of poor quality?
- Sweet smell
 - Dusty and moldy
 - Leafy and green
 - Soft texture
44. The main danger of storing hay with high moisture is:
- High quality feed
 - Spontaneous fire outbreak
 - Easy transportation
 - Reduced labor
45. Which method of harvesting grass ensures uniform cutting for hay?
- Sickle mowing
 - Hand pulling
 - Using animals to graze
 - Cutting with panga
46. Which factor does NOT affect the quality of hay?
- Stage of harvesting
 - Storage method
 - Color of hay barn
 - Drying method
47. A farmer stores his hay in polythene bags. The main danger is:
- Poor ventilation leading to mold growth
 - Increased storage life
 - Better nutrient retention
 - Reduced labor
48. The picture below shows compacted hay tied together.



What is this product called?

- | | |
|-----------------------|----------------------|
| A. Hay bundles | B. Hay bales |
| C. Hay heaps | D. Hay silage |



KITCHEN AND BACKYARD GARDENING



COOKING

49. Which Kenyan county benefits most from hay making due to frequent droughts?

- A. Turkana
- B. Kisumu
- C. Nyeri
- D. Murang'a

50. Which of the following is an environmental benefit of hay making?

- A. Causes soil erosion
- B. Provides ground cover when left on the field
- C. Destroys soil fertility
- D. Increases bush fires

51. A farmer in Kitui conserves grass by cutting, drying, and pressing it into compact blocks for storage. What is this method called?

- A. Stacking
- B. Standing forage
- C. Baled hay making
- D. Silage making

52. Why is baled hay preferred by many Kenyan farmers?

- A. It takes more storage space
- B. It is easier to transport and store
- C. It is always cheaper to make
- D. It contains more weeds

53. A farmer leaves mature grass standing in the field to be grazed during dry periods. This practice is called:

- A. Standing forage
- B. Hay baling
- C. Stacking
- D. Zero grazing

54. Which of the following is a disadvantage of standing forage?

- A. Easy to prepare
- B. Feed available in dry season
- C. Loss of nutrients as grass matures
- D. Reduced labor cost

55. A farmer in Baringo piles dried grass into large heaps covered with thatch to protect it from rain. What method of conservation is this?

- A. Stacking
- B. Silage making
- C. Baling
- D. Grazing



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56. Which of the following is an advantage of stacking hay?

- A. Requires expensive machines
- B. Can be done using simple farm tools
- C. Hay is always lost through fire
- D. Requires large buildings

57. The main purpose of conserving forage using baling, stacking, or standing forage is to:

- A. Provide bedding for animals
- B. Ensure animal feed during drought
- C. Increase soil fertility
- D. Prevent pest attacks in crops

58. Which conservation method is most labor-saving for pastoral farmers in ASAL (Arid and Semi-Arid Lands) areas of Kenya?

- A. Standing forage
- B. Silage making
- C. Baled hay making
- D. Zero grazing

59. Which of the following is a disadvantage of stacking compared to baling?

- A. Requires more labor
- B. Difficult to transport hay
- C. Preserves more nutrients
- D. Needs advanced technology

60. Which of the following combinations are methods of conserving forage?

- A. Mulching, Composting, Manuring
- B. Stacking, Baled hay, Standing forage
- C. Grafting, Irrigation, Weeding
- D. Manure application, Terracing, Ridging

SET 2

GRADE 9 AGRICULTURE REVISION QUESTIONS

STRAND 1.0: CONSERVATION OF RESOURCES

SUB-STRAND 1.1: CONSERVING ANIMAL FEED – HAY

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. During a long dry season, a farmer notices that their cattle are losing weight because of lack of pasture. The farmer remembers learning about feed conservation at an agricultural show. Which method would best help the farmer prepare feed that can last long and is easy to transport?
 - A) Direct grazing of forage
 - B) Making silage in a pit
 - C) Baling hay
 - D) Feeding fresh Napier grass



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2. Learners visit a dairy farm and see big rectangular blocks of dried grass stacked in a shed. The farmer explains they were prepared in the rainy season for use during drought. Why is grass dried before being compacted into blocks?
 - A) To increase protein content
 - B) To reduce bulkiness for storage
 - C) To lower moisture content and prevent spoilage
 - D) To improve taste for livestock
3. In an arid area with frequent droughts, an extension officer advises farmers on how to keep animals alive. Which hay conservation strategy is most effective in preparing for drought?
 - A) Depending only on relief food
 - B) Growing drought-resistant crops for people only
 - C) Conserving extra forage in wet seasons for use in dry seasons
 - D) Selling all livestock when drought begins
4. A farmer harvests Rhodes grass during the rainy season. What is the most important step to make good quality hay right after cutting the grass?
 - A) Adding molasses to improve taste
 - B) Spreading it under the sun to dry
 - C) Stacking it in a barn immediately
 - D) Chopping it into small pieces
5. A farmer wants to conserve maize stalks and leaves after harvest for dry season feeding. Which method is most suitable to reduce spoilage and allow good storage?
 - A) Making silage
 - B) Baling with a box baler
 - C) Feeding fresh stover directly
 - D) Composting the stover
6. A farmer finds that stored hay has a moldy smell and has darkened. What is the most likely reason for this?
 - A) It was stored in a well-ventilated place
 - B) It was harvested at the right maturity
 - C) It was stored with too much moisture
 - D) It was exposed to sunlight in storage
7. Learners discuss the importance of hay. One says it helps animals stay healthy during drought. Which aspect of animal health does hay most directly support?
 - A) Increased milk production
 - B) Prevention of internal parasites
 - C) Maintaining body weight and nutrition
 - D) Stronger immunity against all diseases
8. A community suffers from cattle rustling during drought when animals wander in search of pasture. How can hay conservation help reduce this problem?
 - A) By encouraging more rustling because animals are valuable
 - B) By keeping animals near homesteads since feed is available
 - C) By creating more competition for water
 - D) By forcing farmers to sell animals



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9. A farmer is making hay from lucerne by drying it on the ground. Which weather condition is best for this process?
 - A) Cloudy with light drizzles
 - B) Hot, sunny with a gentle breeze
 - C) Humid with light rain
 - D) Cool and foggy
10. A young farmer notices that some dried grasses lose many leaves while others keep them. Which forage is best at retaining its leaves when dried into hay?
 - A) Napier grass
 - B) Sudangrass
 - C) Oat hay
 - D) Legumes like lucerne or desmodium
11. At what stage should Rhodes grass be harvested to make high-quality hay?
 - A) When very young and leafy, before flowering
 - B) At full bloom with seeds
 - C) When flowering starts but seeds are not mature
 - D) After seeds have dried and matured
12. A relief agency wants feed that is easy to transport, store, and distribute in drought areas. Which conserved feed is best?
 - A) Fresh forage
 - B) Silage in pits
 - C) Baled hay
 - D) Standing forage
13. A farmer fears spontaneous combustion in their hay stack. What usually causes this?
 - A) Hay that is too dry when stacked
 - B) Hay exposed to strong winds
 - C) High moisture in hay leading to microbial heating
 - D) Hay stored in an open field
14. Learners see a simple box baler made from local materials. What is its main advantage to a small farmer?
 - A) Increases nutritional value of hay
 - B) Reduces need for drying
 - C) Compacts hay for easier storage and transport
 - D) Removes need for manual labor
15. To keep hay nutritious, which practice is most important?
 - A) Over-drying until brittle
 - B) Frequent vigorous raking
 - C) Fast drying under good conditions to preserve leaves
 - D) Allowing exposure to dew and light rain
16. What is the main challenge of making hay from maize stover compared to forage grasses?
 - A) Stover has more protein
 - B) Stover is fibrous, less palatable and digestible
 - C) Stover dries too quickly
 - D) Stover attracts more pests
17. How does hay conservation help farms withstand climate change and droughts?
 - A) By increasing water use
 - B) By providing a backup feed source in scarcity
 - C) By encouraging monoculture
 - D) By relying on aid



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18. A farmer with goats plans to conserve hay. What is most important in choosing the forage type?
A) The forage's color when fresh
B) How palatable and nutritious it is for goats
C) The forage's height at harvest
D) Cost of buying a baler
19. Why should hay be stored in a shed instead of an open field?
A) To stop it from becoming too dry
B) To protect from rain, sunlight, and moisture
C) To make transport easier
D) To allow fermentation
20. A learner presents on economic benefits of hay conservation. Which is one direct benefit to farmers?
A) Lower feeding labor costs
B) Higher animal prices in wet season
C) Avoiding costly feeds in drought
D) More government subsidies
21. A farmer wants to protect hay from pests and moisture. Which material is best to cover the stack?
A) Thin cotton cloth
B) Black plastic sheeting
C) Old sacks with holes
D) Iron sheets
22. Learners see a farmer turning drying grass in the field. Why is this done?
A) To mix grass types
B) To dry evenly and avoid mold
C) To compact hay before baling
D) To remove weeds
23. A semi-arid farmer wants hay but has little land for forage. Which strategy is best?
A) Grow only cash crops
B) Use crop residues like maize stover and bean haulms
C) Buy hay from other regions
D) Reduce livestock numbers
24. What is the main advantage of baling hay instead of stacking it loosely?
A) Increases nutritional value
B) Reduces volume for easier storage
C) Improves palatability
D) Removes need for drying
25. A farmer explains to their children that conserving hay in wet seasons for use in dry seasons is an example of:
A) Wasteful management
B) Short-term planning
C) Sustainable use of resources
D) Reliance on aid
26. Why is the maturity stage of forage important when harvesting for hay?
A) Mature forage is always more nutritious
B) Immature forage is hard to dry
C) Nutrition and palatability depend on maturity
D) Hay color changes with maturity
27. A farmer cuts extra forage in the rainy season to dry as hay. Which manual tool is most suitable for cutting?
A) Plough
B) Slasher or sickle
C) Wheelbarrow
D) Watering can



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28. In some regions, farmers dry forage on the stem in the field. What is the major disadvantage of this compared to baling?
- A) Improves nutritional value
 - B) Easily lost to rain, wind, and wild animals
 - C) Requires advanced machines
 - D) Can be stored forever without spoiling
29. A farmer's stored hay is being damaged by rodents. What is the best prevention method?
- A) Painting stacks with repellent
 - B) Storing hay on the ground
 - C) Elevating hay on pallets and keeping storage clean
 - D) Leaving openings for ventilation
30. What is the principle of hay conservation?
- A) Fermenting forage to keep nutrients
 - B) Drying forage to reduce moisture and prevent spoilage
 - C) Storing in airtight conditions
 - D) Adding chemicals to preserve forage

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Instructions: Answer all the questions in the spaces provided. Show all your working where needed.

Question 1 (15 Marks)

Mr. Kimani is a dairy farmer with 10 cows. During the rainy season he has plenty of pasture, but during the dry season his cows lack feed. He has heard about hay making.

a)

- i) State **three advantages** of conserving hay for animals. (3 Marks)

.....

.....

.....

- ii) Explain the meaning of **moisture content** in hay making and why it is important. (3 Marks)

.....

.....

.....

- b) Outline the **steps of making hay** from Napier grass from cutting to storage. (6 Marks)

.....

.....

.....

.....



c) State **three challenges** Mr. Kimani may face when making hay and give a **solution for each**. (3 Marks)

.....

.....

.....

Question 2 (15 Marks)

An agricultural officer trains farmers on hay conservation. One farmer complains that his hay became moldy and lost leaves.

a)

i) Give **two reasons** why hay may lose nutrients during drying. (4 Marks)

.....

.....

ii) State **two features** of good quality hay. (2 Marks)

.....

.....

b) Describe with the help of a **simple sketch or diagram**, two methods of storing hay on the farm. (6 Marks)

.....

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.....

.....

c) State **three other benefits** of hay conservation apart from feeding animals during drought. (3 Marks)

.....

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KITCHEN AND BACKYARD GARDENING



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Question 3 (15 Marks)

Mwangi, a Grade 9 learner, prepares a demonstration using maize stover and a simple box baler.

a)

i) Outline the **steps of preparing maize stover** for hay making. (3 Marks)

.....

.....

.....

ii) State **two reasons** why maize stover is less preferred than pasture grasses for hay. (2 Marks)

.....

.....

b) Draw and label a **simple box baler**. (5 Marks)

c) Explain **two ways** a box baler helps in conserving hay efficiently on a small farm. (3 Marks)

.....

.....

.....

d) Explain how hay conservation shows the idea of **sustainable use of resources**. (2 Marks)

.....

.....



KITCHEN AND BACKYARD GARDENING



COOKING

Question 4 (15 Marks)



a) Identify the feed conservation method shown. (1 Mark)

.....

b) Describe the **main process** of preparing hay. (4 Marks)

.....

.....

.....

.....

c) Give **three advantages** of conserving feed in the form of bales. (3 Marks)

.....

.....

.....

d) Differentiate between **hay making** and **silage making** based on:

i) Moisture content. (2 Marks)

.....

.....

ii) Method of preservation. (2 Marks)

.....

.....



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iii) Type of forage used. (2 Marks)

.....

.....

iv) Physical form of final product. (1 Mark)

.....

Question 5 (15 Marks)

A severe drought causes many animals to die. Farmers who had hay survived.

a) Explain how hay conservation reduces losses during drought. (3 Marks)

i.

ii.

iii.

b) Name **four local forage plants** in your area that can be used to make hay. (4 Marks)

i.

ii.

iii.

c) State **three socio-economic benefits** of hay conservation to a community. (6 Marks)

i.

ii.

iii.

d) A farmer's animals refuse to eat hay. Give **two reasons related to hay quality**. (2 Marks)

i.

ii.

iii.



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Question 6 (15 Marks)

Grade 9 learners visit a farm where the farmer is preparing a site to stack hay.

a) State **three features** of a good site for stacking hay. (3 Marks)

- i.
- ii.
- iii.

b) Describe **two common methods** of stacking hay on small farms. (4 Marks)

- i.
- ii.

c) Suggest **two ways** to protect hay from rain and **two ways** to protect it from rodents. (4 Marks)

- i.
- ii.

d) Explain how **forage conservation** promotes peace in a community during drought. (4 Marks)

- i.
- ii.

Question 7 (10 Marks)

A farmer is comparing sun-drying of hay with modern methods.

a) State **two benefits** of rapid drying during hay making. (2 Marks)

- i.
- ii.

b) State **three challenges** of making hay during unpredictable rainfall. (3 Marks)

- i.
- ii.
- iii.

c) Explain **three roles of community groups** in promoting hay conservation. (5 Marks)

- i.
- ii.
- iii.



KITCHEN AND BACKYARD GARDENING



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SECTION B: STRUCTURED QUESTIONS (100 Marks)

Instructions: Answer all questions in this section.

Question 1 (20 Marks)

A farmer in Machakos wants to prepare hay for his 5 dairy cows during the dry season.

a) Define hay. (2 marks)

.....

.....

.....

b) Explain three advantages of conserving grass as hay. (6 marks)

i.

ii.

iii.

c) List four types of grasses suitable for hay making in Kenya. (4 marks)

i.

ii.

iii.

iv.

d) State four qualities of good hay. (4 marks)

i.

ii.

iii.

iv.

e) Suggest two challenges a farmer may face in hay conservation. (4 marks)

i.

ii.

iii.



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Question 2 (20 Marks)

Study the diagram of hay storage below:



a) Name the structure shown. (2 marks)

.....

b) Give the advantages of silage as a method of forage preservation.

i.

ii.

iii.

c) State four factors that determine the nutrient content of hay.

i.

ii.

iii.

iv.



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d) Grade 9 prepared the following during conservation of forage. Use the picture to answer the questions that follow.

A

B



a.) Name the two methods of conserving hay above. (2mks)

Method A.....

Method B.....

e.) State three advantages of conserving hay using method A compared to method B.

i.

ii.

iii.

f.) List down four materials required when making hay using method A.

i.

ii.

iii.

iv.

g.) Give three factors to consider when selecting the methods of conserving forage

i.

ii.

iii.

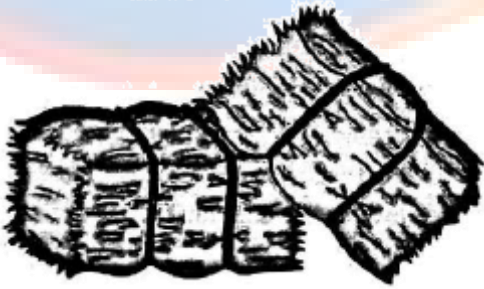


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h) A farmer conserved his forage using the method shown below. use it to answer the questions that follow.



(i) The method used to conserve forage above is known as (1 mark)

(ii) Apart from the method shown above, name other two methods used to conserve forage. (2 marks)

ii) Why is it important for farmers to conserve forages? State two importance. (2 marks)

i) Grade 9 learners want to conserve forage as a class project.

(a) Name two methods of conserving forage the learners can use. (2 marks)

i.

ii.

(b) Give two reasons why farmers should conserve forage. (2 marks)

i.

ii.

iii.

c). Describe the procedure of silage making. (5marks)

i.

ii.

iii.



d).List 3 methods of conserving forage in coping with drought. (3 mks)

- i.
- ii.
- iii.

e) State two advantages of storing hay in this structure. (4 marks)

- i.
- ii.

f) Explain three dangers of storing hay in a poorly ventilated place. (6 marks)

- i.
- ii.
- iii.

g) Outline four precautions to take when storing hay. (4 marks)

- i.
- ii.
- iii.
- iv.

h) Why is baling hay recommended? (4 marks)

- i.
- ii.
- iii.
- iv.



Question 3 (20 Marks)

A group of Grade 9 learners visit a farm where hay is being prepared.

a) State four steps in hay making. (4 marks)

- i.
- ii.
- iii.
- iv.

b) Describe three conditions necessary for hay drying. (6 marks)

- i.
- ii.
- iii.

c) Suggest four tools used in hay making. (4 marks)

- i.
- ii.
- iii.

d) State three signs of good quality hay. (3 marks)

- i.
- ii.
- iii.

e) Explain why hay is important for livestock farmers in arid areas. (3 marks)

- i.
- ii.
- iii.



KITCHEN AND BACKYARD GARDENING



COOKING

Question 4 (20 Marks)

A farmer in Eldoret harvested his hay late when grasses had already seeded.

a) State three disadvantages of harvesting hay late. (6 marks)

- i.
- ii.
- iii.

b) Explain the difference between hay and silage. (6 marks)

- i.
- ii.
- iii.

c) List four problems of storing hay in the open. (4 marks)

- i.
- ii.
- iii.
- iv.

d) Suggest two methods of preserving hay quality. (4 marks)

- i.
- ii.

Question 5 (20 Marks)

The picture below shows hay being used to feed livestock.



KITCHEN AND BACKYARD GARDENING



COOKING

a) Mention two livestock animals that commonly feed on hay. (2 marks)

i.

ii.

b) Explain three benefits of feeding hay to dairy cows. (6 marks)

i.

ii.

iii.

c) State three disadvantages of feeding animals with poor quality hay. (6 marks)

i.

ii.

iii.

d) Suggest two methods of preventing hay wastage during feeding. (4 marks)

i.

ii.

e) Give three reasons why hay is preferred over fresh grass in dry seasons. (2 marks)

i.

ii.

iii.



KITCHEN AND BACKYARD GARDENING



COOKING

STRAND 1.0: CONSERVATION OF RESOURCES
SUB-STRAND 1.2: CONSERVING LEFTOVER FOODS
SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. After a family meal, some rice is left over. The family wants to eat it the next day. What should they do to keep it safe?
A) Leave it on the dining table overnight.
B) Put it in an airtight container and refrigerate once it cools.
C) Place it in a bowl inside the kitchen cabinet.
D) Reheat it immediately then store it.
2. A learner cooked too many mashed potatoes. What is the best way to reuse them for breakfast?
A) Throw them away.
B) Reheat and serve again as they are.
C) Make patties, fry them, and serve with eggs.
D) Mix with raw vegetables and eat cold.
3. A family often has leftover ugali. How can they best reduce wastage?
A) Throw away ugali after one hour.
B) Cook very tiny portions always.
C) Use the leftover ugali to make fried ugali for breakfast.
D) Give all ugali leftovers to pets.
4. Children often leave half-eaten fruits. What should a parent teach them?
A) Always throw away bitten fruit.
B) Finish eating the fruit or store the good part properly for later use.
C) Leave the fruit on the table.
D) Feed the fruit to wild animals.
5. A family cooked too much stew for lunch. How can they store it safely for the next day?
A) Keep it in a warm oven.
B) Freeze it immediately after cooling.
C) Leave it covered on the counter.
D) Add extra salt to it.
6. How does conserving leftover food help a family economically?
A) It raises the electricity bill.
B) It reduces the need to buy more food, saving money.
C) It requires expensive storage containers.
D) It encourages eating out.
7. A family often has leftover chapati. What can they do to make a snack from it?
A) Make fresh chapati again.
B) Cut into strips, fry until crispy, and add salt or sugar.
C) Boil the chapati to make soup.
D) Put it directly into compost.
8. In a hygiene lesson, learners learn how to handle leftover chicken. What should be done to avoid food poisoning?
A) Leave it at room temperature for hours.
B) Reheat it thoroughly before serving.
C) Store it uncovered in the pantry.
D) Wash it with cold water before reheating.



KITCHEN AND BACKYARD GARDENING



COOKING

9. A family often has leftover vegetables. What is the best way to avoid waste?
 - A) Throw away all leftovers.
 - B) Use them in an omelet or frittata the next day.
 - C) Buy fewer vegetables next time.
 - D) Feed them to stray animals.
10. A young chef has some leftover pasta. How can they reuse it creatively?
 - A) Throw it away.
 - B) Serve it plain again.
 - C) Make a pasta salad with vegetables and dressing.
 - D) Boil it again for longer.
11. What is the main environmental benefit of conserving leftover food?
 - A) It increases greenhouse gases.
 - B) It reduces food waste in landfills.
 - C) It increases water use.
 - D) It has no impact.
12. A mother has leftover bean soup. What is the best way to store it for a few days?
 - A) Leave it in the pot on the stove.
 - B) Put it in shallow containers and refrigerate after cooling.
 - C) Keep it warm in a thermos.
 - D) Reheat it slightly every few hours.
13. A family cooked too much chicken stew. Why is freezing it useful?
 - A) It makes food taste fresher.
 - B) It keeps food for a long time.
 - C) It makes food easier to digest.
 - D) It reduces nutrition.
14. Why should hot leftover food be cooled quickly before refrigeration?
 - A) To prevent the fridge from getting too cold.
 - B) To save electricity.
 - C) To reduce the time food stays in the danger zone where bacteria grow.
 - D) To improve food taste.
15. A grandmother uses rice water to water plants. What does this show?
 - A) Food poisoning.
 - B) Food wastage.
 - C) Resourcefulness and reducing waste.
 - D) More cooking needed.
16. A family has leftover vegetables. How can they safely reuse them?
 - A) Leave them on the counter uncovered.
 - B) Blend into soup or sauce and refrigerate/freeze.
 - C) Reheat them again and again.
 - D) Throw away after 2 hours.
17. What is the main health risk of poor leftover handling?
 - A) Weight gain.
 - B) Food poisoning.
 - C) Tooth decay.
 - D) Allergies.
18. A school cafeteria has leftover bread. What can they make from it?
 - A) Breadcrumbs
 - B) Fresh flour
 - C) Cooking oil
 - D) Sugar syrup



KITCHEN AND BACKYARD GARDENING



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19. How can a family successfully use leftovers in meals?
 - A) Cook extra food daily.
 - B) Plan meals to include leftovers in new dishes.
 - C) Avoid leftovers completely.
 - D) Throw away leftovers immediately.
20. Why should rice not be reheated many times?
 - A) It loses flavor.
 - B) It becomes sticky.
 - C) Bacteria may grow if not cooled/stored properly.
 - D) It becomes hard.
21. What should be done with leftover cooked fish?
 - A) Leave it at room temperature.
 - B) Refrigerate immediately, eat within 1–2 days, and reheat well.
 - C) Wash with cold water before storing.
 - D) Give it raw to pets.
22. What is the role of portion control in reducing food wastage?
 - A) Encourages cooking excess food.
 - B) Helps cook/serve only enough food to avoid leftovers.
 - C) Improves food taste.
 - D) Reduces cooking time.
23. Which is good hygiene when handling cooked leftovers?
 - A) Using the same spoon for raw and cooked food.
 - B) Putting hot food directly in a deep container in the fridge.
 - C) Washing hands before handling leftovers.
 - D) Leaving leftovers uncovered.
24. A family freezes leftover meat sauce. What method of food conservation is this?
 - A) Dehydration
 - B) Canning
 - C) Freezing
 - D) Fermentation
25. A Grade 9 learner is making a poster on food wastage. What impact should they highlight?
 - A) More food security.
 - B) Economic losses for families and the country.
 - C) Better environmental health.
 - D) Reduced farming needs.
26. How can leftover potatoes be reused?
 - A) Make potato juice.
 - B) Mash with other ingredients to make potato cakes.
 - C) Dry them in the sun.
 - D) Make potato wine.
27. A family adds leftover boiled greens into scrambled eggs. What does this show?
 - A) Lack of nutrition.
 - B) Resourcefulness and waste reduction.
 - C) Excessive cooking.
 - D) Love for cold food.



KITCHEN AND BACKYARD GARDENING



COOKING

28. How does conserving leftover food improve food security?
- Increases food imports.
 - Ensures more food is available during scarcity.
 - Makes food expensive.
 - Reduces food variety.
29. During a birthday party, a family has a lot of cake left. The mother stores it in the fridge for the next day. Why is this important?
- To prevent it from spoiling
 - To make it heavier
 - To change its colour
 - To waste electricity
30. A learner observes that his grandmother keeps leftover ugali in a clean container instead of leaving it uncovered. What is the main reason?
- To keep away dust and flies
 - To make it look attractive
 - To waste time
 - To spoil it faster
31. A group of learners in school notices that many learners leave food on their plates in the dining hall. What is the result of this practice?
- Increased food wastage
 - Better food conservation
 - Reduced garbage
 - Improved hygiene
32. A family has leftover beans. Instead of throwing them away, they are reheated and eaten the next day. This practice helps to:
- Reduce wastage and save resources
 - Spoil food quickly
 - Waste fuel and time
 - Increase hunger
33. Why is it better to prepare new recipes from leftover foods instead of throwing them away?
- To avoid food wastage and encourage creativity
 - To waste cooking fuel
 - To spoil food intentionally
 - To increase garbage at home
34. A mother prepares soup using leftover vegetables from the previous day. This shows the importance of:
- Reducing food wastage through reuse
 - Throwing away food carelessly
 - Mixing spoilt food with fresh food
 - Keeping food uncovered



35. During Agriculture class, learners are asked to make porridge from leftover ugali. What skill are they developing?
 - A. Creativity in food preparation
 - B. Laziness in food handling
 - C. Ignorance of hygiene
 - D. Wasting resources
36. Why should leftover food not be left uncovered overnight?
 - A. It attracts flies and germs, making it unsafe
 - B. It becomes tastier
 - C. It becomes stronger
 - D. It increases food security
37. A family covers leftover food with a lid and stores it in the fridge. Which food handling value are they practicing?
 - A. Responsibility and hygiene
 - B. Laziness and ignorance
 - C. Dishonesty and greed
 - D. Carelessness and wastefulness
38. Which of the following best explains the importance of conserving leftover foods at home?
 - A. Saves money, reduces wastage, and promotes food security
 - B. Increases garbage, spreads disease, and wastes fuel
 - C. Reduces appetite, spoils food, and wastes energy
 - D. Creates hunger, promotes greed, and wastes resources

SET 2

STRAND 1.0: CONSERVATION OF RESOURCES

SUB-STRAND 1.2: CONSERVING LEFTOVER FOODS

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Read each situation carefully and choose the best answer.

Questions 1–10: Importance of Conserving Leftover Foods

1. During Christmas, a family cooks more ugali than they can finish. They keep the remaining ugali for the next day's meal. Why is this important?
 - A. To save money and reduce food wastage
 - B. To increase cooking time
 - C. To spoil the food faster
 - D. To reduce family members' appetite
2. Jane stores leftover rice in a clean airtight container inside the fridge after supper. Why is this practice important?
 - A. To make it look attractive
 - B. To keep it safe and prevent contamination
 - C. To increase its weight
 - D. To change its taste



3. A learner observes that her parents always keep leftover food well covered instead of throwing it away. What is the main reason for this?
 - A. To waste fewer resources
 - B. To impress visitors
 - C. To practice new recipes only
 - D. To avoid buying more utensils
4. At school, learners are taught to keep food safely and avoid waste. Conserving leftover food helps them develop which value?
 - A. Integrity and responsibility
 - B. Laziness and wastefulness
 - C. Greed and selfishness
 - D. Ignorance and untidiness
5. In the evening, a family warms leftover chapati on a pan before eating. What is the benefit of this practice?
 - A. To reduce wastage and make food safe to eat
 - B. To make chapati harder to chew
 - C. To make chapati lose nutrients
 - D. To waste cooking gas
6. When leftover foods are conserved properly, families benefit by:
 - A. Preventing foodborne diseases
 - B. Reducing food wastage
 - C. Saving money
 - D. All of the above
7. A learner finds his mother covering leftover food with a clean lid before storage. What does this show?
 - A. Covering and storing them properly
 - B. Leaving them uncovered in the kitchen
 - C. Giving them to pets only
 - D. Throwing them away immediately
8. Leftover beans are put in the fridge by a family in Eldoret. Why is this important?
 - A. To make them spoil quickly
 - B. To preserve them for later use
 - C. To reduce their nutritional value
 - D. To avoid eating them
9. When families conserve leftover food, it contributes to:
 - A. Hunger in homes
 - B. Food security
 - C. Pollution
 - D. Dishonesty
10. A dustbin at home is full of food remains like rice, ugali, and vegetables. What does this show about the family?
 - A. Proper use of resources
 - B. Food wastage and poor conservation
 - C. Healthy eating practices
 - D. Responsible food storage



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11. A mother finds extra ugali left from lunch. She cuts it into pieces and fries it with onions. What is the main purpose of this preparation?
- A. To create ugali chips and reduce wastage
 - B. To keep it in the sun
 - C. To throw it away
 - D. To give it to animals immediately
12. A learner observes fried leftover rice with vegetables served for supper. This is an example of:
- A. Food wastage
 - B. A new recipe from leftovers
 - C. Poor hygiene in food handling
 - D. Spoilt food
13. To make sure leftover food warms evenly before eating, the best method is:
- A. Microwaving or covering while heating
 - B. Leaving food uncovered overnight
 - C. Boiling without covering
 - D. Throwing it away
14. Vegetables remain after dinner. The next morning, the family fries them with eggs. This is an example of:
- A. Preparing leftovers for reuse
 - B. Throwing them into the dustbin
 - C. Leaving them outside overnight
 - D. Mixing them with spoilt food
15. A learner wants to eat leftover meat from the fridge. What should he do first?
- A. Reheat it properly to kill germs
 - B. Eat it directly when cold
 - C. Mix it with spoiled vegetables
 - D. Leave it uncovered
16. A family has leftover ugali. Which new recipe can they make from it?
- A. Porridge
 - B. Rice pilau
 - C. Boiled potatoes
 - D. Chapati
17. Leftover chapati is cut into small pieces, fried with stew, and served. This food is called:
- A. Chapati chips with stew
 - B. Compost manure
 - C. Pet food
 - D. Waste in the bin



18. When beans left overnight are reheated properly before eating, the main benefit is:
- A. The food becomes safe for eating
 - B. Food loses nutrients
 - C. Food spoils faster
 - D. Food looks unattractive
19. A learner mixes leftover rice with eggs, fries it, and serves it for lunch. This practice shows:
- A. Creativity and imagination
 - B. Waste of resources
 - C. Laziness in food handling
 - D. Irresponsibility
20. After supper, leftover food is stored in the fridge before reuse. Why is this step important?
- A. Refrigeration keeps food safe until prepared again
 - B. Leaving food uncovered prevents germs
 - C. Mixing with raw foods makes it last longer
 - D. Pouring water on food preserves it
21. A family uses leftover spinach to make vegetable soup the next day. This promotes:
- A. Food wastage
 - B. Resourcefulness
 - C. Poor hygiene
 - D. Laziness
22. In a homestead, githeri left from lunch is fried with onions and tomatoes for supper. This practice demonstrates:
- A. Food wastage
 - B. Reuse of leftovers creatively
 - C. Poor cooking methods
 - D. Bad food storage
23. Learners in a school competition are asked to prepare meals using leftover food. Which values are they likely to show?
- A. Hygiene and integrity
 - B. Greed and selfishness
 - C. Wastage and carelessness
 - D. Dirtiness
24. A teacher encourages learners to make new recipes from leftover foods. This promotes:
- A. Creative cooking methods
 - B. Throwing away food
 - C. Spoiling food intentionally
 - D. Wasting cooking fuel



26. When learners prepare meals from leftover foods during practical lessons, they mainly develop:
- A. Creativity and experimentation skills
 - B. Laziness in cooking
 - C. Ignorance of hygiene
 - D. Poor attitudes towards food
27. A group of learners presents dishes made from leftovers during Agriculture class. What skill are they practicing?
- A. Communication and teamwork skills
 - B. Wasting resources
 - C. Dirty cooking habits
 - D. Bad table manners
28. Why is it important for families to embrace the use of leftover foods?
- A. To reduce wastage and save resources
 - B. To spoil food faster
 - C. To increase garbage collection
 - D. To waste family money
29. Four pictures are shown to learners:

(a) Food in dustbin,



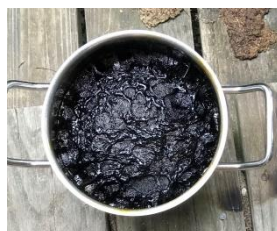
(b) Covered food in fridge,



(c) Spoilt food on the floor,



(d) Burnt food in pan.



Which picture shows proper use of leftover foods?

A. A

B. B

C. C

D. D



30. Learners are encouraged to create new recipes from leftover foods. Why is this important?
- A. It helps them become innovative in food preparation
 - B. It wastes time and food
 - C. It avoids learning hygiene skills
 - D. It increases garbage in homes
31. After supper, Mary covers the leftover fish with a clean plate before putting it in the fridge. What is the main reason for covering the food?
- A. To keep away dust and germs
 - B. To increase its weight
 - C. To make it look shiny
 - D. To spoil it quickly
32. A learner eats leftover rice that was left uncovered in the kitchen overnight. He later suffers from stomachache. What caused this?
- A. Proper food storage
 - B. Food contamination
 - C. Safe handling of food
 - D. Creativity in cooking
33. When handling leftover foods, which of the following practices is MOST important to prevent food poisoning?
- A. Washing hands before touching food
 - B. Talking while cooking
 - C. Using dirty utensils
 - D. Leaving food on the floor
34. A mother advises her children never to mix leftover cooked food with raw meat in the same container. Why is this advice important?
- A. Raw meat can contaminate the cooked food
 - B. To make food heavy
 - C. To increase food quantity
 - D. To save on utensils
35. A learner warms leftover githeri but fails to heat it thoroughly. What could happen if eaten?
- A. Food may cause illness
 - B. Food becomes tastier
 - C. Food grows faster
 - D. Food lasts longer
36. Which of the following leftover foods is most likely to spoil quickly if not refrigerated?
- A. Rice
 - B. Ugali
 - C. Dried maize
 - D. Millet flour



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37. Why should leftover food be eaten within 2–3 days of storage?
- To avoid spoilage and food poisoning
 - To waste more food
 - To use more electricity
 - To reduce appetite
38. A boy packs yesterday's leftover sukuma wiki for school lunch. Which food handling step is MOST important?
- Storing it in a clean container
 - Leaving it uncovered
 - Mixing it with spoilt food
 - Putting it in a paper bag
39. Which ONE shows good hygiene in handling leftovers?
- Food left uncovered on a table
 - Food stored in clean containers in a fridge
 - Food thrown on the floor
 - Food eaten by flies
- U
 - V
 - W
 - X
40. A learner wants to keep leftover ugali safe. Which of the following is the BEST storage method?
- Refrigerating in a clean container
 - Placing it on the floor
 - Leaving it uncovered in the kitchen
 - Putting it in a dirty polythene bag
41. A teacher asks learners to prepare new dishes from leftover foods. What skill are they practicing?
- Creativity and imagination
 - Wastage of resources
 - Laziness in cooking
 - Ignorance of hygiene
42. A family prepares leftover ugali with milk to make porridge. This is an example of:
- Innovation in food preparation
 - Food wastage
 - Poor cooking practice
 - Dirty food handling
43. A learner makes a fruit salad using leftover bananas and mangoes. Which value is being promoted?
- Creativity and proper resource use
 - Wasting food carelessly
 - Dishonesty in food handling
 - Ignorance of hygiene



44. Why is it important for schools to teach learners how to use leftover foods?
- A. To encourage innovation and reduce wastage
 - B. To waste more food
 - C. To spoil food quickly
 - D. To increase hunger
45. A group of learners makes presentations of recipes created from leftover foods in class. Which core competency are they practicing?
- A. Communication and collaboration
 - B. Food wastage
 - C. Laziness in cooking
 - D. Ignorance
46. A mother uses leftover potatoes to make mashed potato cakes for breakfast. Which competency does this show?
- A. Creativity and imagination
 - B. Wastage of food
 - C. Dishonesty in cooking
 - D. Ignorance of resources
47. When a family avoids wasting leftover food, which national value are they showing?
- A. Integrity and responsibility
 - B. Laziness and wastefulness
 - C. Greed and selfishness
 - D. Untidiness
48. During Agriculture practicals, learners are asked to reheat and prepare safe meals from leftovers. What key value are they promoting?
- A. Hygiene in food handling
 - B. Food wastage
 - C. Carelessness
 - D. Laziness
49. Why should learners apply creativity when preparing leftover foods?
- A. To discover new recipes and reduce waste
 - B. To waste family resources
 - C. To make food unsafe
 - D. To ignore hygiene
50. A household uses all leftover food by preparing soups, salads, and fried dishes. What is the main outcome?
- A. Reduced food wastage and better food security
 - B. Increased hunger and poor health
 - C. More garbage at home
 - D. Wastage of money



SECTION B: STRUCTURED QUESTIONS (100 Marks)

Question 1 (20 marks)

A family in Kericho prepares a lot of food during Christmas celebrations. After the feast, some ugali, rice, and beans remain. The mother decides to keep them safely for later use.

a) Explain **three reasons** why it is important to conserve leftover foods at home. (6 marks)

- i.
- ii.
- iii.

b) State **two values** promoted when a family avoids wasting leftover foods. (4 marks)

- i.
- ii.

c) Identify **two risks** of eating leftover food that is not well stored. (4 marks)

- i.
- ii.

d) Suggest **two ways** the family can prepare leftover ugali for reuse. (4 marks)

- i.
- ii.

e) Explain how conserving leftover foods supports **food security in Kenya**. (2 marks)

- i.
- ii.

Question 2 (20 marks)

At a boarding school, learners often remain with some cooked food after meals. The Agriculture teacher encourages them to practice proper conservation of leftovers during practical lessons.

a) Describe **two safe storage methods** for leftover foods. (4 marks)

- i.
- ii.



b) Explain **three advantages** of reheating leftover food before eating. (6 marks)

- i.
- ii.
- iii.

c) Suggest **two creative recipes** that can be made from leftover chapati. (4 marks)

- i.
- ii.

d) Identify **two hygiene practices** learners should observe when handling leftover food. (4 marks)

- i.
- ii.

e) State **one core competency** developed when learners prepare meals from leftovers. (2 marks)

- i.

Question 3 (20 marks)

A family in Mombasa prepares pilau for lunch. Some rice remains, and the next day the daughter decides to fry it with eggs and vegetables for supper.

a) Explain the importance of preparing leftover foods into new recipes. (4 marks)

- i.
- ii.
- iii.
- iv.

b) Identify **three dangers** of eating leftover food without reheating. (6 marks)

- i.
- ii.
- iii.



c) Suggest **two leftover foods** that can be used to prepare porridge. (4 marks)

i.

ii.

d) State **two ways** in which leftover vegetables can be reused. (4 marks)

i.

ii.

e) Mention **one national value** shown when a family uses leftover food wisely. (2 marks)

i.

ii.

Question 4 (20 marks)

Learners in Grade 9 Agriculture are asked to present creative meals made from leftover foods. Some prepare fried githeri, others vegetable soup, and others chapati chips.

a) Explain **three benefits** of embracing the use of leftover foods in schools and homes. (6 marks)

i.

ii.

iii.

b) State **two environmental problems** caused by throwing away leftover foods. (4 marks)

i.

ii.

c) Identify **two creative skills** learners practice when making new recipes from leftovers. (4 marks)

i.

ii.

d) Suggest **two examples** of leftover foods that can be turned into soup. (4 marks)

i.

ii.



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COOKING

e) Mention **one value** promoted when learners share recipes made from leftover foods. (2 marks)

i.

Question 5 (20 marks)

During a wedding ceremony, a lot of food remains after guests are served. The event organizers decide to distribute some to needy families and preserve some for later use.

a) Give **three reasons** why conserving leftover foods is important in such events. (6 marks)

i.

ii.

iii.

b) Suggest **two safe methods** of transporting leftover foods to needy families. (4 marks)

i.

ii.

c) Identify **two challenges** that may arise when handling leftover foods during such events. (4 marks)

i.

ii.

d) Explain **two ways** leftover potatoes can be reused at home. (4 marks)

i.

ii.

e) State **one core competency** developed when learners learn to conserve leftover foods. (2 marks)

i.

ii.



SET 2

SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all the questions in the spaces provided. Show your working where needed.

Question 1 (10 Marks)

A family often remains with cooked food after meals.

a) Give three reasons why leftover foods should be conserved. (3 Marks)

i.

ii.

iii.

b) Explain the meaning of the danger zone in food safety. (2 Marks)

i.

ii.

c) State two ways cooling leftover food quickly helps to keep it safe. (2 Marks)

i.

ii.

d) Suggest three practical methods of preparing leftover foods to avoid wastage. Give one example of food for each method. (3 Marks)

i.

ii.

iii.

Question 2 (10 Marks)

Jane notices that many edible foods are thrown away in a community kitchen.

a) State two hygiene practices important when handling leftover food. (2 Marks)

i.

ii.



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b) Give three new dishes that can be made from leftover rice and vegetables. (6 Marks)

- i.
- ii.
- iii.

c) Explain how the value of *integrity* helps in reducing food wastage. (2 Marks)

- i.
- ii.

Question 3 (10 Marks)

A refrigerator has neatly packed and labeled containers of cooked food.

a) What method of food conservation is shown in the refrigerator? (1 Mark)

- i.

b) State three benefits of storing leftover food in labeled containers. (3 Marks)

- i.
- ii.
- iii.

c) Give two factors that affect how long cooked food can be stored in a fridge. (2 Marks)

- i.
- ii.

d) Why is it important to heat leftover food evenly before eating? (2 Marks)

- i.
- ii.

e) Give two ways of reducing the amount of leftover food before cooking. (2 Marks)

- i.
- ii.
- iii.



Question 4 (10 Marks)

A health clinic advises families to embrace the use of leftover foods.

a) What does the phrase mean? (2 Marks)

.....

.....

b) Give three creative uses of leftover bread. (3 Marks)

- i.
- ii.
- iii.

c) Explain two ways conserving leftover food can improve a country's food security. (3 Marks)

- i.
- ii.
- iii.

d) Why should families check taste and texture when reusing leftover food? (2 Marks)

- i.
- ii.

Question 5 (10 Marks)

Kevin wants to prepare a snack using leftover boiled potatoes and cooked sausages.

a) Write down steps he can follow to make a new snack from these leftovers. (4 Marks)

- i.
- ii.
- iii.
- iv.



b) State three competencies Kevin is gaining from this activity. (3 Marks)

i.

ii.

iii.

c) Explain the link between conserving leftover foods and controlling food-borne diseases. (3 Marks)

i.

ii.

iii.



KITCHEN AND
BACKYARD GARDENING



COOKING

STRAND 1.0: CONSERVATION OF RESOURCES

SUB-STRAND 1.3: INTEGRATED FARMING

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. A farmer has a small piece of land and wants to maximize resource use. They keep fish in a pond, and the pond water is used to irrigate vegetables. This practice is called:
A) Monocropping
B) Shifting cultivation
C) Integrated farming
D) Pastoralism
2. In an integrated farm, chicken droppings are used as fertilizer in the garden. Which resource is mainly conserved?
A) Water
B) Land
C) Nutrients
D) Sunlight
3. A community adopts integrated farming to conserve the environment. How does this system help most?
A) By using more chemical fertilizers
B) By reducing waste and reusing resources on the farm
C) By cutting down forests for new land
D) By encouraging single-crop farming
4. A farmer keeps rabbits and uses their manure to fertilize vegetables. This helps conserve:
A) Pest control
B) Animal feed
C) Fertilizers
D) Water for irrigation
5. A learner designs a farm with ducks, fish, and rice paddies. Ducks eat insects and weeds, and their droppings fertilize rice and fish ponds. This shows the principle of:
A) Specialization
B) Linear production
C) Synergistic relationships and recycling
D) Dependence on external inputs
6. Which is a key component in integrated farming?
A) A separate industrial plant
B) Interconnected enterprises where outputs from one are inputs for another
C) Growing only one cash crop
D) Relying on non-renewable resources
7. What is the main benefit of integrated farming for a small-scale farmer?
A) It needs heavy machines
B) It gives diverse income and saves resources
C) It limits crop and animal choices
D) It always increases disease outbreaks



KITCHEN AND BACKYARD GARDENING



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8. In an integrated farm, leftover vegetables are fed to rabbits, and their manure fertilizes the garden. This helps reduce:
 - A) Need for manual labor
 - B) Food production
 - C) Waste and use of outside inputs
 - D) Water use
9. A Grade 9 class builds an integrated farm model with biogas from manure. What resource is conserved?
 - A) Water
 - B) Animal feed
 - C) Energy for cooking and lighting
 - D) Building materials
10. A farmer collects rainwater for fish farming and later uses it to irrigate crops. This conserves:
 - A) Land
 - B) Labor
 - C) Water
 - D) Animal feed
11. Why do learners design models of integrated farming?
 - A) For fun
 - B) To show how farm parts connect and recycle resources
 - C) To show farming is difficult
 - D) To display animals
12. A farmer keeps ducks under fruit trees. Ducks eat pests and weeds, reducing use of:
 - A) Water
 - B) Chemical pesticides
 - C) Sunlight
 - D) Fertilizer
13. Which example best shows conservation of resources through integrated farming?
 - A) Buying fertilizer from shops
 - B) Dumping manure into a river
 - C) Using compost and animal manure on fields
 - D) Clearing forest for new farms
14. A learner presents a farm where poultry droppings feed Black Soldier Fly larvae, which are used to feed fish. This conserves:
 - A) Land for housing
 - B) Commercial animal feed
 - C) Water for fish
 - D) Sunlight
15. Why is respecting others' creative work important when making farm models?
 - A) It makes work faster
 - B) It encourages teamwork and new ideas
 - C) It makes all models look the same
 - D) It is not important
16. A farmer feeds pigs vegetable remains, and the manure is used in the garden. This cycle mainly addresses:
 - A) Pest control
 - B) Waste management and nutrient recycling
 - C) Water shortage
 - D) Labor intensity



17. How does integrated farming improve community food security?
 - A) By producing only one type of food
 - B) By producing many foods and making farms stronger to shocks
 - C) By depending on imported food
 - D) By requiring huge land
18. A farmer uses fish pond water to irrigate bananas. This is called:
 - A) Aquaponics
 - B) Hydroponics
 - C) Monoculture
 - D) Organic farming
19. What is the role of relationships between fish, rabbits, and vegetables in an integrated farm?
 - A) They increase competition
 - B) They make each unit work alone
 - C) They show how outputs become inputs for others
 - D) They show poor planning
20. A farmer wants to reduce synthetic fertilizer use. Which farm component helps most?
 - A) Storage shed
 - B) Animal husbandry
 - C) High-yield crops only
 - D) External water sources
21. What does resource cycling mean in integrated farming?
 - A) Throwing away resources
 - B) Moving resources to another farm
 - C) Reusing and transforming resources within the farm
 - D) Using only new resources
22. Which environmental benefit is linked directly to integrated farming?
 - A) More fossil fuel use
 - B) Less pollution from farm waste
 - C) More land degradation
 - D) Single-crop ecosystems
23. Poultry waste is composted and used in a vegetable garden. This conserves:
 - A) Water
 - B) Labor
 - C) Soil nutrients
 - D) Seeds
24. Integrated farming is called holistic because:
 - A) It solves one problem at a time
 - B) It views the farm as one ecosystem where all parts interact
 - C) It specializes in one product
 - D) It ignores the environment
25. A farmer keeps fish and pigs. How can pigs benefit the fish pond?
 - A) Let pigs swim in the pond
 - B) Use pig manure as fish feed
 - C) Keep pigs far away
 - D) Feed pigs with fish waste



KITCHEN AND BACKYARD GARDENING



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26. Which value is promoted by integrated farming?
 - A) Wastefulness
 - B) Competition
 - C) Prudence
 - D) Extravagance
27. A Grade 9 learner wants to show conservation in their farm model. Which element shows this best?
 - A) Separate plots for each crop
 - B) Recycling waste from one unit into another
 - C) Unused land
 - D) Using only synthetic inputs
28. How should animal waste be managed in an integrated farm?
 - A) Dump into rivers
 - B) Collect, compost, or use anaerobic digestion
 - C) Leave untreated
 - D) Burn openly
29. In an arid area, a farmer grows drought-tolerant crops and harvests water while keeping livestock. This conserves:
 - A) Labor
 - B) Water resources
 - C) Building materials
 - D) Animal houses
30. A farmer integrates beekeeping with crop farming. Bees pollinate crops while providing honey. This mainly conserves:
 - A) Water resources
 - B) Pollination services
 - C) Farm labor
 - D) Fertilizer use
31. A Grade 9 learner visits a farm where goat manure is used in biogas production, and the slurry is spread in the garden. This cycle shows conservation of:
 - A) Energy and soil nutrients
 - B) Water and land
 - C) Seeds and pesticides
 - D) Time and labor
32. In an integrated farm, fish waste is used to fertilize rice fields, while rice husks feed poultry. What is being conserved?
 - A) Crop diversity only
 - B) Nutrient resources and animal feed
 - C) Sunlight only
 - D) Water only
33. A farmer wants to lower farming costs by using integrated systems. Which factor is most reduced?
 - A) Need for external inputs like chemical fertilizers and feeds
 - B) Farm labor requirements
 - C) Access to farm markets
 - D) Need for rainfall



34. Integrated farming helps protect the soil because:
- It avoids recycling of waste
 - Manure and compost improve soil fertility
 - It requires clearing new land each season
 - It uses more chemical fertilizers
35. A group of learners is told to design a small integrated farm model in school. Which example is most appropriate?
- Maize only plot
 - Poultry house, vegetable garden, and compost pit linked together
 - A store for farm inputs
 - A tree nursery only
36. In an integrated farm, crop residues are fed to cows, and cow manure is used in crop fields. This mainly reduces:
- Cost of fertilizers
 - Rainfall shortages
 - Pest invasions
 - Cost of farm machinery
37. Which of the following is a *social benefit* of integrated farming?
- It promotes teamwork and sharing of ideas among farmers
 - It increases waste on farms
 - It discourages collaboration
 - It focuses on one enterprise only
38. Why is integrated farming considered *sustainable*?
- It depends heavily on external resources
 - It creates closed loops of recycling resources
 - It wastes natural resources
 - It promotes monocropping
39. A farmer introduces earthworms in a compost system to break down manure and crop waste. This practice is known as:
- Hydroponics
 - Vermicomposting
 - Aquaponics
 - Silviculture
40. A farmer keeps dairy cows and uses the manure to produce biogas. The slurry is spread on maize fields, and maize stalks are later fed back to the cows. This cycle is an example of:
- Resource wastage
 - Open system farming
 - Resource recycling and integration
 - Monocropping



KITCHEN AND BACKYARD GARDENING



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SET 2

STRAND 1.0: CONSERVATION OF RESOURCES

SUB-STRAND 1.3: INTEGRATED FARMING

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

(Answer ALL questions. Each question carries 1 mark.)

Q1. After lunch, a farmer collects vegetable leftovers and gives them to rabbits. This practice is part of:

- A. Wastage of food
- B. Integrated farming
- C. Poor feeding
- D. Monocropping

Q2. A farmer uses water from a fish pond to irrigate sukuma wiki. This practice shows:

- A. Soil erosion
- B. Integrated farming
- C. Misuse of water
- D. Food wastage

Q3. Which of the following is NOT a component of integrated farming?

- A. Poultry keeping
- B. Fish farming
- C. Vegetable growing
- D. Burning plastics

Q4. A Grade 9 learner notices that rabbit droppings are being used to fertilize vegetables. This shows:

- A. Overgrazing
- B. Recycling of farm waste
- C. Soil erosion
- D. Crop failure

Q5. A farmer keeps fish, rabbits, and vegetables on the same land. This system is called:

- A. Monoculture
- B. Integrated farming
- C. Crop rotation
- D. Mixed grazing

Q6. A homestead farmer keeps hens whose droppings are used in a fish pond. The droppings act as:

- A. Fish feed
- B. Farm poison
- C. Weeds
- D. Pesticides

Q7. A family uses cow dung in a biogas digester to cook meals. This helps them to:

- A. Increase pollution
- B. Produce clean energy and manure
- C. Waste resources
- D. Reduce farm yields

Q8. A school farm grows maize, keeps poultry, and rears fish. The main aim of such integration is:

- A. Conservation of resources
- B. Destruction of soil
- C. Increasing waste
- D. Overuse of land



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Q9. A farmer sells eggs, vegetables, and fish from the same plot. This is an example of:

- A. Resource wastage
- B. Diversified income
- C. Monocropping
- D. Overgrazing

Q10. Integrated farming mainly benefits:

- A. Farmers with little land
- B. Only large-scale farmers
- C. People with no farms
- D. Urban landlords

Q11. The waste from a fish pond can be used to:

- A. Irrigate and fertilize crops
- B. Poison vegetables
- C. Kill soil organisms
- D. Reduce soil fertility

Q12. Rabbit droppings applied on kale farms help in:

- A. Decreasing yields
- B. Improving soil fertility
- C. Reducing soil nutrients
- D. Polluting the farm

Q13. Integrated farming conserves resources because it:

- A. Reuses farm waste
- B. Encourages deforestation
- C. Pollutes rivers
- D. Reduces biodiversity

Q14. A poultry house is built above a fish pond. The main reason is:

- A. To let poultry droppings feed fish
- B. To allow hens to drink pond water
- C. To help hens swim in ponds
- D. To stop fish from escaping

Q15. A farmer chops maize stalks after harvest to feed cows. This practice is called:

- A. Burning waste
- B. Recycling crop residues
- C. Food wastage
- D. Soil erosion

Q16. A farmer keeps bees, goats, and grows maize. This system helps the farmer to:

- A. Depend on one product
- B. Earn income from different sources
- C. Waste farm resources
- D. Practice monocropping



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Q17. If vegetables fail due to pests, a farmer in integrated farming can still earn from:

- A. Weeds
- B. Poultry and fish
- C. Land leasing
- D. Burning trees

Q18. Vegetables irrigated using fish pond water are usually:

- A. Poor in quality
- B. Rich in nutrients
- C. Unsuitable for markets
- D. Less productive

Q19. Integrated farming contributes to food security because it:

- A. Provides different types of food
- B. Promotes soil erosion
- C. Encourages pollution
- D. Reduces soil fertility

Q20. The combination of crops and animals in integrated farming results in:

- A. Lower production
- B. Balanced diet and higher income
- C. More waste
- D. Increased hunger

Q21. A learner observes poultry feathers used in making pillows. This is an example of:

- A. Food wastage
- B. Resource recycling
- C. Poor farming
- D. Overgrazing

Q22. Vegetable peelings can be used as:

- A. Manure or rabbit feed
- B. Plastic waste
- C. Fuel for cars
- D. Soil poison

Q23. The use of cow urine in vegetable farming is important because it:

- A. Improves soil fertility
- B. Wastes resources
- C. Reduces crop yields
- D. Pollutes soil

Q24. A farmer making compost manure from farm waste shows:

- A. Integrated farming practice
- B. Soil erosion
- C. Deforestation
- D. Water pollution



Q25. A Grade 9 learner suggests that poultry droppings should not be thrown away but used to grow vegetables. This shows:

- A. Good understanding of resource conservation
- B. Careless use of manure
- C. Pollution
- D. Poor planning

Q26. Integrated farming reduces pollution because:

- A. Wastes are reused instead of being dumped
- B. Chemicals are overused
- C. Farmers cut all trees
- D. Crops are burnt

Q27. Biodiversity is promoted in integrated farming since:

- A. Only one crop is grown
- B. Different crops and animals coexist
- C. Land is abandoned
- D. Forests are cleared

Q28. A farmer integrates tree planting with poultry. The trees mainly help in:

- A. Preventing erosion and giving shade
- B. Wasting land
- C. Stopping farming
- D. Polluting air

Q29. A farm with fish, poultry, rabbits, and vegetables is an example of:

- A. Environmental conservation
- B. Pollution
- C. Overuse of resources
- D. Monocropping

Q30. Integrated farming helps to control soil erosion by:

- A. Maintaining ground cover with crops
- B. Burning all crop residues
- C. Cutting trees carelessly
- D. Leaving soil bare

Q31. A Grade 9 learner draws a model showing a fish pond, poultry house, and vegetable garden. This shows:

- A. An integrated farming design
- B. Soil erosion
- C. Monocropping
- D. Deforestation

Q32. The main reason for making integrated farming models in class is to:

- A. Understand resource conservation
- B. Waste time
- C. Destroy creativity
- D. Encourage pollution



Q33. A model showing rabbit hutches near a vegetable garden demonstrates:

- A. Use of rabbit waste as manure
- B. Overgrazing
- C. Deforestation
- D. Poor planning

Q34. During group work, learners make integrated farm models using clay and sticks. This helps them to:

- A. Show creativity and teamwork
- B. Waste resources
- C. Destroy the environment
- D. Avoid learning

Q35. A model with a fish pond near a vegetable garden shows the importance of:

- A. Recycling pond water for irrigation
- B. Soil erosion
- C. Monocropping
- D. Food wastage

Q36. During a school trip, learners observe a farm where poultry waste is used in ponds. This demonstrates:

- A. Integration of enterprises
- B. Soil erosion
- C. Pollution
- D. Poor farming

Q37. Learners take notes from an excursion on integrated farming. The main benefit is:

- A. Gaining real-life knowledge
- B. Forgetting classroom work
- C. Wasting school time
- D. Destroying crops

Q38. After an excursion, learners present group findings on integrated farming. This helps them to:

- A. Share knowledge and build confidence
- B. Waste class time
- C. Forget teamwork
- D. Avoid participation

Q39. A teacher asks learners to design integrated farm sketches after research. This improves:

- A. Creativity and application of knowledge
- B. Careless farming
- C. Soil erosion
- D. Pollution

Q40. When learners visit a farm with fish ponds and vegetables, they are most likely learning about:

- A. Integrated farming systems
- B. Desertification
- C. Overgrazing
- D. Pollution



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Q41. Integrated farming ensures food security because it provides:

- A. Different types of food
- B. Only one type of food
- C. Less food variety
- D. Poor nutrition

Q42. Keeping poultry and vegetables together helps a farmer to:

- A. Earn more and feed the family
- B. Waste land
- C. Overuse chemicals
- D. Pollute the farm

Q43. Integrated farming improves family nutrition because:

- A. It provides proteins, vitamins, and energy foods
- B. It reduces food diversity
- C. It wastes resources
- D. It produces only starch

Q44. A farmer earns from selling fish, eggs, and cabbages. This shows that integrated farming:

- A. Improves income sources
- B. Reduces farm profits
- C. Causes food shortage
- D. Encourages wastage

Q45. Integrated farming is important to the environment mainly because it:

- A. Promotes recycling and reduces pollution
- B. Encourages soil erosion
- C. Reduces crop diversity
- D. Promotes food wastage

Q46. Learners clap after a group presents a farm model. This shows:

- A. Respect for creativity
- B. Pollution
- C. Poor teamwork
- D. Carelessness

Q47. Designing an integrated farming model encourages learners to:

- A. Be creative and conserve resources
- B. Waste school time
- C. Destroy the environment
- D. Forget farming practices

Q48. Working in groups to create farm models helps learners to develop:

- A. Teamwork and respect for ideas
- B. Laziness
- C. Pollution
- D. Monocropping



Q49. Learners who appreciate each other's models show:

- A. Respect and cooperation
- B. Disrespect
- C. Poor performance
- D. Wastage

Q50. Integrated farming models in class help learners to:

- A. Apply knowledge practically
- B. Forget what they learnt
- C. Waste materials
- D. Pollute the environment



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COOKING

SET 3

STRAND 2.0: FOOD PRODUCTION PROCESSES

SUB-STRAND 2.1: ORGANIC GARDENING

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. A small-scale farmer notices that their vegetables are often attacked by aphids. Instead of using synthetic chemicals, they spray an extract made from neem leaves. This practice is an example of which organic gardening principle?
A) Use of inorganic fertilizers
B) Mechanical weed control
C) Organic pest control
D) Crop rotation with chemicals
2. A Grade 9 learner is preparing soil for a vegetable patch in an organic gardening project. Which material is most suitable for enriching the soil organically?
A) Synthetic NPK fertilizer
B) Well-decomposed compost
C) Fresh wood ash in large amounts
D) Chemical pesticides
3. A farmer notices that cabbages have yellowing leaves due to nutrient deficiency. They prepare a liquid feed by fermenting animal waste and plant materials such as Mexican sunflower, then apply it to the crops. This is an example of:
A) Synthetic foliar feeding
B) Organic foliar feeding
C) Inorganic soil amendment
D) Chemical pesticide application
4. A school starts an organic garden to grow sukuma wiki. Learners remove weeds by hand or using a jembe. This method of weed control is called:
A) Chemical weeding
B) Biological weed control
C) Mechanical weed control
D) Herbicidal application
5. Why is **honesty (integrity)** an important value when practicing organic gardening, especially for produce meant for sale?
A) It helps to hide chemical use.
B) It ensures adherence to organic standards and builds consumer trust.
C) It makes crops grow faster.
D) It reduces the need for labeling.



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6. A family wants to eat healthy food free from chemical residues. They convert their backyard into an organic garden. What is the main benefit they seek?
 - A) Higher crop yields
 - B) Production of chemical-free healthy foods
 - C) Faster crop growth
 - D) Reduced labor needs
7. A farmer plants marigolds around tomatoes to repel nematodes. This is an example of:
 - A) Using synthetic nematicides
 - B) Biological pest control
 - C) Mechanical pest removal
 - D) Soil sterilization
8. Before planting, a farmer mixes well-rotted cow dung into the soil. The main purpose is:
 - A) To introduce harmful pathogens
 - B) To improve soil structure and fertility
 - C) To make the soil acidic
 - D) To repel beneficial insects
9. Grade 9 learners growing beans organically notice some pods attacked by borers. They pick and destroy affected pods and larvae. This is an example of:
 - A) Chemical control
 - B) Biological control
 - C) Cultural control (organic practice)
 - D) Systemic pesticide use
10. A gardener notices that soil fertility is declining. Which organic practice restores and maintains soil health?
 - A) Continuous monocropping
 - B) Annual use of synthetic fertilizers
 - C) Applying compost and practicing crop rotation
 - D) Leaving soil bare after harvest
11. Why is **food health and safety** an important issue in organic gardening?
 - A) Organic foods are always less nutritious.
 - B) Organic gardening avoids agro-chemicals, giving safer food.
 - C) Organic foods need more processing.
 - D) Organic gardening increases foodborne diseases.
12. A farmer makes organic pesticide using chili, garlic, and soap solution. Before spraying widely, they test it on a small area. This shows:
 - A) Carelessness
 - B) Risk-taking behavior
 - C) Careful experimentation and application
 - D) Chemical dependence
13. Which of the following is **not allowed** in organic gardening?
 - A) Compost use
 - B) Crop rotation
 - C) Synthetic herbicides
 - D) Manual weeding



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14. How does organic gardening help protect the environment?
 - A) By increasing chemical runoff into rivers
 - B) By promoting biodiversity and reducing chemical pollution
 - C) By depleting soil nutrients quickly
 - D) By encouraging GMOs
15. A farmer sets up beer traps to catch and drown slugs in their organic garden. This is an example of:
 - A) Chemical control
 - B) Mechanical/Physical control
 - C) Biological control
 - D) Genetic control
16. Which crop would be most suitable for a short-season Grade 9 organic gardening project?
 - A) Sugarcane
 - B) Maize
 - C) Spinach
 - D) Avocado tree
17. A farmer uses organic mulch (straw, dry leaves) around vegetables. What is the main benefit?
 - A) It attracts pests
 - B) Suppresses weeds, conserves moisture, and adds organic matter
 - C) Reduces soil fertility
 - D) Requires frequent watering
18. Instead of spraying pesticides, a gardener introduces ladybugs to control aphids. This is:
 - A) Cultural control
 - B) Biological control
 - C) Chemical control
 - D) Physical control
19. A learner makes a compost heap mixing green (nitrogen-rich) and brown (carbon-rich) materials. The purpose is to:
 - A) Produce synthetic fertilizer
 - B) Create fertile organic soil amendment
 - C) Compact the soil
 - D) Encourage weed growth
20. A farmer wants to improve soil aeration and water infiltration without machinery. Which manual practice is best?
 - A) Compacting the soil with heavy steps
 - B) Digging deeply with a fork (double digging)
 - C) Leaving soil undisturbed for long
 - D) Covering soil with cement
21. What is the main advantage of organic manure over synthetic fertilizers?
 - A) Provides highly concentrated nutrients
 - B) Improves soil structure, water retention, and microbial activity
 - C) Always cheaper to buy
 - D) Works faster than synthetic fertilizers
22. How does crop rotation benefit organic gardening?
 - A) Depletes specific nutrients
 - B) Breaks pest/disease cycles and improves fertility
 - C) Requires constant pesticide use
 - D) Limits farming to one crop only



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23. How does organic gardening promote **health** in food production?
 - A) Encourages processed food
 - B) Produces food free from chemical residues
 - C) Makes food expensive
 - D) Depends on fast-food chains
24. Traditional farming and modern organic gardening share which feature?
 - A) Heavy chemical use
 - B) Reliance on natural processes and inputs
 - C) Exclusive use of hybrid seeds
 - D) Large-scale monocultures
25. Why does organic gardening increase biodiversity?
 - A) Promotes single crop variety
 - B) Creates habitats for beneficial organisms without harsh chemicals
 - C) Encourages only one type of plant
 - D) Sterilizes the soil
26. A farmer harvests vegetables from their organic garden and considers them healthy. Why?
 - A) High sugar content
 - B) No synthetic fertilizers or pesticides used
 - C) Unusual color
 - D) Grown indoors
27. Why are cover crops (e.g., legumes) used in organic gardens during fallow periods?
 - A) To deplete nutrients
 - B) To prevent erosion, suppress weeds, and add nitrogen/organic matter
 - C) To attract pests
 - D) To require extensive tilling
28. Which amendment improves both drainage in clay soil and water retention in sandy soils?
 - A) Fine sand
 - B) Pure gravel
 - C) Well-rotted compost or manure
 - D) Chemical lime
29. Why should we practice organic gardening?
 - A) For maximum short-term profits only
 - B) To produce safe, healthy, and eco-friendly food
 - C) To increase dependency on imports
 - D) To make farming harder and less productive



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STRAND 2.0 FOOD PRODUCTION PROCESSES

Sub-Strand 2.1: Organic Gardening

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

Instructions: Choose the best answer.

1. A farmer in Kisii decides to grow vegetables without using synthetic fertilizers or chemicals. This method is known as:
A. Conventional farming
B. Mixed farming
C. Organic gardening
D. Zero grazing
2. Which of the following is an example of organic manure used in crop production?
A. Urea
B. Farmyard manure
C. DAP fertilizer
D. Ammonium nitrate
3. A Grade 9 learner is asked to control weeds in their organic garden by hand-pulling. This practice is called:
A. Chemical control
B. Mechanical control
C. Cultural control
D. Biological control
4. The main reason farmers practice organic gardening is to:
A. Increase chemical use
B. Produce healthy foods
C. Reduce labor
D. Reduce soil fertility
5. A teacher demonstrates how to prepare organic foliar feed from animal urine and Mexican sunflower. This mixture is mainly used to:
A. Kill weeds
B. Increase soil acidity
C. Provide nutrients to crops
D. Preserve harvested crops
6. Which one of the following practices is not part of organic gardening?
A. Composting
B. Crop rotation
C. Use of synthetic pesticides
D. Mulching
7. A farmer uses neem leaves extract to control pests in kale farming. This is an example of:
A. Chemical pesticide
B. Organic pesticide
C. Biological pesticide
D. Synthetic foliar feed



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8. Mulching in organic gardening mainly helps to:
 - A. Add stones to the soil
 - B. Conserve soil moisture
 - C. Increase soil erosion
 - D. Kill earthworms
9. The most suitable short-season crop for organic gardening among the following is:
 - A. Sugarcane
 - B. Tomatoes
 - C. Coffee
 - D. Tea
10. A Grade 9 learner prepares compost manure from maize stalks, kitchen waste, and cow dung. This practice improves soil by:
 - A. Reducing fertility
 - B. Improving aeration and nutrients
 - C. Hardening the soil
 - D. Removing soil organisms
11. A learner applies well-decomposed cow manure to the vegetable garden instead of chemical fertilizer. This practice is an example of:
 - A. Conventional farming
 - B. Organic gardening
 - C. Greenhouse farming
 - D. Irrigation farming
12. A farmer sprays neem extract on kale leaves to control aphids. This is an example of using:
 - A. Organic pesticide
 - B. Chemical pesticide
 - C. Fertilizer
 - D. Manure
13. The picture below shows a farmer adding kitchen waste into a pit. What is being prepared?



- A. Silage
- B. Compost manure
- C. Mulch
- D. Baled hay



14. Which of the following is the main advantage of using compost manure in organic gardening?

- A. It adds harmful chemicals to the soil
- B. It improves soil fertility naturally
- C. It requires a lot of money to buy
- D. It causes soil erosion

15. A learner pulls out weeds by hand in a vegetable garden instead of spraying chemicals. This method of weed control is called:

- A. Organic foliar feeding
- B. Mechanical weed control
- C. Organic spraying
- D. Fertilizer application

16. The diagram below shows a farmer covering the soil around tomato plants with dry grass. What practice is this?



- A. Manuring
- B. Mulching
- C. Irrigation
- D. Weeding

17. The use of Mexican sunflower in organic gardening is important because it provides:

- A. Water
- B. Organic foliar feed
- C. Shade
- D. Pest habitat

18. One benefit of organic gardening to human health is that it produces food that is:

- A. Sprayed with chemicals
- B. Free from harmful residues
- C. Imported from other countries
- D. Very expensive



19. A farmer prepares a mixture of garlic and pepper to spray on vegetables against caterpillars. This is an example of:
- A. Organic pesticide
 - B. Fertilizer
 - C. Mulching
 - D. Compost manure
20. Which of the following is NOT an organic gardening practice?
- A. Applying compost manure
 - B. Using synthetic fertilizers
 - C. Spraying neem extract
 - D. Mulching
21. A learner observes that vegetables grown with organic manure have stronger stems and greener leaves compared to those grown without manure. This is because organic manure:
- A. Improves soil nutrients
 - B. Attracts weeds
 - C. Destroys soil structure
 - D. Increases pests
22. The picture below shows learners preparing raised vegetable beds. Raised beds are useful in organic gardening because they:



- A. Reduce soil erosion
- B. Increase weeds
- C. Attract pests
- D. Reduce yields



23. Which of the following organic gardening practices helps to conserve soil moisture?

- A. Composting
- B. Mulching
- C. Weeding
- D. Crop rotation

24. A farmer plants beans after harvesting maize in the same garden. This practice is an example of:

- A. Mulching
- B. Crop rotation
- C. Weeding
- D. Fertilizer application

25. Which of the following is a short season crop suitable for organic gardening?

- A. Mango
- B. Kale (sukuma wiki)
- C. Avocado
- D. Coconut

26. A farmer makes a liquid fertilizer by soaking fresh leaves of Mexican sunflower in water for 2 weeks. This type of fertilizer is called:

- A. Compost
- B. Organic foliar feed
- C. Mulch
- D. Chemical fertilizer

27. One advantage of organic gardening over conventional farming is that:

- A. It requires expensive inputs
- B. It conserves the environment
- C. It reduces soil fertility
- D. It uses harmful pesticides

28. The picture below shows pests attacking vegetable leaves. In organic gardening, the farmer should control them by:



- A. Spraying chemical pesticide
- C. Burning the plants

- B. Spraying organic pesticide
- D. Adding chemical fertilizer



29. Which of the following materials can be used for mulching in organic gardening?

- A. Plastic bags
- B. Dry grass
- C. Stones
- D. Metal sheets

30. In organic gardening, cow dung, poultry droppings, and compost are used mainly for:

- A. Pest control
- B. Soil fertility improvement
- C. Weeding
- D. Harvesting

31. The picture below shows learners mixing compost manure into the soil. This practice improves:



- A. Soil nutrients and structure
- B. Pests and diseases
- C. Soil erosion
- D. Water wastage

32. Which of the following weeds can be used to make organic pesticides?

- A. Black jack
- B. Eucalyptus
- C. Cactus
- D. Pine

33. A farmer adds ash to the garden after harvesting. Ash is useful in organic gardening because it:

- A. Attracts pests
- B. Adds potassium to the soil
- C. Causes soil erosion
- D. Reduces fertility

34. In organic gardening, crop rotation helps to:

- A. Spread soil diseases
- B. Break pest and disease cycles
- C. Reduce soil fertility
- D. Increase weeds



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35. The picture below shows healthy cabbages grown organically. Which of the following practices helps achieve this?



- A. Use of synthetic fertilizer
 - B. Use of organic manure and mulch
 - C. Spraying chemicals
 - D. Over-irrigation
36. Which of the following is an example of a spice crop that can be grown using organic gardening?
- A. Ginger
 - B. Maize
 - C. Sorghum
 - D. Wheat
37. A farmer practices intercropping of maize and beans. One main benefit of this practice is that:
- A. Beans add nitrogen to the soil
 - B. Beans reduce maize yield
 - C. Maize controls pests
 - D. Beans use up all the nutrients
38. In organic gardening, covering the soil with dry maize stalks prevents:
- A. Sunlight penetration
 - B. Soil erosion and water loss
 - C. Growth of vegetables
 - D. Soil fertility improvement
39. Which of the following is an example of mechanical weed control in organic gardening?
- A. Spraying herbicides
 - B. Hand weeding
 - C. Using pesticides
 - D. Applying fertilizer



40. A farmer applies compost manure every season. Continuous use of compost manure helps to:

- A. Destroy soil structure
- B. Improve soil fertility
- C. Reduce soil organisms
- D. Increase chemical residues

41. The picture below shows a farmer using a watering can in a vegetable garden. Watering is important in organic gardening because it:



- A. Helps crops grow well
- B. Increases pests
- C. Reduces soil nutrients
- D. Stops mulching

42. Which of the following is a characteristic of food produced through organic gardening?

- A. Contains chemical residues
- B. Is fresh and healthy
- C. Is harmful to humans
- D. Cannot be sold in markets

43. A farmer decides to plant onions instead of continuously planting kale. This practice is called:

- A. Crop rotation
- B. Composting
- C. Mulching
- D. Manuring

44. The s learners applied mulch on young tomato plants. One main benefit of mulching is that it:

- A. Attracts pests
- B. Conserves soil moisture
- C. Increases weeds
- D. Destroys roots

45. Which of the following is a social benefit of organic gardening?

- A. Promotes food security
- B. Causes soil pollution
- C. Encourages use of chemicals
- D. Increases health risks



46. A farmer sprays a mixture of ash and water on vegetables to control pests. This is an example of:

- A. Organic pesticide
- B. Mulching
- C. Compost manure
- D. Irrigation

47. Which of the following crops is NOT suitable for short-season organic gardening?

- A. Spinach
- B. Kale
- C. Mango
- D. Cowpeas

48. The picture below shows a farmer in her shamba removing weeds. Which organic gardening practice is being carried out?



- A. Irrigation
- B. Weeding
- C. Mulching
- D. Manuring

49. In organic gardening, using animal droppings improves soil because it:

- A. Increases acidity
- B. Adds organic matter and nutrients
- C. Increases chemical residues
- D. Causes erosion

50. Which of the following BEST explains why organic gardening promotes health?

- A. It uses chemicals to control pests
- B. It produces food free from harmful chemicals
- C. It increases soil erosion
- D. It reduces food quality



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SECTION B: STRUCTURED QUESTIONS (100 Marks)

(Answer ALL questions. Marks are indicated.)

1. (a) Describe the components of integrated farming. (20 marks)

i. Explain what integrated farming means. (4 marks)

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ii. Identify four common components of integrated farming. (8 marks)

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iii. Explain how poultry keeping supports vegetable farming in integration. (4 marks)

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iv. Describe how fish farming is linked with crop production. (4 marks)

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1. (b) Making a model of integrated farming. (20 marks)

i. Explain why it is important to design models of integrated farming. (5 marks)

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ii. State four materials that can be used to make an integrated farm model in class. (4 marks)

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iii. Draw a simple sketch showing fish ponds, poultry houses, and vegetable gardens in integration. (6 marks)

iv. Explain two skills learners gain when making farm models. (5 marks)

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1. (c) Importance of integrated farming in conserving resources. (20 marks)

i. Explain how integrated farming reduces waste on the farm. (5 marks)

ii. State four ways integrated farming conserves soil fertility. (8 marks)

iii. Explain how integrated farming contributes to environmental protection. (4 marks)

iv. Give three reasons why integrated farming improves family income. (3 marks)



1. (d) Excursion or field study on integrated farming. (20 marks)

i. State two objectives of taking learners for an excursion on integrated farming. (4 marks)

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ii. List four observations learners can make during a visit to an integrated farm. (8 marks)

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iii. Explain two benefits of learners presenting their findings after an excursion. (4 marks)

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iv. Describe two ways in which learners can record findings during the field trip. (4 marks)

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1. (e) Creativity, appreciation, and conservation. (20 marks)

i. Explain two ways learners can show creativity when making integrated farm models. (4 marks)

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ii. State two values learners gain when appreciating others' creative work. (4 marks)

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iii. Explain how integrated farming promotes respect for the environment. (4 marks)

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iv. Suggest two ways teachers can encourage learners to be creative in agriculture. (4 marks)

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v. State two reasons why creativity is important in modern farming. (4 marks)

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SET 2

SECTION B: Structured Questions (100 Marks)

Instructions: Answer all the questions in the spaces provided. Show your working where necessary.

Question 1 (10 Marks)

Mr. Otieno has a small farm. He wants to practice integrated farming by keeping poultry, fish, and growing vegetables.

a)

i) Name **three components** of integrated farming that Mr. Otieno wants to use. (3 Marks)

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ii) Explain how combining these enterprises helps to **conserve farm resources**. (3 Marks)

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b) Mr. Otieno wants to use poultry droppings in vegetable production. Describe **steps** of how this can be done in an integrated farm. (4 Marks)

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Question 2 (10 Marks)

Study the image below



a) What type of farming system is shown in the diagram? (1 Mark)

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b) Give **three ways** in which this system helps to conserve resources. (3 Marks)

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c) State **four benefits** that come from linking different parts of the farm. (4 Marks)

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d) Explain why **creativity** is important when planning and running an integrated farm. (2 Marks)

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Question 3 (10 Marks)

A Grade 9 group designed a farm model with rabbits, a kitchen garden, and a compost pit.

a) Draw a **well-labelled sketch** of this farm model. (5 Marks)

b) Explain how this model helps to **conserve resources**, using examples. (3 Marks)

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c) What is the importance of the **compost pit** in this model? (2 Marks)

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Question 4 (10 Marks)

Ms. Chebet, an agricultural officer, is teaching farmers in a dry area about integrated farming.

a) Explain how integrated farming can help to **save water** in dry regions. Give one example. (3 Marks)

b) Apart from water, name **three other resources** that integrated farming saves. (3 Marks)

c) Give **two examples** of how integrated farming helps to **protect the environment**. (4 Marks)

Question 5 (10 Marks)

During a class presentation, a learner said integrated farming gives more than just income.

a) State three benefits of integrated farming. (3 Marks)

b) Give **two ways** integrated farming can provide a farmer with different sources of income. (2 Marks)



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c) Explain how integrated farming reduces the risk of **food shortage** during disasters. (3 Marks)

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d) Outline two benefits of integrated farming? (2 Marks)

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Set 3

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Instructions: Answer all questions in the spaces provided.

Question 1

A small-scale farmer in Kisii has one acre of land. They rear poultry, grow vegetables, and keep fish in a pond.

a) Define integrated farming. (2 marks)

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b) Explain how poultry rearing contributes to vegetable production in this system. (3 marks)

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c) State two ways in which the fish pond conserves farm resources. (4 marks)

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d) Identify one environmental benefit of this integrated system. (2 marks)

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e) Suggest one challenge this farmer may face when practicing integrated farming. (4 marks)

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Question 2

A Grade 9 Agriculture Club sets up a school demonstration farm with rabbits, maize, and a compost pit.

a) Describe how rabbit keeping supports maize farming in this system. (3 marks)

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b) State two by-products from rabbits that can be recycled within the farm. (4 marks)

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c) Explain why composting is important in conserving resources. (4 marks)

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d) Identify two examples of a natural manure from the farm. (2 marks)

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e) Give two items from a local environment that can be used to make an integrated farming model. (2 marks)

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Question 3

A farmer in Makueni County faces frequent droughts. They set up an integrated farm with goats, water harvesting tanks, and drought-tolerant crops.

a) Explain why water harvesting is important in integrated farming. (3 marks)

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b) State two products from goats that can benefit crop farming. (4 marks)

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c) Describe how drought-tolerant crops contribute to resource conservation. (4 marks)

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d) Identify one way this system can help improve community food security. (2 marks)

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e) Suggest one improvement the farmer can make to strengthen the system. (2 marks)

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Question 4

A farmer in Kiambu keeps pigs, grows bananas, and produces biogas from manure.

a) Explain how pigs contribute to the sustainability of the farm. (3 marks)

b) State two uses of biogas in the homestead. (4 marks)

c) Describe how biogas production helps conserve the environment. (4 marks)

d) Identify one challenge a farmer may face when managing pig waste. (2 marks)

e) Suggest one practice to ensure safety in using biogas. (2 marks)

Question 5

A Grade 9 learner presents a model of an integrated farm with poultry, Black Soldier Fly larvae, and fish.

a) Explain how poultry droppings support fish production in this model. (3 marks)



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b) State two benefits of using Black Soldier Fly larvae as animal feed. (4 marks)

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c) Mention two importance of resource cycling in the farm (4 marks)

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d) Identify two importance of keeping chicken in the farm. (2 marks)

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e) Give two reasons why farmers plant tree in the farm. (2 marks)

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Question 6

A farmer in Kakamega integrates beekeeping with vegetable farming.

a) Explain how bees support vegetable farming. (3 marks)

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b) State two products the farmer can obtain from beekeeping. (4 marks)

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c) Describe one way beekeeping helps conserve resources. (3 marks)

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d) Identify one risk that the farmer may face in beekeeping. (2 marks)

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e) Suggest one safety measure to reduce the risk in (d). (3 marks)

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SECTION B: STRUCTURED QUESTIONS (100 Marks)

Instructions: Answer all questions.

1. Organic Gardening Practices

a) Define organic gardening. (2 marks)

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b) List four examples of organic gardening practices. (4 marks)

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c) Explain how compost manure is prepared. (6 marks)

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d) Give three differences between organic gardening and conventional farming. (6 marks)

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e) State three challenges farmers face in practicing organic gardening. (2 marks)

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2. Crop Production Using Organic Methods

A Grade 9 learner is asked to grow sukuma wiki using organic practices.

a) Name three types of organic manure they can apply. (3 marks)

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b) Describe how they can control pests organically. (5 marks)

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c) Explain how mulching helps in crop growth. (4 marks)

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d) Outline the steps in planting sukuma wiki organically. (5 marks)

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e) Mention two short-season crops apart from sukuma wiki suitable for organic gardening. (3 marks)

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3. Importance of Organic Gardening

a) Explain four benefits of organic gardening to human health. (8 marks)

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b) Describe how organic gardening conserves the environment. (6 marks)

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c) State two ways organic gardening promotes integrity among learners. (2 marks)

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d) Explain how organic gardening contributes to soil fertility. (4 marks)

e) Suggest three ways the government can encourage organic gardening in Kenya. (5 marks)

4. Organic Fertilizer Preparation

A farmer in Kericho prepares foliar feed from cow urine and Mexican sunflower.

a) Describe the process of making this foliar feed. (6 marks)

b) Explain how the foliar feed is applied on crops. (4 marks)



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c) Give three advantages of using organic foliar feed. (6 marks)

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d) State two disadvantages of using organic foliar feed. (2 marks)

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e) Suggest three other plants that can be used in preparing organic foliar feed. (2 marks)

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5. Practical Skills in Organic Gardening

a) Identify three tools used in organic gardening. (3 marks)

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b) Explain two safety practices learners should observe when preparing organic pesticides. (4 marks)

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c) Give four methods of organic weed control. (4 marks)

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d) Describe the role of crop rotation in organic gardening. (6 marks)

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e) Write short notes on each of the following:

i. Mulching

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ii. Composting

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iii. Use of cover crops (9 marks)

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SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all questions in the spaces provided. Show your working where necessary.

Question 1 (10 Marks)

Mr. John is a small farmer. He wants to stop using expensive fertilizers and chemicals. He has heard about organic gardening.

a) What is meant by *organic gardening practices* in crop production? (2 Marks)

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b) Explain four organic gardening practices that Mr. John can use to improve his soil without chemicals. (4 Marks)

- i.
- ii.
- iii.
- iv.

c) Mr. John notices that caterpillars are eating his sukuma wiki leaves. Suggest two organic ways he can control the caterpillars. (2 Marks)

- i.
- ii.

d) Why is it important for Mr. John to follow organic rules if he wants to sell his vegetables? (2 Marks)

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Question 2 (10 Marks)

A Grade 9 class decides to start an organic garden in school. They plant carrots, spinach, and beans.

a) i) Write down the steps the learners would take to prepare their garden bed using organic methods before planting. (3 Marks)

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ii) The learners learn that Mexican sunflower can be used as organic foliar feed. Explain how they can prepare and use it on crops. (3 Marks)

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b) Give four advantages of organic gardening compared to farming with chemicals. (4 Marks)

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Question 3 (10 Marks)

Study the picture below



a) From the image, which type of gardening is being practiced? Give two reasons for your answer. (3 Marks)

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b) The gardener wants to control weeds without using herbicides. Describe three organic weed control methods he can use. (3 Marks)

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c) How does growing spinach organically help in food health and safety? (2 Marks)

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d) Explain how organic gardening increases biodiversity on a farm. (2 Marks)

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Question 4 (10 Marks)

A community group is discussing whether to switch to organic gardening. Some people say it needs too much labour.

a) i) Give two examples of mechanical weed control used in organic gardening. (2 Marks)

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ii) Why is manual weeding important in organic gardening even though it is hard work? (1 Mark)

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b) Explain four ways organic gardening protects the environment. (4 Marks)

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Question 5 (10 Marks)

Mwangi, a Grade 9 learner, wants to start an organic vegetable garden at home. He knows that good soil is very important.

a) i) State two benefits of adding compost or farmyard manure to the soil. (2 Marks)

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ii) Why is crop rotation important in organic gardening? (2 Marks)

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b) Mwangi chooses to grow beans organically. Write the steps he should follow from planting to harvesting using organic methods. (4 Marks)

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COOKING

STRAND 2.0: FOOD PRODUCTION PROCESSES

SUB-STRAND 2.2: STORAGE OF CROP PRODUCE

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. A farmer has harvested a lot of maize. Before storing it in a granary, they inspect the structure. Which step is most important to stop pest attack?
A) Painting the walls outside.
B) Cleaning and repairing the granary.
C) Leaving doors open for air.
D) Storing old produce inside.
2. A family keeps beans in a store. They keep checking the beans for mold and insects. This practice is an example of:
A) Ignoring spoilage. B) Proper ventilation.
C) Regular inspection. D) Adding chemicals to cover spoilage.
3. A farmer notices powdery green-yellow mold on stored maize. This shows aflatoxin contamination. What mainly causes aflatoxin?
A) Very low moisture content.
B) Very cold storage.
C) High moisture and poor ventilation that encourage mold.
D) Exposure to sunlight.
4. Before storing wheat, a farmer sun-dries it. Why must cereals like wheat be dried before storage?
A) To make them lighter to carry.
B) To increase protein.
C) To stop mold and insect attack.
D) To make milling easier.
5. A learner prepares a bag for storing beans. The bag is clean and has no holes. Why should storage containers be sealed and hole-free?
A) To make them look neat.
B) To block pests like rodents and insects, and control moisture.
C) To make the bag heavier.
D) To make it harder to open.



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6. A farmer keeps turning maize in a granary. Why do they do this?
- A) To mix different maize types.
 - B) To allow even aeration, avoid heating, and notice spoilage early.
 - C) To make maize easy to remove.
 - D) To speed up ripening.
7. A household wants to store groundnuts. Their store is prone to rodents. Which measure is best to control rodents?
- A) Putting open food near groundnuts.
 - B) Sealing cracks, raising stores, and setting traps.
 - C) Leaving garbage near the store.
 - D) Allowing stray cats in the store.
8. A farmer wants to use an old granary for new maize. What should be done with old produce inside?
- A) Leave it there.
 - B) Remove it and clean the granary well.
 - C) Mix old maize with new.
 - D) Spray water on it.
9. A school club prepares a store for beans. They see cracks in the walls. Why should they seal cracks?
- A) To make the store look nice.
 - B) To stop entry of insects and rodents.
 - C) To make the store darker.
 - D) To increase humidity.
10. Why is **food safety** important in managing stored crops with aflatoxin?
- A) Aflatoxin improves taste.
 - B) Aflatoxin is poisonous and dangerous to health.
 - C) Aflatoxin makes food easy to digest.
 - D) Aflatoxin preserves food.
11. A farmer sees insect damage in stored beans and removes spoilt beans. Why is this important?
- A) To increase spoilage of the rest.
 - B) To stop spread of pests and diseases to good beans.
 - C) To create more space.
 - D) To add more work.



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12. Which is a good condition of a storage structure for cereals?
- A) High humidity with changing temperatures.
 - B) Poor air flow and easy entry for pests.
 - C) Dry, cool, well-ventilated, and safe from pests and moisture.
 - D) Made from absorbent material.
13. A farmer notices a leaking roof in their sorghum store. Why must it be repaired?
- A) To make the store look new.
 - B) To stop moisture that causes mold and spoilage.
 - C) To increase air circulation.
 - D) To attract useful insects.
14. Why do farmers dust a granary with insecticide or diatomaceous earth before storage?
- A) To improve smell.
 - B) To protect the produce against storage pests.
 - C) To make walls shiny.
 - D) To attract rodents.
15. A farmer uses a moisture meter to check maize before storage. What skill is being applied?
- A) Artistic skills.
 - B) Critical thinking and careful management.
 - C) Communication skills.
 - D) Physical strength.
16. If cereals are stored with high moisture, what happens?
- A) More nutrients.
 - B) Faster growth of molds and insects.
 - C) Reduced weight.
 - D) Better taste.
17. Stored beans develop a musty smell. What causes this?
- A) Beans too dry.
 - B) Poor ventilation, moisture, and mold.
 - C) Store too cold.
 - D) Beans harvested too early.



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18. Why should crops be stored on pallets or raised platforms?
A) To make them taller. B) To avoid floor moisture and allow air flow.
C) To make counting easier. D) To hide them.
19. A farmer removes rotting potatoes from storage. Why?
A) To reduce total volume. B) To stop rot from spreading to good potatoes.
C) To soften potatoes. D) To improve skin colour.
20. What does ensuring **ventilation** in crop storage mean?
A) Closing all openings. B) Allowing air flow to control heat and moisture.
C) Painting the store often. D) Opening the store only once a year.
21. A farmer lines the inside of a granary with mats. Why?
A) To beautify the granary.
B) To block rodents and make surfaces smooth.
C) To absorb all moisture.
D) To block air flow.
22. A farmer sees holes and droppings around stored beans. This shows:
A) Bird infestation. B) Rodent infestation.
C) Insect infestation. D) Helpful insect activity.
23. What is the main aim of managing stored produce?
A) To increase quantity.
B) To reduce spoilage and keep quality for use or sale.
C) To change colour.
D) To encourage pests.
24. An agricultural officer advises farmers to clean stores before use. Why?
A) To make it look beautiful.
B) To remove old grains, dust, and pests that can damage new produce.
C) To increase heat in the store.
D) To attract new pests.
25. A farmer wants to store new groundnuts, but the shed smells bad from last season. What should they do?
A) Spray air fresheners.
B) Clean and air the shed well to remove smell and contaminants.
C) Store new nuts on top of old ones.
D) Add chemicals to mask the smell.



26. Why must cereals like maize and beans be dried before storage?
- To make them soft.
 - To cook faster.
 - To stop spoilage by molds and insects in wet grains.
 - To increase size.
27. Why do farmers use raised traditional granaries for maize?
- To increase insect activity.
 - To keep away rodents and moisture from ground.
 - To make maize hard to reach.
 - To encourage mold.
28. A farmer finds a small batch of millet with weevils. What should they do?
- Ignore it.
 - Dispose of it well to stop spread of weevils.
 - Mix it with healthy millet.
 - Leave it to dry.
29. Which action shows responsibility in managing stored crops?
- Storing all crops together whether spoilt or not.
 - Regular inspection, cleaning, and corrective action.
 - Waiting until most are spoilt before acting.
 - Blaming external causes for losses.
30. Nancy saw some storage pest that fed on the maize store in a sack. Which one is it?

A	B	C	D
			



Set 2

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

Choose the best answer for each.

1. A farmer notices that the granary roof is leaking during the rainy season. Which practice should be done before storing maize?
 - A) Applying fertilizer
 - B) Repairing the roof
 - C) Drying the grains inside
 - D) Spreading grains on the floor

2. Before storing beans in a sack, the farmer ensures they are well dried. Why is this important?
 - A) To make them shiny
 - B) To prevent germination
 - C) To reduce moisture and avoid moulds
 - D) To make them heavy

3. Which storage structure is common in Kenyan homes for storing cereals like maize?
 - A) Greenhouse
 - B) Granary
 - C) Cowshed
 - D) Seedbed

4. A farmer sweeps and dusts a store before filling it with dry maize. This practice helps to:
 - A) Increase maize sweetness
 - B) Reduce contamination
 - C) Improve seed germination
 - D) Add nutrients to maize



KITCHEN AND BACKYARD GARDENING



COOKING

5. A family has stored maize in sacks stacked against the wall. What risk does this pose?

A) Moisture and mould

B) Faster drying

C) Pest control

D) Proper aeration

6. Which pest is most common in stored cereals?

A) Aphids

B) Stem borers

C) Weevils

D) Termites

7. Which method helps to prevent rodents from entering a granary?

A) Painting walls

B) Raising the granary on stilts

C) Sprinkling water

D) Planting trees near the store

8. A farmer mixes dry ash with maize before storing. What is the main purpose?

A) To make maize white

B) To control storage pests

C) To improve taste

D) To make maize soft

9. A store has cracks on the walls. Before storage, the farmer should:

A) Cover them with posters

B) Seal the cracks with mud/cement

C) Put grains inside immediately

D) Pour water into the cracks



KITCHEN AND BACKYARD GARDENING



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10. Checking moisture content in cereals before storage is important to prevent:
- A) Germination
 - B) Aflatoxin contamination
 - C) Shining of grains
 - D) Weight loss
11. A farmer places wooden pallets on the floor of the store before stacking sacks of maize. The main reason for this practice is to:
- A) Make maize dry faster
 - B) Allow air circulation and prevent dampness
 - C) Reduce the weight of the sacks
 - D) Make the store look neat
12. Why should previous harvest remains be removed from the store before storing new produce?
- A) To create space only
 - B) To prevent pests and diseases from spreading
 - C) To make cleaning difficult
 - D) To mix old and new produce
13. Which of the following is NOT a correct method of controlling rodents in stores?
- A) Using traps
 - B) Sealing holes in the store
 - C) Leaving food remains on the floor
 - D) Keeping cats
14. A farmer regularly checks the maize in storage. This helps in:
- A) Early detection of spoilage and pests
 - B) Increasing maize germination
 - C) Reducing maize size
 - D) Making maize sweeter
15. Which of the following produces aflatoxin in maize and groundnuts?
- A) Bacteria
 - B) Fungi
 - C) Viruses
 - D) Weevils



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16. Why should spoilt produce be removed from the store immediately?
A) To reduce workload
B) To prevent spread of spoilage
C) To increase weight
D) To make the store tidy only
17. What is the best moisture content for storing maize safely?
A) 25%
B) 20%
C) 13%
D) 5%
18. A granary is raised above the ground mainly to:
A) Prevent theft
B) Allow air circulation and keep off rodents
C) Make it attractive
D) Reduce building costs
19. Which one is a modern storage facility for large-scale farmers?
A) Granary
B) Silo
C) Pot
D) Sack
20. A farmer fumigates maize in a store. What is the main aim of fumigation?
A) To increase grain weight
B) To kill storage pests
C) To polish grains
D) To dry maize
21. Which of the following is a sign of weevil attack on maize?
A) Shiny grains B) Holes in grains
C) Fresh smell D) Bigger grain size
22. Which of the following pests damages stored pulses like beans?
A) Bean bruchid B) Locusts
C) Termites D) Armyworms



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23. Turning maize in storage from time to time helps to:
- A) Mix with new grains
 - B) Ensure uniform aeration
 - C) Reduce grain size
 - D) Increase sweetness
24. Why should sacks of grains be arranged leaving space from the wall?
- A) For decoration
 - B) To allow proper ventilation
 - C) To make sweeping easier
 - D) To increase grain weight
25. A farmer uses neem leaves mixed with maize. This practice is used to:
- A) Make maize shiny
 - B) Repel pests
 - C) Reduce moisture
 - D) Improve colour
26. Which of these is NOT a safe storage practice?
- A) Using clean sacks
 - B) Keeping the store well-ventilated
 - C) Mixing wet and dry grains
 - D) Controlling rodents
27. Aflatoxin in food is dangerous because it can cause:
- A) Headaches only
 - B) Cancer and liver damage
 - C) Stronger immunity
 - D) Faster digestion
28. A learner at home helps to clean a store before beans are stored. This shows:
- A) Responsibility
 - B) Laziness
 - C) Lack of knowledge
 - D) Selfishness
29. Which material is most suitable for making grain storage sacks?
- A) Polythene
 - B) Sisal/jute
 - C) Cotton clothes
 - D) Paper



KITCHEN AND BACKYARD GARDENING



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30. One advantage of using metal silos in grain storage is:
- A) They are easy for pests to enter
 - B) They prevent moisture entry
 - C) They are very weak
 - D) They cannot last long
31. Which of the following should be done LAST when preparing a store?
- A) Repairing cracks
 - B) Dusting and sweeping
 - C) Storing crop produce
 - D) Sealing leakages
32. A farmer notices rat droppings in a store. What should be done immediately?
- A) Add more maize
 - B) Set traps or seal holes
 - C) Ignore it
 - D) Mix grains with sand
33. Why should farm produce be sorted before storage?
- A) To remove spoilt produce
 - B) To make grains shiny
 - C) To increase workload
 - D) To add weight
34. Which crop is most prone to aflatoxin contamination?
- A) Maize
 - B) Rice
 - C) Millet
 - D) Sorghum
35. Proper drying of cereals before storage mainly prevents:
- A) Germination
 - B) Fungal attack
 - C) Weight loss
 - D) Seed colour change
36. Which digital device can be used to search for crop storage methods?
- A) Smartphone
 - B) Hoe
 - C) Spade
 - D) Sack



37. Online agricultural videos can help farmers learn about:
- A) Cleaning storage structures
 - B) Dancing styles
 - C) Football matches
 - D) Painting houses
38. Which of the following apps can be useful for farmers in managing storage?
- A) WhatsApp
 - B) Tiktok
 - C) Netflix
 - D) Music apps
39. Use of moisture meters in storage management helps farmers to:
- A) Measure grain size
 - B) Check dryness of grains
 - C) Sort grains by colour
 - D) Count sacks
40. Which of the following is a responsibility of learners in storage management?
- A) Adding pests
 - B) Reporting signs of spoilage
 - C) Mixing spoilt and good grains
 - D) Ignoring stored crops
41. Which of the following is the BEST material for lining a grain silo?
- A) Cardboard
 - B) Metal
 - C) Paper
 - D) Cloth
42. Which crop is best stored in ventilated baskets?
- A) Tomatoes
 - B) Maize
 - C) Beans
 - D) Sorghum
43. Which practice prevents theft of stored produce in rural areas?
- A) Locking stores
 - B) Leaving stores open
 - C) Mixing grains
 - D) Using weak sacks



44. Which of the following pests is controlled by using rat guards on granary posts?
A) Birds B) Rats C) Weevils D) Termites
45. When storing Irish potatoes, farmers avoid plastic bags mainly because:
A) They are too light
B) They reduce ventilation and cause rotting
C) They are attractive
D) They are reusable
46. Proper storage of produce helps in:
A) Food wastage B) Food security
C) Pest multiplication D) Poor hygiene
47. A farmer spreads groundnuts on a raised wire mesh for drying. Why is this method good?
A) Prevents soil contamination
B) Makes groundnuts heavier
C) Increases size of nuts
D) Adds nutrients
48. Which one is NOT an example of a storage container?
A) Gunny bag B) Silo C) Watering can D) Granary
49. A farmer sorts out rotten tomatoes before storage. The main purpose is:
A) To reduce spread of rot
B) To reduce size
C) To make tomatoes shiny
D) To increase weight
50. Which of the following best describes responsibility in managing stored produce?
A) Ignoring storage pests B) Reporting and helping in controlling spoilage
C) Mixing wet and dry grains D) Eating stored crops secretly
51. During a field trip to a local village, learners saw the following. It's an old storage model. It's called a?



A. Granary

B. Mansion

C. Apartment

D. Cottage



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SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all questions in the spaces provided. Show all your working where necessary.

Question 1 (10 Marks)

Mr. Kiprop has harvested a lot of maize. He wants to use his old storeroom to keep part of the harvest for home use and sale.

a) State three steps Mr. Kiprop should take to prepare the storeroom before storing the maize. (3 Marks)

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b) Explain the importance of:

i) Checking moisture content in maize. (2 Marks)

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ii) Ensuring good ventilation in the store. (2 Marks)

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c) Mr. Kiprop sees holes and droppings on the store floor, showing rodent presence. Suggest three ways he can control rodents in his storeroom. (3 Marks)

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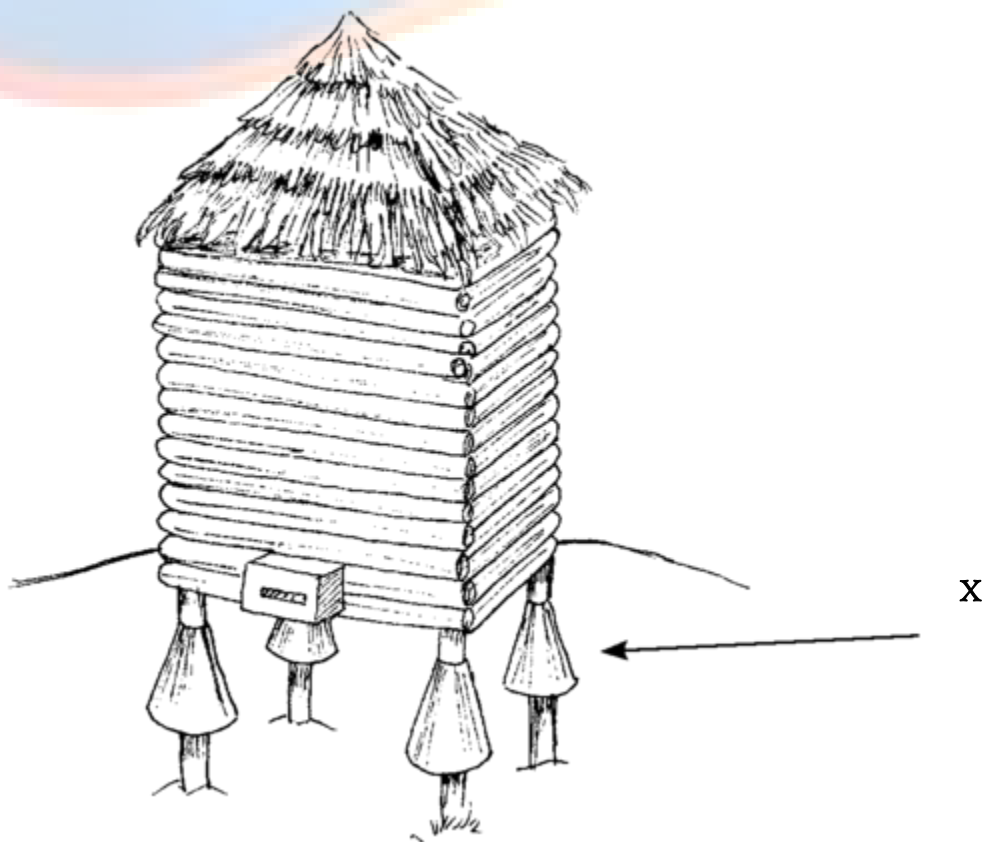
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Question 2 (10 Marks)

Study the picture below.



a) Name the type of storage structure shown. (1 Mark)

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b) Identify the structure labelled X

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c) Explain four features of this granary that help to reduce spoilage of stored produce. (4 Marks)

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d) State three ways a farmer can actively manage maize after storing it in the granary. (3 Marks)

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e) Why is food safety important when handling stored produce, especially in preventing fungal contamination such as aflatoxin? (2 Marks)

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Question 3 (10 Marks)

A Grade 9 learner is researching post-harvest losses in cereals. They discover that moisture and pests cause most spoilage.

a) Explain how high moisture content causes spoilage in stored cereals. (3 Marks)

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b) State two ways insects damage stored produce. (2 Marks)

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c) Suggest three ways farmers can reduce spoilage caused by moisture in stored maize or beans. (3 Marks)

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d) Explain how being responsible helps a farmer manage stored produce better. (2 Marks)

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Question 4 (10 Marks)

Farmers in a community are losing harvests due to poor storage. An agricultural officer trains them on proper storage.

a) Why should farmers remove old produce from the store before adding new harvests? (2 Marks)

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b) State three steps to ensure cleanliness of a storage area before use. (3 Marks)

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c) Mention three signs that show stored produce is spoiling. (3 Marks)

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d) Explain how proper storage helps improve a family's food security. (2 Marks)

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Question 5 (10 Marks)

Mrs. Akinyi has harvested beans. She wants to store them in old jute bags inside a dry room.

a) State three measures she should take to prepare the jute bags before use. (3 Marks)

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b) Why is ventilation important when storing beans? (2 Marks)

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c) Some beans develop mold. Explain what Mrs. Akinyi should do to stop further spoilage and keep the beans safe. (3 Marks)

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d) Explain how critical thinking skills (like being open-minded) help a farmer manage storage structures better. (2 Marks)

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Set 2

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Answer all questions. Each question has parts a–e.

Question 1 (20 Marks)

A farmer wants to store maize after harvesting.

a) Explain **five ways** of preparing the storage structure before storing maize. (5m)

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KITCHEN AND BACKYARD GARDENING



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b) Describe how you would **prepare an existing granary** in readiness for storage. (5m)

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c) Explain **three practices** a farmer should carry out to manage stored maize. (6m)

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d) Suggest **two responsibilities** of a learner in helping manage stored produce at home. (2m)

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e) State **two effects of poor storage practices** on crop produce. (2m)

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Question 2 (20 Marks)

A household has harvested beans and plans to keep them for future use.

a) Describe **three reasons** for drying beans before storage. (3m)

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b) List **four ways** of ensuring the storage bags are ready for beans. (4m)

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c) Explain **three methods** of controlling pests in stored beans. (6m)

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d) Outline **two ways** of disposing off spoilt produce responsibly. (2m)

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e) Give **two reasons** why learners should practice responsibility in food storage. (2m)

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f) Suggest **three dangers** of consuming food contaminated with aflatoxin. (3m)

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KITCHEN AND BACKYARD GARDENING



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Question 3 (20 Marks)

A teacher takes learners to observe different farm storage facilities.

a) Identify **five types** of storage facilities used in Kenya. (5m)

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b) Explain **three reasons** for repairing leakages in a store before storage. (3m)

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c) Describe **three practices** of ensuring proper ventilation in storage structures. (6m)

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d) Suggest **two ways** learners can use digital devices to find storage information. (2m)

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e) State **four benefits** of managing stored crop produce properly. (4m)

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Question 4 (20 Marks)

A farmer has 10 sacks of maize damaged by weevils.

a) Explain **three signs** of pest damage in stored cereals. (3m)

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b) State **four methods** of controlling weevils in stored produce. (4m)

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c) Describe **three reasons** for turning stored grains regularly. (6m)

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d) Explain **two dangers** of consuming pest-damaged cereals. (2m)

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e) Suggest **two ways** learners can help reduce pest attacks at home. (2m)

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f) Mention **three modern digital technologies** that help farmers in managing crop storage. (3m)

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Question 5 (20 Marks)

A family stores maize in a store without cleaning it. After one month, most grains spoil.

a) Explain **five mistakes** the family made in preparing the storage structure. (5m)

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b) Suggest **three measures** the family should take to prevent spoilage. (3m)

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c) State **three responsibilities** of learners in promoting food safety at home. (3m)

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d) Explain **four ways** aflatoxin affects human health. (4m)

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e) Suggest **five ways** to improve food safety in crop storage. (5m)

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COOKING

STRAND 2.0: FOOD PRODUCTION PROCESSES

SUB-STRAND 2.3: COOKING: USING FLOUR MIXTURES

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. A Grade 9 learner is learning to bake. They are working with a mixture of flour, water, yeast, salt, and a little sugar, which they knead vigorously until it becomes elastic. This type of flour mixture is best described as a:

A) Thin batter

B) Pouring batter

C) Dough

D) Dropping batter
2. A family is preparing pancakes for breakfast. They mix flour, milk, eggs, and a leavening agent to form a runny consistency that can be easily poured. This type of flour mixture is known as a:

A) Dough

B) Stiff batter

C) Pouring batter

D) Paste
3. A baker wants to make light and airy mandazi. They know that proper mixing of the flour, sugar, eggs, milk, and leavening agent is crucial. Which specific type of flour mixture is required for mandazi?

A) Thin batter

B) Dough

C) Stiff batter

D) Liquid mixture



4. A chef is preparing chapati. They combine wheat flour, water, and a little oil to form a firm, pliable mass that they will then knead. The product they are aiming for falls under which category of flour mixtures?
- A) Batter
 - B) Dough
 - C) Roux
 - D) Slurry
5. Why is safety in the use of tools and equipment a crucial aspect when preparing food using flour mixtures (e.g., using a hot pan for pancakes or kneading dough)?
- A) It speeds up the cooking process.
 - B) It prevents accidents like burns, cuts, or electric shocks.
 - C) It makes the food taste better.
 - D) It is only necessary for large-scale production.
6. A learner is making a dropping batter for muffins. What characteristic best describes a dropping batter?
- A) Very runny, easily poured.
 - B) Stiff and kneadable.
 - C) Too thick to pour, but can be dropped from a spoon.
 - D) Hard and brittle.
7. A group of friends is baking a cake. They use flour, sugar, eggs, butter, and baking powder. The resulting mixture before baking is mainly a:
- A) Dough
 - B) Pouring batter
 - C) Dropping batter
 - D) Stiff dough



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8. When making chapati, the dough needs to be kneaded until it is smooth and elastic. What is the primary purpose of kneading chapati dough?
- A) To incorporate more air, making it chewy.
 - B) To develop the gluten, giving the chapati its characteristic texture and elasticity.
 - C) To make it stickier.
 - D) To remove all the moisture.
9. A young cook is trying to distinguish between different flour mixtures. They observe that one mixture is very sticky and almost liquid, while another is firm and can be rolled out. The sticky, liquid one is most likely a:
- A) Dough
 - B) Batter
 - C) Paste
 - D) Crumb mixture
10. A family is preparing to make fried doughnuts (mandazi). They are discussing the ratio of liquid to flour. For mandazi, the mixture should form a:
- A) Very thin batter.
 - B) A soft, pliable dough.
 - C) A pouring batter.
 - D) A very stiff, crumbly mixture.
11. During preparation of dough for bread, why is it important to control the temperature of water added to yeast?
- A) Cold water gives better flavor.
 - B) Hot water kills the yeast.
 - C) Warm water activates yeast for rising.
 - D) Any temperature works the same.



12. A learner is preparing dough for bread. They notice that adding too much water makes the dough very sticky and difficult to handle. This highlights the importance of:
- A) The type of flour used.
 - B) The correct ratio of ingredients, especially liquid to flour.
 - C) The color of the flour.
 - D) The temperature of the kitchen.
13. Which of the following ingredients is primarily responsible for the 'rise' or lightness in products like mandazi and pancakes when baking powder is used?
- A) Flour
 - B) Sugar
 - C) Leavening agent (e.g., baking powder, yeast)
 - D) Salt
14. A culinary learner is asked to demonstrate how to make a pouring batter. Which of these products would they most likely prepare?
- A) Bread
 - B) Chapati
 - C) Crepes
 - D) Scones
15. Why is sifting flour an important step before mixing it with other ingredients?
- A) It adds flavor to the flour.
 - B) It removes lumps and aerates the flour for lighter mixtures.
 - C) It makes the flour sweeter.
 - D) It changes the flour color.



16. A family often makes different types of flatbreads. For a very thin, flexible flatbread like a crepe, the flour mixture needed would be a:
- A) Stiff dough
 - B) Dropping batter
 - C) Thin pouring batter
 - D) Crumbly mixture
17. After mixing ingredients for a cake, a learner observes that the batter is too thick to pour smoothly. To achieve the correct consistency, what should they most likely add?
- A) More flour
 - B) More liquid
 - C) More sugar
 - D) More baking powder
18. A farmer's spouse wants to make mandazi. They combine flour, water, sugar, and yeast. The yeast is crucial because it:
- A) Adds color to the mandazi.
 - B) Acts as a leavening agent, producing gas that makes the mandazi rise.
 - C) Makes the dough stickier.
 - D) Prevents the mandazi from frying.
19. When making products from flour mixtures, such as pancakes or mandazi, what is the primary role of cooking them (e.g., frying or baking)?
- A) To make them raw.
 - B) To transform the raw mixture into an edible product through heat.
 - C) To make them extremely tough.
 - D) To remove all the nutrients.

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20. Why is pre-heating the oven important before baking products like cakes or bread?
- A) It shortens cooking time only.
 - B) It ensures even heat distribution for proper rising and baking.
 - C) It burns off excess flour.
 - D) It makes the oven smell nice.
21. What is the best way to handle hot cooking oil safely when frying products like mandazi?
- A) Pouring water into it to cool it down.
 - B) Using a shallow pan and filling it to the brim with oil.
 - C) Keeping a lid nearby and never leaving hot oil unattended.
 - D) Using bare hands to place mandazi into the oil.
22. A recipe calls for a stiff dough. Which product would mainly require a stiff dough for its preparation?
- A) Waffles B) Biscuits C) Crepes D) Porridge
23. A learner finds that their chapati dough is too dry and crumbly. What is the most appropriate adjustment to make?
- A) Add more flour. B) Add more water, a little at a time, until the desired consistency.
- C) Add more salt. D) Knead it harder without adding liquid.
24. A culinary instructor emphasizes the importance of precise measurements when working with flour mixtures. Why are precise measurements particularly crucial for achieving desired results in baking or cooking with flour?
- A) To make the process faster.
 - B) To ensure the correct balance of ingredients for proper texture, rise, and flavor.
 - C) To use up all available ingredients.
 - D) To make the food cheaper.



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25. Which of the following is a product mainly made from a yeast-leavened dough?
- A) Pancake
 - B) Chapati
 - C) Bread
 - D) Waffle
26. A young chef is making thin pancakes. They notice the batter is too thick, resulting in heavy pancakes. What adjustment should they make to the batter for a lighter product?
- A) Add more flour.
 - B) Add more liquid
 - C) Add more sugar.
 - D) Add more salt.
27. A learner is experimenting with different types of flour to make mandazi. They observe differences in how each flour absorbs water and forms dough. This experience enhances their understanding of:
- A) The cost of flour.
 - B) The properties of different flours and their impact on mixtures.
 - C) The history of mandazi.
 - D) The best color for mandazi.
28. When preparing food from flour mixtures, what is the importance of proper hygiene (e.g., washing hands, clean utensils)?
- A) To make the food taste better.
 - B) To prevent food contamination and ensure food safety.
 - C) To make the cooking process slower.
 - D) To use more water.



29. A family is preparing to make pastries like samosa wrappers. They need a very thin, flexible dough that can be rolled out. This type of dough is essential for:
- A) Making a thick stew.
 - B) Creating thin, delicate wrappers that hold shape when cooked.
 - C) Frying as a pouring batter.
 - D) Forming a solid block.
30. Which of the following products is mainly not made using a flour mixture?
- A) Ugali
 - B) Chapati
 - C) Cake
 - D) Doughnut

TOPICAL REVISION SET 2

Strand 2.0: Food Production Processes

Sub-strand 2.3: Cooking – Using Flour Mixtures

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

Instructions: Answer ALL questions.

1. Learners are shown a picture of a **pourable flour mixture** used to make pancakes. This type of mixture is called:



- A. Paste
- B. Batter
- C. Soft dough
- D. Stiff dough



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2. A learner is kneading dough that can be rolled into flat discs for cooking. Which product will most likely result?



- A. Pancake
B. Chapati
C. Doughnut
D. Bhajia
3. A teacher displays two bowls: one with **thin batter** and another with **soft dough**. Which correct matching is TRUE?
- A. Thin batter – Mandazi, Soft dough – Pancake
B. Thin batter – Chapati, Soft dough – Bhajia
C. Thin batter – Pancake, Soft dough – Chapati
D. Thin batter – Doughnut, Soft dough – Sauce
4. Which flour mixture is **too firm** to pour but can be shaped into small balls before frying?
- A. Soft dough B. Stiff dough C. Thin batter D. Paste
5. A group of learners prepares **bhajia** by dipping potatoes in a mixture. Which flour mixture was used?
- A. Soft dough
B. Thin batter
C. Stiff dough
D. Paste



6. A learner notices that pancake mixture spreads easily when poured on a pan. This is because it is:
- A. Stiff dough
 - B. Thin batter
 - C. Soft dough
 - D. Thick paste
7. In a class demo, the teacher asks: Which flour mixture requires fermentation when yeast is added?
- A. Batter
 - B. Stiff dough
 - C. Soft dough
 - D. Paste
8. Which mixture is BEST described as elastic and stretchable when kneaded?
- A. Paste
 - B. Batter
 - C. Dough
 - D. Custard
9. A learner pours a mixture into hot oil, forming round snacks. Which flour mixture was this?
- A. Soft dough
 - B. Batter
 - C. Stiff dough
 - D. Paste
10. Which mixture can be described as runny, smooth and pourable?
- A. Batter
 - B. Dough
 - C. Paste
 - D. Cream



11. When preparing mandazi, milk is added to the dough. Its main function is:
- A. To increase volume only
 - B. To add nutrients and improve flavor
 - C. To act as a raising agent
 - D. To make dough stiff
12. A teacher asks learners to demonstrate **kneading**. Which tool should be used?
- A. Fork
 - B. Knife
 - C. Hands
 - D. Whisk
13. Which ingredient makes mandazi brownish during frying?
- A. Flour
 - B. Sugar
 - C. Water
 - D. Salt
14. Why should flour be sieved before mixing?
- A. To make it sweet
 - B. To remove dirt and aerate it
 - C. To add flavor
 - D. To reduce elasticity
15. In chapati preparation, warm water is often used to:
- A. Harden gluten
 - B. Soften the dough
 - C. Act as a raising agent
 - D. Add sweetness



16. Which of these ingredients is NOT essential in pancake preparation?
- A. Flour
 - B. Milk
 - C. Eggs
 - D. Yeast
17. While mixing batter, a learner adds too much water. The batter becomes:
- A. Very thick
 - B. Very thin
 - C. Very elastic
 - D. Very sticky
18. When preparing chapati, adding oil during kneading helps to:
- A. Make it layered and soft
 - B. Harden the dough
 - C. Increase fermentation
 - D. Reduce flavor
19. A learner uses baking soda in pancake preparation. The main role is:
- A. To make pancakes fluffy
 - B. To add color
 - C. To reduce sweetness
 - D. To add oiliness
20. Which ingredient acts as the main raising agent in mandazi?
- A. Water
 - B. Oil
 - C. Baking powder
 - D. Flour



KITCHEN AND
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21. A learner fries mandazi until golden brown. What cooking method is this?
- A. Deep frying
 - B. Boiling
 - C. Baking
 - D. Steaming
22. Chapati is usually cooked on a flat pan called:
- A. Oven
 - B. Grill
 - C. Sufuria
 - D. Pan
23. Pancakes are prepared by cooking batter on a lightly oiled pan. This is an example of:
- A. Deep frying
 - B. Shallow frying
 - C. Dry frying
 - D. Boiling
24. Which product is rolled flat before cooking?
- A. Pancake
 - B. Chapati
 - C. Mandazi
 - D. Doughnut
25. A teacher asks learners to prepare a product that requires deep frying stiff dough. Which product should they choose?
- A. Chapati
 - B. Pancake
 - C. Mandazi
 - D. Bhajia



26. Which flour mixture product is often eaten with beans, beef, or stew?
- A. Pancake
 - B. Mandazi
 - C. Chapati
 - D. Doughnut
27. A snack made from thin batter, fried until light brown, is:
- A. Chapati
 - B. Pancake
 - C. Mandazi
 - D. Doughnut
28. When yeast is used in dough, which product is most likely made?
- A. Chapati
 - B. Mandazi
 - C. Pancake
 - D. Bread
29. A learner cooks chapatis until burnt. The problem was:
- A. Low heat
 - B. Too much oil
 - C. Excess heat
 - D. Lack of rolling
30. A learner prepares a snack by coating bananas with batter and frying. The product is:
- A. Mandazi
 - B. Chapati
 - C. Bhajia
 - D. Pancake



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COOKING

31. Which nutrient is mainly provided by chapati?
- A. Carbohydrates
 - B. Proteins
 - C. Vitamins
 - D. Minerals
32. Why are pancakes a quick energy food?
- A. They contain a lot of proteins
 - B. They are high in carbohydrates
 - C. They are deep fried
 - D. They take long to cook
33. Which ingredient adds tenderness to dough?
- A. Flour
 - B. Oil
 - C. Salt
 - D. Baking powder
34. Mandazi is considered a snack because:
- A. It is eaten in the morning only
 - B. It is fried and eaten between meals
 - C. It has less flour
 - D. It is expensive
35. Gluten is important in chapati preparation because it:
- A. Adds sweetness
 - B. Makes dough elastic and easy to roll
 - C. Makes chapati sweet
 - D. Adds oiliness



KITCHEN AND
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COOKING

36. Baking powder is important in pancake preparation because it:
- A. Adds elasticity
 - B. Makes pancakes rise and soft
 - C. Acts as preservative
 - D. Adds flavor
37. Mandazi provide energy mainly because of:
- A. High protein content
 - B. Oil and carbohydrates
 - C. High mineral content
 - D. Vitamins
38. Why are chapatis considered filling foods?
- A. They are light
 - B. They provide both energy and bulk
 - C. They cook fast
 - D. They are sweet
39. A learner chooses pancakes instead of mandazi for a low-fat snack. This is because:
- A. Pancakes are shallow fried, mandazi deep fried
 - B. Pancakes use more sugar
 - C. Pancakes require kneading
 - D. Mandazi has no oil
40. Preparing flour mixtures in class helps learners to:
- A. Only eat food
 - B. Develop teamwork, skills, and hygiene practices
 - C. Waste flour
 - D. Pass time



41. Before starting cookery, learners tie headscarves. This prevents:
- A. Sweating
 - B. Hair from falling into food
 - C. Burning
 - D. Dirt on apron
42. A learner uses a knife to cut chapati dough. What is the main safety rule?
- A. Cut away from the body
 - B. Cut toward the body
 - C. Hold knife loosely
 - D. Use knife with oily hands
43. When frying mandazi, learners should use a cooking spoon to:
- A. Reduce sugar
 - B. Turn mandazi safely
 - C. Add flavor
 - D. Remove gluten
44. A learner licks a spoon and uses it to stir batter again. This is wrong because:
- A. It adds water
 - B. It is unhygienic
 - C. It increases flavor
 - D. It makes batter elastic
45. Which tool is used to roll chapati evenly?
- A. Spoon
 - B. Rolling pin
 - C. Fork
 - D. Knife



46. If oil spills on the floor during frying, the FIRST action should be:

- A. Ignore it
- B. Mop immediately
- C. Step over it
- D. Call another learner

47. Which item of clothing prevents burns from hot oil?

- A. Headscarf
- B. Gloves
- C. Apron
- D. Dust coat

48. Using expired flour to make chapati is wrong because:

- A. It makes chapati elastic
- B. It can cause food poisoning
- C. It saves money
- D. It adds flavor

49. Identify an item used to stir ingredient when preparing a cake.

A	B	C	D
			

50. Why should learners switch off gas stoves immediately after cooking?

- A. To save time
- B. To prevent fire accidents
- C. To save flour
- D. To add flavor



Strand 2.0: Food Production Processes

Sub-strand 2.3: Cooking – Using Flour Mixtures

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

Instructions: Answer all questions.

1. A group of Grade 9 learners are asked to name flour mixtures used in preparing foods like chapati, mandazi, and pancakes. Which TWO main types of flour mixtures should they mention?
 A. Creams and sauces B. Batters and doughs
 C. Soups and stews D. Oils and fats

2. When preparing pancake batter in class, the teacher explains that the mixture is runny and can be poured. This type of mixture is best described as:
 A. Stiff dough B. Soft dough C. Batter D. Paste

3. Chapati dough is soft, elastic and needs kneading before rolling. This shows that chapati is prepared from:
 A. Batter B. Soft dough C. Stiff dough D. Thick batter

4. Learners observe that mandazi dough is firmer compared to chapati dough and is shaped before deep frying. This type of flour mixture is called:
 A. Stiff dough B. Batter C. Soft dough D. Thin paste

5. A learner mixes flour, milk and eggs until smooth. The mixture is used for frying thin pancakes. This type of mixture is:
 A. Soft dough B. Thick batter C. Thin batter D. Stiff dough

6. A Grade 9 learner wants to prepare bhajia. The flour mixture used must coat potato slices before frying. The suitable flour mixture is:
 A. Thin batter B. Soft dough
 C. Stiff dough D. Cream paste



7. If a mixture is too runny to hold shape but can easily be poured, it is most likely:
- A. Soft dough
 - B. Thin batter
 - C. Thick dough
 - D. Stiff dough
8. Which flour mixture requires the most kneading before shaping into chapatis?
- A. Batter
 - B. Paste
 - C. Soft dough
 - D. Thin batter
9. A farmer's wife wants to prepare doughnuts (maandazi). Which flour mixture should she use?
- A. Soft dough B. Stiff dough C. Thin batter D. Thick paste
10. A learner is given two mixtures: one is pourable like pancake, the other firm like mandazi. Which correct classification is true?
- A. Pancake = Soft dough, Mandazi = Batter
 - B. Pancake = Batter, Mandazi = Soft dough
 - C. Pancake = Stiff dough, Mandazi = Soft dough
 - D. Pancake = Paste, Mandazi = Batter
11. Before preparing chapati, a learner mixes flour, salt, oil and water. The process of combining these ingredients is called:
- A. Kneading
 - B. Mixing
 - C. Rolling
 - D. Proofing



12. While preparing pancakes, why is whisking of the batter important?
- A. To increase elasticity
 - B. To introduce air for lightness
 - C. To make dough stiff
 - D. To reduce gluten development
13. Which of the following is NOT a correct ingredient for chapati?
- A. Wheat flour
 - B. Salt
 - C. Yeast
 - D. Cooking oil
14. During preparation of mandazi, sugar is added to the mixture. Its function is mainly to:
- A. Make the dough stiff
 - B. Add sweetness and color
 - C. Provide elasticity
 - D. Improve rolling
15. In preparation of chapati, oil is applied on the dough before rolling. Why is this step important?
- A. To add flavor only
 - B. To make chapati softer and layered
 - C. To reduce gluten development
 - D. To prevent frying
16. A learner forgets to add baking powder when preparing mandazi. What will most likely happen?
- A. Mandazi will rise and become fluffy
 - B. Mandazi will remain flat and dense
 - C. Mandazi will taste sour
 - D. Mandazi will turn black



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17. Which method of mixing is mainly used when preparing chapati?
- A. Folding in
 - B. Kneading
 - C. Creaming
 - D. Whisking
18. In pancake preparation, oil is added to the pan before cooking. What is its main purpose?
- A. To make the batter thin
 - B. To add sweetness
 - C. To prevent sticking
 - D. To speed up mixing
19. A learner makes dough that is too stiff. What should be added to soften it?
- A. More oil
 - B. More water
 - C. More flour
 - D. More sugar
20. Which of the following ingredients acts as a raising agent in pancakes?
- A. Yeast
 - B. Baking powder
 - C. Oil
 - D. Water
21. Chapatis are cooked on a hot flat pan. This cooking method is called:
- A. Boiling
 - B. Deep frying
 - C. Shallow frying
 - D. Dry frying



22. Mandazi is cooked in hot oil, fully covering the dough. This cooking method is:
- A. Steaming
 - B. Deep frying
 - C. Roasting
 - D. Baking
23. Pancakes are cooked using:
- A. Dry frying
 - B. Deep frying
 - C. Shallow frying
 - D. Steaming
24. Which of the following foods is made from batter?
- A. Pancake
 - B. Chapati
 - C. Mandazi
 - D. Doughnut
25. A learner rolls dough into thin circles and cooks on a pan without oil. Which product is this?
- A. Mandazi
 - B. Pancake
 - C. Chapati
 - D. Doughnut
26. A visitor is served snacks prepared from stiff dough, deep fried, and sprinkled with sugar. These snacks are most likely:
- A. Chapatis
 - B. Mandazi
 - C. Pancakes
 - D. Doughnuts



27. Which flour mixture product is commonly served with stew in Kenyan homes?
- A. Pancake
 - B. Chapati
 - C. Mandazi
 - D. Doughnut
28. A Grade 9 learner uses thin batter to coat bananas before frying. The product is:
- A. Chapati
 - B. Mandazi
 - C. Bhajia
 - D. Doughnut
29. If a teacher asks learners to prepare a product that requires kneading, which product should they choose?
- A. Pancake
 - B. Mandazi
 - C. Chapati
 - D. Bhajia
30. Which food product requires fermentation when yeast is used as a raising agent?
- A. Chapati B. Pancake
 - C. Bread D. Bhajia
31. Why is gluten important in chapati dough?
- A. It adds sweetness
 - B. It gives elasticity for rolling
 - C. It adds flavor
 - D. It prevents sticking



32. Oil in pancake batter helps to:
- A. Add elasticity
 - B. Reduce sticking to the pan
 - C. Act as a raising agent
 - D. Improve fermentation
33. Baking powder in mandazi acts as:
- A. Flavoring
 - B. Preservative
 - C. Raising agent
 - D. Sweetener
34. Why are pancakes considered a quick snack?
- A. They are boiled
 - B. They require no kneading or fermentation
 - C. They are baked
 - D. They require long preparation
35. Why is chapati important in a balanced diet?
- A. It provides mainly vitamins
 - B. It provides proteins only
 - C. It provides energy from carbohydrates
 - D. It provides fats only
36. Mandazi is often served with tea in Kenya because:
- A. It is light and easy to eat
 - B. It has high protein
 - C. It has no flavor
 - D. It cooks without oil



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COOKING

37. Which flour mixture product can be prepared fastest in class practical?

- A. Mandazi
- B. Chapati
- C. Pancake
- D. Doughnut

38. Which product provides more energy due to high oil content?

- A. Pancake
- B. Chapati
- C. Mandazi
- D. Bhajia

39. Which product is most suitable for learners who want a low-fat diet?

- A. Mandazi B. Chapati
- C. Doughnut D. Pancake

40. Flour mixtures help learners to:

- A. Only make snacks
- B. Develop cookery skills, teamwork and safety
- C. Avoid using stoves
- D. Only pass exams

41. Before preparing chapati, why should hands be washed?

- A. To make rolling easier
- B. To maintain hygiene and prevent contamination
- C. To soften flour
- D. To add moisture



42. A learner tastes pancake batter with a finger then continues mixing. This is wrong because:
- A. It wastes batter
 - B. It is unhygienic
 - C. It makes batter thicker
 - D. It adds sweetness
43. Which tool is most suitable for rolling chapati dough?
- A. Spoon
 - B. Rolling pin
 - C. Knife
 - D. Fork
44. When deep frying mandazi, why is it important to regulate oil temperature?
- A. To make mandazi absorb too much oil
 - B. To ensure even cooking and avoid burning
 - C. To make dough harder
 - D. To reduce sweetness
45. Which personal protective wear should learners use when cooking in school?
- A. Dust coat only
 - B. Apron and headscarf
 - C. Gumboots
 - D. Gloves only
46. A learner spills oil on the floor during cooking. The FIRST safe action should be:
- A. Ignore it
 - B. Call the teacher
 - C. Wipe it immediately to prevent accidents
 - D. Continue frying



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COOKING

47. Why should tools and equipment be cleaned immediately after use?
- A. To make kitchen attractive only
 - B. To avoid rust, dirt, and food poisoning
 - C. To waste time
 - D. To reduce costs
48. Using spoiled flour to prepare chapati is wrong because:
- A. It increases flavor
 - B. It can cause food poisoning
 - C. It makes dough elastic
 - D. It saves money
49. Which behavior shows integrity during cooking practical?
- A. Using spoiled ingredients secretly
 - B. Sharing duties honestly and preparing food correctly
 - C. Eating before others
 - D. Ignoring safety rules
50. During class practical, learners are told to switch off gas stoves after cooking. The MAIN reason is:
- A. To save flour
 - B. To save time
 - C. To prevent fire accidents
 - D. To make food taste good



SECTION B: STRUCTURED QUESTIONS (100 Marks)

Question 1.

a) Define batter and dough. (4 marks)

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b) Differentiate between soft dough and stiff dough. (4 marks)

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c) State FOUR examples of products from batter. (4 marks)

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d) State FOUR examples of products from dough. (4 marks)

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e) Give FOUR factors that affect the quality of flour mixtures. (4 marks)

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Question 2: (20 marks)

a) List FIVE steps in preparing chapati. (5 marks)

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b) State FOUR functions of kneading dough. (4 marks)

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c) Explain THREE functions of water in dough preparation. (6 marks)

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d) Give TWO reasons for sieving flour before use. (2 marks)

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e) State THREE effects of adding too much flour in dough. (3 marks)

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Question 3: (20 marks)

a) Describe the procedure of preparing mandazi. (6 marks)

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b) Give THREE cooking methods for flour mixture products. (3 marks)

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c) State FOUR common faults in pancake making. (4 marks)

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d) Explain THREE differences between deep frying and shallow frying. (3 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

e) Suggest FOUR ways of serving chapatis attractively. (4 marks)

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Question 4: Importance & Functions (20 marks)

a) State THREE functions of baking powder in flour mixtures. (3 marks)

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b) Explain THREE nutritional benefits of chapatis. (6 marks)

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c) State TWO importance of pancakes to school learners. (2 marks)

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d) Explain FOUR importance of flour mixture practicals in CBC learning. (8 marks)

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e) State ONE reason why mandazi is best served with tea. (1 mark)

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Question 5: Safety & Integrity (20 marks)

a) State FOUR safety rules when using gas stoves to cook. (4 marks)

b) Explain THREE importance of cleaning kitchen tools after use. (6 marks)

c) State TWO dangers of frying with overheated oil. (2 marks)

d) Give THREE personal hygiene practices when preparing food. (3 marks)

e) Explain FIVE ways learners can show integrity in food preparation. (5 marks)



SET 2

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Instructions: Answer ALL questions.

Question 1: (20 Marks)

a) Define a flour mixture. (2 marks)

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b) Distinguish between batter and dough, giving one example of each. (6 marks)

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c) Explain TWO characteristics of batter. (4 marks)

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d) Explain TWO characteristics of dough. (4 marks)

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e) List TWO examples of products made from batter and TWO from dough. (4 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

Question 2: (20 Marks)

a) List THREE ingredients used in making chapati. (3 marks)

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b) State the function of each of the following ingredients: (6 marks)

i. Flour

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ii. Oil

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iii. Baking powder

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c) Outline FIVE steps in the preparation of chapati. (5 marks)

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d) Explain TWO reasons why kneading is important in dough preparation. (4 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

e) State TWO effects of using excess water in chapati dough. (2 marks)

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Question 3: (20 Marks)

a) Describe the procedure of preparing pancakes. (6 marks)

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b) Give FOUR cooking methods used in making flour mixture products. (4 marks)

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c) Differentiate between shallow frying and deep frying. (4 marks)

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d) State THREE common products prepared from stiff dough. (3 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

e) Explain THREE factors to consider when selecting flour for baking. (3 marks)

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Question 4: (20 Marks)

a) State THREE nutritional values of chapati. (3 marks)

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b) Give THREE importance of mandazi in Kenyan meals. (3 marks)

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c) Explain THREE functions of oil in flour mixtures. (6 marks)

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d) Discuss TWO importance of flour mixture practicals in schools. (4 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

e) State FOUR functions of raising agents in flour mixtures. (4 marks)

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Question 5: Safety & Ethics (20 Marks)

a) Give FOUR safety rules to observe when frying mandazi. (4 marks)

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b) Explain THREE hygienic practices learners should follow before cooking. (6 marks)

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c) State TWO consequences of ignoring safety measures in cookery practicals. (4 marks)

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d) List THREE personal protective equipment (PPE) used during cooking. (3 marks)

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e) Explain the importance of integrity when preparing food mixtures in school. (3 marks)

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SET 3

SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all questions in the spaces provided. Show your working clearly where necessary.

Question 1 (10 Marks)

Jane is learning how to prepare different products using flour mixtures.

a) Name and describe **two main types of flour mixtures** used in food production. For each, give one example of a product made from it. (4 Marks)

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b) Jane wants to make pancakes.

i) List the main ingredients needed for pancake batter. (2 Marks)

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ii) State the type of batter used for pancakes. (1 Mark)

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KITCHEN AND BACKYARD GARDENING



COOKING

c) Jane's pancakes turn out hard and flat. Give **three reasons** why this may have happened. (3 Marks)

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Question 2 (10 Marks)

Mwangi's family enjoys chapati, and he wants to learn how to make them.

a) i) What type of flour mixture is used to make chapati? (1 Mark)

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ii) Write the **steps of preparing chapati dough** from mixing to rolling. (4 Marks)

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b) Explain why kneading is important when making chapati or bread dough. (3 Marks)

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c) State two **safety precautions** Mwangi should take when using a hot pan with oil. (2 Marks)

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Question 3 (10 Marks)

Susan served a certain product as shown.



a) Identify the product shown in the image. (1 Mark)

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b) State the type of flour mixture used to make mandazi. (1 Mark)

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c) Write down **four ingredients** commonly used when making mandazi. (4 Marks)

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d) Explain the role of a raising agent (baking powder or yeast) in making mandazi soft and spongy. (2 Marks)

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e) State **two changes** that show mandazi are well cooked. (2 Marks)

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Question 4 (10 Marks)

A school's home science club is having a cooking competition using flour mixtures.

a) A team decides to make muffins using a dropping batter.

i) Describe the consistency of a dropping batter. (1 Mark)

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KITCHEN AND BACKYARD GARDENING



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ii) Write the steps followed when preparing muffin batter. (3 Marks)

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b) State and explain **two common problems** that can occur when preparing flour mixtures. For each, give one solution. (4 Marks)

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c) Give two differences between a batter and a dough. (2 Marks)

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Question 5 (10 Marks)

Sarah is learning about different flour mixtures.

a) Describe the characteristics of a **thin batter** and give one example of a product made from it. (2 Marks)

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b) State three factors that affect the thickness of a flour mixture. (3 Marks)

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c) Why is it important to store flour properly before using it in mixtures? (2 Marks)

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d) State three qualities of well-prepared chapati dough. (3 Marks)

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e) A teacher asked the learners to write down the uses of the baking tools. Write down the uses of the following.

(a) Casserole

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(b) Loaf tin

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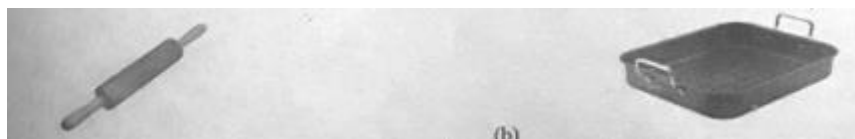
(c) Baking tray

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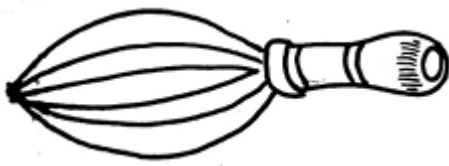
f) Name the following equipment used for baking. (2 mks)



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g) The following picture shows an equipment used when baking. Name the equipment.



h) Emma wanted to bake a cake but she did not have a baking tin and an electric oven. Name improvised tools and equipment she can make and use in baking. (2 mks)

i) Outline **three** advantages of baking as a method of cooking. (3 marks)

j) Grade 7 learners prepared a birthday cake for their classmate. Which cooking method were they likely to use during preparation?

k) Explain why it's important to leave a batter mixture to stand for at least half an hour before cooking



l) Name and give one use of any five ingredients used in flour mixtures

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m) List four general rules to observe when making flour mixtures.

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n) Give two reasons for folding in flour in creamed cake mixtures.

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KITCHEN AND BACKYARD GARDENING



COOKING

PRACTICAL (15 MARKS)

Task 2: Preparing a Simple Flour Mixture (Pancake Batter)

Materials Needed

1. Wheat flour
2. Sugar
3. Salt
4. Baking powder
5. Milk or water
6. Eggs (optional)
7. Mixing bowl
8. Spoon or whisk
9. Measuring cups and spoons
10. Frying pan or hot plate (for demonstration only if available)
11. Apron and head cover
12. Clean working surface

Instructions to the learner

1. Follow the correct steps to make a smooth pancake batter using the ingredients provided.
2. Maintain cleanliness and good hygiene throughout the activity.
3. Ask for help if needed and keep your working space neat.

Procedure (10 marks)

1. Wash your hands and wear an apron and head cover.
2. Measure 1 cup of flour into a bowl.
3. Add 1 tablespoon of sugar, a pinch of salt, and 1 teaspoon of baking powder.



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4. Add 1 egg (optional).
5. Gradually add $\frac{1}{2}$ cup of milk or water while stirring.
6. Mix until the batter is smooth and free from lumps.
7. Ensure the batter has a pourable consistency.
8. Clean your tools and work area after preparation.
9. Present your batter for inspection and explain the steps taken.

Questions (5 marks)

1. Why is it important to wash your hands before preparing pancake batter?
2. State why we add baking powder to the flour mixture.
3. Give a reason why the batter should be free from lumps.
4. What is the importance of keeping the working area clean?
5. Explain why we stir the mixture gradually while adding milk or water.



STRAND 3.0: HYGIENE PRACTICES

SUB-STRAND 3.1: CLEANING WASTE DISPOSAL FACILITIES

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. A household uses a waste bin in their kitchen. Over time, they notice an unpleasant odor even after emptying it. What is the most likely cause?
 - A) The bin is too small.
 - B) Residual food particles and moisture allowing bacterial growth.
 - C) The bin is placed in a cool area.
 - D) It is an old bin.
2. A learner cleans the kitchen sink after preparing food. Why is regular cleaning of the sink important?
 - A) To make it shine for visitors.
 - B) To prevent buildup of grime, odors, and harmful microorganisms.
 - C) To increase water consumption.
 - D) To change the color of the sink.
3. A community faces frequent outbreaks of cholera due to poor drainage. How does cleaning open drains help prevent such diseases?
 - A) It encourages water stagnation.
 - B) It removes waste, preventing breeding grounds for vectors.
 - C) It makes water flow faster, spreading pathogens.
 - D) It has no effect on disease spread.



4. A family washes their waste bin with soap and water after garbage collection. This practice shows:
- A) Ignoring hygiene.
 - B) Proper household waste management.
 - C) Increasing waste production.
 - D) Overuse of chemicals.
5. Why is environmental hygiene promoted through regular cleaning of waste disposal facilities?
- A) It increases pollution.
 - B) It reduces odors, pests, and spread of pathogens.
 - C) It has no impact on the environment.
 - D) It contributes to global warming.
6. A household uses an open pit for biodegradable waste. Flies gather around it. What should they do?
- A) Add more waste to the pit.
 - B) Cover the pit properly and manage it with soil.
 - C) Leave the pit open.
 - D) Spray the flies with water.
7. A learner scrubs a dirty waste bin with detergent. Which type of skill is applied here?
- A) Theoretical skill.
 - B) Practical cleaning skill.
 - C) Communication skill.
 - D) Research skill.
8. A family wants to reduce odors from their kitchen waste bin. Which practice is most helpful?
- A) Throwing liquids directly into the bin.
 - B) Wrapping food scraps before discarding.
 - C) Leaving the lid open.
 - D) Using a lidless bin.



9. A kitchen sink is clogged with grease and food particles. What is the best initial cleaning step?
- A) Pour corrosive chemicals immediately.
 - B) Remove debris and flush with hot water and detergent.
 - C) Ignore the clog.
 - D) Add more food waste.
10. A community drain is blocked with plastics, leading to stagnant water. What health risk does this cause?
- A) Improved water quality.
 - B) Mosquito breeding, causing malaria and dengue.
 - C) Enhanced air circulation.
 - D) Reduced skin infections.
11. A household uses small bins in each room. How does cleaning them regularly improve hygiene?
- A) It increases dust.
 - B) It prevents pests and odors.
 - C) It makes bins heavier.
 - D) It wears bins out faster.
12. What is a direct result of not cleaning waste facilities regularly?
- A) Improved air quality.
 - B) Fewer pests.
 - C) Attraction of rats, cockroaches, and flies.
 - D) Cleaner surfaces.
13. A public market introduces daily cleaning of waste bins. What is the main benefit?
- A) Customers stop visiting.
 - B) Reduced odors, improved health, and a cleaner environment.
 - C) More water consumption.
 - D) Higher waste collection costs.



KITCHEN AND BACKYARD GARDENING



COOKING

14. When cleaning an open drain, which protective gear is most suitable?
- A) Clean clothes only.
 - B) Gloves, boots, and a mask.
 - C) No protective gear.
 - D) A hat only.
15. Which personal health benefit is most directly linked to clean waste bins and drains?
- A) Better eyesight.
 - B) Reduced exposure to germs.
 - C) Stronger bones.
 - D) Improved sleep.
16. After cleaning a bin with detergent, what should be done next?
- A) Leave detergent on.
 - B) Rinse with clean water and dry completely.
 - C) Immediately add waste.
 - D) Wipe with a dry cloth only.
17. Why should a waste bin have a tight-fitting lid?
- A) To make it decorative.
 - B) To prevent odors and keep pests out.
 - C) To make it hard to open.
 - D) To increase capacity.
18. A household uses a refuse pit. How should they manage waste added to it?
- A) Leave it exposed.
 - B) Cover it with soil after adding waste.
 - C) Burn waste daily in the open.
 - D) Add water constantly.



KITCHEN AND BACKYARD GARDENING



COOKING

19. A dustbin attracts ants. What waste disposal practice can stop this?
- A) Leave food waste uncovered.
 - B) Securely bag food scraps before disposal.
 - C) Fill the bin with water.
 - D) Add sweet substances.
20. What is the best way to unclog a greasy kitchen sink drain?
- A) Pour cold water.
 - B) Use baking soda and vinegar, then hot water.
 - C) Use a sharp object to poke it.
 - D) Wait until it blocks completely.
21. How does cleaning waste disposal facilities support a clean environment?
- A) By polluting more.
 - B) By removing sources of contamination and vectors.
 - C) By promoting bacteria growth.
 - D) By reducing need for clean water.
22. Which tool is commonly used to clear solid waste from open drains?
- A) Feather duster.
 - B) Shovel or rake.
 - C) Paintbrush.
 - D) Cooking spoon.
23. A family adopts waste separation at home. How does this improve hygiene?
- A) It complicates disposal.
 - B) It makes recycling easier and reduces pests.
 - C) It increases waste generation.
 - D) It has no effect on hygiene.



24. A learner notices sticky residue at the bottom of a waste bin. Why is scrubbing necessary?
- A) To make it look nice.
 - B) Residue harbors bacteria and pests.
 - C) To make the bin heavier.
 - D) To help drying faster.
25. Why should waste bins be emptied regularly, even when not full?
- A) To make space for new waste.
 - B) To prevent decomposition, odor, and pests.
 - C) To increase waste volume.
 - D) To allow longer storage.
26. A family uses a pit latrine. How can they maintain its hygiene?
- A) Leave the hole open.
 - B) Keep the seat clean and cover the hole after use.
 - C) Throw food waste in it.
 - D) Avoid cleaning the area.
27. What is the most immediate benefit of a clean waste facility at home?
- A) Bigger-looking room.
 - B) A pleasant and healthy environment.
 - C) Higher cost of living.
 - D) More insects attracted.
28. Which example shows problem-solving in cleaning waste facilities?
- A) Always cleaning the same way.
 - B) Identifying odor/blockage causes and fixing them.
 - C) Waiting for someone else.
 - D) Complaining about the dirt.



29. What is the main reason for cleaning waste disposal facilities?

- A) To generate more waste.
- B) For appearance only.
- C) To prevent diseases, odors, and pests.
- D) To make cleaning difficult.

30. Which simple step can reduce foul smells in outdoor bins?

- A) Keeping the lid closed and bin washed regularly.
- B) Sprinkling food waste directly.
- C) Ignoring the bin.
- D) Using bins without covers.

SET 2

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

(Each question carries 1 mark. Choose the best answer.)

1. A waste bin in the kitchen has been left uncleaned for several days. It produces a bad smell.



What is the best way to solve this problem?

- A. Spray air freshener
- B. Empty and wash the bin with water and detergent
- C. Cover the bin with a cloth
- D. Leave the window open



KITCHEN AND
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COOKING

2. Why is it important to clean a sink regularly?
 - A. To make the kitchen look shiny
 - B. To prevent clogging and growth of germs
 - C. To reduce the amount of water used
 - D. To make washing dishes easier
3. A household has open drains around the compound. If not cleaned, these drains may cause:
 - A. Flooding and mosquito breeding
 - B. Faster water flow
 - C. Easier waste disposal
 - D. Better crop growth nearby
4. Which cleaning material is most suitable for removing grease from a sink?
 - A. Sand
 - B. Detergent
 - C. Soil
 - D. Sawdust
5. A farmer cleans his dustbin every week. This practice mainly helps to:
 - A. Reduce water consumption
 - B. Prevent foul smell and pests
 - C. Save on soap
 - D. Increase kitchen size



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COOKING

6. What is the first step when cleaning a waste bin?
- A. Add water before emptying it
 - B. Empty all the waste inside
 - C. Spray perfume inside
 - D. Cover it tightly
7. Why should waste disposal facilities be covered after cleaning?
- A. To keep the waste warm
 - B. To prevent entry of flies and rodents
 - C. To help garbage rot faster
 - D. To make the bin heavier
8. Which disease is most likely to spread from dirty open drains?
- A. Malaria
 - B. Diabetes
 - C. Asthma
 - D. Anemia
9. Which tool is best for cleaning open drains at household level?
- A. Hoe
 - B. Mop
 - C. Stick and brush
 - D. Broom only



10. When cleaning a kitchen sink, why should running water be used instead of stagnant water?
- A. Saves more water
 - B. Removes dirt faster and keeps it hygienic
 - C. Helps store water in sink
 - D. Makes kitchen look modern
11. A household leaves food waste in an uncovered bin. What problem is most likely to occur?
- A. Flies and bad smell
 - B. Fresh air circulation
 - C. Faster composting
 - D. Increased storage space
12. Which is the safest way of disposing dirty water after cleaning a waste bin?
- A. Pouring it into an open drain
 - B. Pouring it into a toilet or soak pit
 - C. Storing it in a container
 - D. Throwing it on the road
13. Which material helps to remove germs when cleaning a dustbin?
- A. Soap or detergent
 - B. Stones
 - C. Ash only
 - D. Sand



14. Why is it advisable to wear gloves when cleaning waste bins?
- A. To look smart
 - B. To protect hands from dirt and germs
 - C. To wash faster
 - D. To use less water
15. A kitchen sink has blocked because of food particles. Which method is most suitable to unblock it?
- A. Pouring hot water and detergent
 - B. Adding more food waste
 - C. Covering the sink
 - D. Ignoring the problem
16. The main reason why children should avoid playing near dirty open drains is because:
- A. It makes them untidy
 - B. They may get injured or infected
 - C. They can fall inside
 - D. They may spill water
17. Which of the following is a proper hygiene practice at household level?
- A. Leaving the dustbin uncovered
 - B. Cleaning waste disposal facilities regularly
 - C. Throwing waste on the ground
 - D. Ignoring clogged drains
18. Dirty waste disposal facilities are breeding grounds for:
- A. Germs and pests
 - B. Fertilizers
 - C. Food crops
 - D. Clean air



19. Which chemical can be used to kill germs in waste bins?
- A. Disinfectant
 - B. Fertilizer
 - C. Fuel
 - D. Cooking oil
20. A farmer pours animal waste directly into an open drain. What problem can this cause?
- A. Water contamination
 - B. Food preservation
 - C. Increased water supply
 - D. Cleaner drains
21. Which is the most important reason for covering a pit latrine after use?
- A. To make it warm
 - B. To prevent flies and bad smell
 - C. To decorate it
 - D. To save space
22. A household uses different bins for plastics, food waste, and bottles. This practice is called:
- A. Composting
 - B. Waste segregation
 - C. Recycling
 - D. Drainage cleaning



23. The best tool to scrub the inside of a waste bin is:
- A. Wire brush or stiff brush
 - B. Knife
 - C. Stick
 - D. Broom only
24. Why should waste be disposed of properly instead of being dumped in open spaces?
- A. To prevent diseases and keep the environment clean
 - B. To reduce work
 - C. To make waste decompose slowly
 - D. To save time
25. Which of the following is NOT a waste disposal facility?
- A. Kitchen sink
 - B. Compost pit
 - C. Dustbin
 - D. Dining table
26. A clogged drain in a household compound can best be cleaned using:
- A. Stick or hoe to remove dirt
 - B. Hands only
 - C. Adding more dirt
 - D. Pouring clean drinking water



27. Why should detergents be rinsed off properly after cleaning waste bins?

- A. To avoid bad smell from soap
- B. To prevent chemicals from contaminating food waste
- C. To make the bin shine
- D. To use more water

28. A clean household dustbin should always be kept:

- A. Covered and in a clean place
- B. Open and near the kitchen
- C. Far away and unwashed
- D. In the sitting room

29. Which of the following diseases is linked to poor waste disposal?

- A. Typhoid
- B. Rickets
- C. Goitre
- D. Scurvy

30. The best way to prevent mosquitoes from breeding in drains is to:

- A. Keep drains clean and flowing
- B. Cover them with paper
- C. Leave stagnant water
- D. Pour food waste inside



31. Which of the following is the LAST step when cleaning a dustbin?
- A. Scrubbing
 - B. Emptying waste
 - C. Rinsing and drying
 - D. Adding soap
32. A school encourages learners to clean sinks after use. The main aim is to:
- A. Promote good hygiene habits
 - B. Reduce water use
 - C. Punish learners
 - D. Avoid cooking
33. Which of these is the best practice in handling waste?
- A. Using covered bins and cleaning them regularly
 - B. Leaving bins dirty for many weeks
 - C. Burning waste inside the house
 - D. Pouring waste in nearby rivers
34. When cleaning an open drain, one should wear:
- A. Gumboots and gloves
 - B. Bare hands and sandals
 - C. Socks only
 - D. Open shoes



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35. Food leftovers should be disposed of in:
- A. Covered waste bin
 - B. Bedroom drawer
 - C. Sitting room table
 - D. On the ground
36. Which practice helps in recycling household waste?
- A. Separating organic waste for composting
 - B. Mixing all waste together
 - C. Burning plastics
 - D. Throwing bottles in drains
37. Why is it important to dry a dustbin after washing?
- A. To prevent rusting and growth of mould
 - B. To make it lighter
 - C. To keep it shiny
 - D. To save on detergent
38. Which of these wastes should NOT be poured into a kitchen sink?
- A. Cooking oil and fats
 - B. Water
 - C. Soap foam
 - D. Tea



39. Which community practice helps keep open drains clean?
- A. Organizing regular clean-up exercises
 - B. Pouring rubbish in drains
 - C. Blocking water flow
 - D. Ignoring drains
40. Which is the correct order of cleaning a waste bin?
- A. Empty → Scrub with soap → Rinse → Dry
 - B. Rinse → Dry → Empty → Scrub
 - C. Scrub → Empty → Dry → Rinse
 - D. Dry → Empty → Scrub → Rinse
41. Waste disposal facilities should be cleaned regularly because:
- A. They attract pests if left dirty
 - B. They increase house size
 - C. They make waste smell nice
 - D. They are hard to use
42. Which waste disposal facility is used to dispose of wastewater in households?
- A. Sink
 - B. Dining table
 - C. Cupboard
 - D. Stool



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COOKING

43. A child throws plastic bottles into a drain. Which advice should be given?
- A. Place bottles in a recycling bin
 - B. Add more bottles in the drain
 - C. Burn them inside the house
 - D. Use them as toys only
44. Why is disinfectant added when cleaning a sink?
- A. To kill germs and remove bad smell
 - B. To make water taste good
 - C. To polish metal
 - D. To thicken soap
45. Which household waste is best taken to a compost pit?
- A. Vegetable peels
 - B. Plastic bags
 - C. Old shoes
 - D. Metal tins
46. The correct way to handle sharp objects in waste is:
- A. Wrap and place in separate container
 - B. Throw directly into open bin
 - C. Leave on the floor
 - D. Burn in open fire
47. A family cleans drains every Saturday. This is important because it:
- A. Prevents water stagnation and disease spread
 - B. Wastes time and energy
 - C. Increases cost of living
 - D. Makes drains look dark



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48. Which practice should be avoided when disposing of waste?
- A. Dumping in open spaces
 - B. Using covered bins
 - C. Composting organic waste
 - D. Recycling plastics
49. Which type of water is recommended for rinsing cleaned waste disposal facilities?
- A. Clean running water
 - B. Dirty water from drains
 - C. Water mixed with oil
 - D. Stagnant pond water
50. Which of the following is the MAIN benefit of cleaning waste disposal facilities at household level?
- A. Promotes good health and hygiene
 - B. Provides more rubbish
 - C. Reduces water supply
 - D. Produces more waste

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Answer all questions.

I. (20 marks)

a) State three reasons why it is important to clean household waste bins. (3 marks)

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b) List four materials used in cleaning waste bins. (4 marks)

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c) Describe the steps followed when cleaning a waste bin. (5 marks)

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d) Explain three dangers of failing to clean waste bins regularly. (6 marks)

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e) Suggest two ways of encouraging family members to use clean waste bins. (2 marks)

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2. (20 marks)

a) State three common problems caused by dirty sinks. (3 marks)

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KITCHEN AND BACKYARD GARDENING



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b) Identify four cleaning agents used to clean sinks. (4 marks)

c) Describe how to properly clean a greasy sink. (5 marks)

d) Explain three benefits of cleaning sinks regularly. (6 marks)

e) Suggest two household practices that prevent sink blockage. (2 marks)

3. (20 marks)

a) Give three reasons why open drains should be cleaned. (3 marks)



KITCHEN AND BACKYARD GARDENING



COOKING

b) Mention four tools used to clean open drains. (4 marks)

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c) Describe the procedure of cleaning an open drain. (5 marks)

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d) Explain three dangers of not cleaning open drains. (6 marks)

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e) Suggest two ways of preventing dirt from entering drains. (2 marks)

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4. (20 marks)

a) Explain three reasons why proper hygiene should be maintained in waste disposal. (3 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

b) List four diseases caused by poor waste disposal. (4 marks)

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c) Describe how waste disposal facilities promote environmental hygiene. (5 marks)

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d) Explain three ways in which poor hygiene in waste disposal affects family health. (6 marks)

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e) Suggest two measures households can adopt to maintain clean waste disposal facilities. (2 marks)

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5. (20 marks)

a) State three examples of waste disposal facilities at household level. (3 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

b) List four advantages of using covered dustbins. (4 marks)

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c) Describe how a family can ensure waste disposal facilities are always clean. (5 marks)

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d) Explain three roles of learners in promoting household hygiene. (6 marks)

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e) Suggest two practices that promote environmental cleanliness in the community. (2 marks)

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KITCHEN AND BACKYARD GARDENING



COOKING

SET 2

SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all questions in the spaces provided. Show your working where necessary.

Question 1 (10 Marks)

Mrs. Adhiambo's kitchen waste bin often smells bad and attracts flies, even though she empties it daily.

a) Give three reasons why it is important to clean waste bins regularly. (3 Marks)

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b) Write down the steps Mrs. Adhiambo should follow to clean her kitchen waste bin properly. (4 Marks)

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c) Suggest three ways she can reduce bad smell from the waste bin between cleanings. (3 Marks)

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Question 2 (10 Marks)

Study the image below.



KITCHEN AND BACKYARD GARDENING



COOKING

a) Name the waste disposal facility shown in the picture. (1 Mark)

b) State three health problems that may arise if the facility remains dirty. (3 Marks)

c) List four steps a community clean-up team should follow to clean the drainage. (4 Marks)

d) Explain two ways cleaning such a facility improves hygiene in the community. (2 Marks)

Question 3 (10 Marks)

A Grade 9 learner is preparing a project on Household Hygiene. Part of the work is about taking care of sinks and drains.

a) Why should a kitchen sink be cleaned regularly, even after washing dishes? (2 Marks)

b) State three ways to prevent a kitchen sink drain from blocking. (3 Marks)

c) Mention three good practices for keeping an outdoor refuse pit clean. (3 Marks)

d) Explain two benefits of using clean waste disposal facilities in a home. (2 Marks)



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Question 4 (10 Marks)

Mr. Kimani wants to improve the cleanliness of waste disposal facilities in his home, starting with dustbins.

a) (i) Give two reasons why a dustbin should be washed with soap and water, not just emptied. (2 Marks)

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(ii) Why is it important to dry the dustbin before putting new waste? (1 Mark)

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b) Mr. Kimani's area has many rodents. Explain three measures he can take when managing waste to keep away rodents. (3 Marks)

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c) Bad smell from sink drains is common in homes. State two causes of the smell and give a simple cleaning method for each. (4 Marks)

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Question 5 (10 Marks)

Some households neglect cleaning their waste disposal facilities.

a) State three problems that may occur when waste bins and drains are not cleaned. (3 Marks)

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b) Explain how poor waste management can affect the health of children in a home. (4 Marks)

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KITCHEN AND BACKYARD GARDENING



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c) Give three ways in which unclean waste facilities can spoil the appearance of a home or a community. (3 Marks)

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STRAND 3.0: HYGIENE PRACTICES

SUB-STRAND 3.2: DISINFECTING CLOTHING AND HOUSEHOLD ARTICLES

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. After a family member recovers from a common cold, their handkerchiefs are washed thoroughly. To ensure complete disinfection and kill any remaining germs, what common household method would be most effective for these fabric items?
A) Soaking in cold water.
B) Hanging them indoors to dry slowly.
C) Boiling them in hot water.
D) Air drying them in the shade.
2. A farmer's protective gloves and dustcoat are often exposed to dirt and potential pathogens from the farm environment. To disinfect these items after washing, especially on a sunny day, Narok County, what natural method would be highly effective?
A) Freezing them overnight.
B) Soaking them in a sugary solution.
C) Spreading them out under direct sunlight.
D) Keeping them in a dark, damp corner.
3. A kitchen sponge is frequently used to wipe surfaces and clean dishes. To prevent the spread of bacteria from the sponge to food preparation areas, what simple disinfecting method can be regularly applied to the sponge?
A) Rinsing with cold water only.
B) Microwaving a wet sponge for a short period or soaking in a disinfectant solution.
C) Leaving it damp on the counter.
D) Adding more soap to it.
4. A mother is concerned about germs on her baby's clothes and cloth diapers. After washing, she adds a small amount of salt to the final rinse water for disinfection. This method of disinfection is known as:
A) Chemical disinfection.
B) Thermal disinfection.
C) Salting.
D) Air drying.



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5. Why is personal hygiene promoted through the regular disinfection of personal items like handkerchiefs, socks, and towels?
- A) It makes items last longer.
 - B) It reduces the spread of germs and prevents skin infections or diseases.
 - C) It adds a pleasant scent to the items.
 - D) It increases the water bill.
6. A learner is helping to clean the household articles. They are advised to use a disinfectant solution for wiping down frequently touched surfaces like doorknobs and light switches. What is the main purpose of disinfecting these surfaces?
- A) To make them shiny.
 - B) To kill germs that can be transferred from hands, preventing disease transmission.
 - C) To make them sticky.
 - D) To add a protective layer.
7. A family iron their clothes after washing and drying. Beyond smoothing out wrinkles, how does ironing contribute to the disinfection of clothing?
- A) It introduces new germs.
 - B) The high heat from the iron helps to kill remaining bacteria and viruses.
 - C) It makes the clothes damp.
 - D) It removes dust.
8. Which of the following household articles would most likely require regular disinfection due to frequent handling and potential for germ transfer?
- A) Decorative picture frames.
 - B) Remote controls, phone covers, and cutting boards.
 - C) Unused books.
 - D) Ceiling fans.
9. A farmer has some reusable face masks that they wear while working. After each use, they wash the masks thoroughly. To ensure they are fully disinfected, what simple method can they use, especially on a bright sunny day?
- A) Soaking them in cold water.
 - B) Drying them completely in direct sunlight.
 - C) Putting them in a dark cupboard.
 - D) Rinsing them quickly.



10. A Grade 9 learner is learning about different methods of disinfection. They are taught that boiling is a highly effective method. For which type of article would boiling be most suitable for disinfection?
- A) Delicate silk clothing.
 - B) Metal cooking utensils or baby bottles.
 - C) Electronic devices.
 - D) Leather shoes.
11. Which of the following items should **never** be disinfected by boiling because it could be damaged?
- A) Cotton handkerchiefs.
 - B) Plastic baby bottles.
 - C) Woolen sweaters.
 - D) Metal utensils.
12. A family uses separate cleaning cloths for different areas of the house (e.g., kitchen, bathroom). To prevent cross-contamination, these cloths are disinfected after each use. Why is this practice important?
- A) To make the cloths last longer.
 - B) To prevent the transfer of germs from one area to another.
 - C) To make cleaning more difficult.
 - D) To save water.
13. Which of the following is a chemical commonly used as a disinfectant for household surfaces?
- A) Cooking oil.
 - B) Chlorine bleach (diluted), rubbing alcohol, or hydrogen peroxide.
 - C) Sugar syrup.
 - D) Plain water.
14. A learner is carrying out disinfection of their personal towel. They decide to wash it with hot water and then hang it out to dry thoroughly. What is the role of hot water in this disinfection process?
- A) It makes the towel rough.
 - B) The heat helps to kill germs.
 - C) It makes the towel smell bad.
 - D) It has no effect on germs.
15. What is the primary purpose of disinfecting clothing and household articles?
- A) To make them look new.
 - B) To kill or inactivate disease-causing microorganisms (germs) on their surfaces.
 - C) To change their color.
 - D) To increase their weight.



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16. A farmer uses specific tools for handling sick animals. After use, these tools must be thoroughly disinfected. What method would be most appropriate for disinfecting durable metal tools?
- A) Wiping with a dry cloth.
 - B) Soaking in boiling water or a strong disinfectant solution.
 - C) Leaving them in a damp shed.
 - D) Burying them in the ground.
17. A household has a child with a skin infection. To prevent the spread of the infection, their bed linens are washed and then disinfected. What is a highly effective method for disinfecting bed linens after washing?
- A) Placing them in a cold, dark room.
 - B) Using a hot wash cycle and/or ironing.
 - C) Just shaking them out.
 - D) Leaving them unwashed for a week.
18. Why is sunlight an effective natural disinfectant for certain clothing and household articles?
- A) It fades colors.
 - B) Ultraviolet (UV) radiation in sunlight helps to kill bacteria and viruses.
 - C) It makes things colder.
 - D) It makes things sticky.
19. A learner is cleaning their school uniform, which they know might have come into contact with various surfaces during the day. After washing, they decide to hang it outside in direct sunlight to dry. This practice aligns with which disinfection method?
- A) Chemical disinfection.
 - B) Thermal disinfection (from sun's heat).
 - C) Salting.
 - D) Freezing.
20. A family uses a wooden cutting board in their kitchen. After cutting raw meat, they scrub the board thoroughly. To further disinfect it and prevent bacterial growth, what natural method can be applied?
- A) Rubbing with cooking oil.
 - B) Washing with soap and water, then possibly using diluted vinegar or salt solution, and drying thoroughly.
 - C) Soaking in sugary water.
 - D) Leaving it unwashed.



21. Which of the following disinfecting methods is **NOT** suitable for delicate fabrics like silk and wool?
- A) Sun drying.
 - B) Boiling.
 - C) Using mild disinfectant solutions.
 - D) Gentle ironing.
22. A learner wants to disinfect their sports shoes, which have a persistent odor. Which of the following methods, specifically involving heat, could help?
- A) Soaking them in cold water.
 - B) Placing them in a tumble dryer on a hot cycle, if suitable for the material, or exposing to strong sunlight.
 - C) Burying them in soil.
 - D) Just spraying with water.
23. When using chemical disinfectants (e.g., bleach) for household articles, what is a crucial safety precaution to observe?
- A) Mixing different chemicals together for stronger effect.
 - B) Wearing gloves and ensuring good ventilation.
 - C) Drinking a small amount to test its potency.
 - D) Using them without diluting.
24. A household has a family member with athlete's foot (a fungal infection). To prevent its spread, what specific item should be disinfected regularly, in addition to feet hygiene?
- A) Remote controls.
 - B) Socks and bath mats.
 - C) Curtains.
 - D) Wall hangings.
25. Which method of disinfection relies purely on very high temperatures to kill microorganisms?
- A) Salting.
 - B) Boiling.
 - C) Air drying.
 - D) Wiping with a wet cloth.
26. A Grade 9 learner is making a class presentation on The Importance of Disinfecting. They want to emphasize preventing the spread of diseases. Which of the following diseases is directly preventable through proper disinfection of surfaces and clothing?
- A) Diabetes.
 - B) Common cold, flu, and stomach infections.
 - C) Heart disease.
 - D) Allergies to pollen.



KITCHEN AND BACKYARD GARDENING



COOKING

27. A farmer uses a watering can. To ensure it doesn't harbor algae or fungal spores that could harm plants, they periodically clean and disinfect it. What simple household item, known for its antiseptic properties, could be used for this purpose?
- A) Sugar.
 - B) Vinegar or baking soda solution.
 - C) Plain sand.
 - D) Cooking oil.
28. After a sick child uses certain toys, the parent disinfects them. For plastic toys, what would be a safe and effective disinfection method?
- A) Boiling in very hot water.
 - B) Wiping with a diluted bleach solution or disinfectant wipes.
 - C) Freezing for an hour.
 - D) Just wiping with a dry cloth.
29. Why is it generally not advisable to disinfect colored clothing with strong chlorine bleach?
- A) It makes the clothing stronger.
 - B) It can cause discoloration or damage to the fabric.
 - C) It makes the clothing warmer.
 - D) It makes the clothing smell better.
30. Which of the following items requires the most frequent disinfection in a typical household due to direct contact with human skin and potential for germ transfer?
- A) Curtains.
 - B) Door handles.
 - C) Rarely used books.
 - D) Wall paintings.



KITCHEN AND BACKYARD GARDENING



COOKING

SET 2

Strand 3.0: Hygiene Practices

Sub-strand 3.2: Disinfecting Clothing and Household Articles

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

Instructions: Choose the best answer for each question.

1. A farmer washes their aprons and then hangs them outside under direct sunlight. Which method of disinfection is being applied?
 - A. Boiling
 - B. Salting
 - C. Sunlight
 - D. Ironing
2. A learner adds disinfectant to water before soaking handkerchiefs. What is the purpose of the disinfectant?
 - A. To remove dust
 - B. To kill germs
 - C. To make the cloth soft
 - D. To remove stains
3. Which method of disinfecting clothing involves using high temperature steam to kill microorganisms?
 - A. Salting
 - B. Boiling
 - C. Sun drying
 - D. Dusting
4. When socks are soaked in salty water for some hours, the salt helps by:
 - A. Removing dust
 - B. Killing microorganisms
 - C. Whitening the cloth
 - D. Making them soft
5. Which article is most suitable for ironing as a disinfection method?
 - A. Towels
 - B. Gloves
 - C. Aprons
 - D. Socks



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6. What is the main reason for disinfecting handkerchiefs?



- A. To prevent spread of diseases
 - B. To make them colorful
 - C. To make them soft
 - D. To save time
7. A farmer boils water and dips gloves in it for 10 minutes. Which method of disinfection is this?
- A. Salting
 - B. Boiling
 - C. Ironing
 - D. Sunlight
8. Why should towels be disinfected regularly?
- A. They are cheap
 - B. They absorb sweat and moisture
 - C. They are colorful
 - D. They are made of cotton
9. Which method is most effective for disinfecting metallic household articles like spoons and forks?
- A. Sunlight
 - B. Ironing
 - C. Boiling
 - D. Salting
10. Which of the following is NOT a method of disinfecting clothing?
- A. Sunlight
 - B. Salting
 - C. Boiling
 - D. Painting



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11. Which is the cheapest method of disinfecting socks at home?
- A. Boiling
 - B. Sunlight
 - C. Ironing
 - D. Use of disinfectants
12. A learner irons their school dustcoat before class. What is the main reason for this?
- A. To make it look neat
 - B. To kill germs with heat
 - C. To make it colorful
 - D. To remove stains
13. Which household article is best disinfected by soaking in disinfectants like Dettol?
- A. Metallic utensils
 - B. Handkerchiefs
 - C. Cooking pans
 - D. Buckets
14. Why is boiling considered effective in disinfecting towels?
- A. It softens the fabric
 - B. It kills bacteria and germs
 - C. It makes towels whiter
 - D. It adds fragrance
15. Which item is not usually disinfected by ironing?
- A. Aprons
 - B. Socks
 - C. Dustcoats
 - D. Uniform shirts
16. The main reason for disinfecting gloves used in farming is to:
- A. Prevent spread of infections
 - B. Change their color
 - C. Make them soft
 - D. Remove dust



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17. Which disinfecting method does not require water?
- A. Boiling
 - B. Salting
 - C. Soaking in disinfectant
 - D. Washing with soap
18. A towel used after swimming should be disinfected mainly because:
- A. It looks dirty
 - B. It may contain germs from shared use
 - C. It changes color when wet
 - D. It feels heavy
19. Why is sunlight considered an environmentally friendly disinfection method?
- A. It is free and uses no chemicals
 - B. It takes less time
 - C. It makes clothes smell nice
 - D. It works on all metals
20. Which article can be disinfected both by boiling and ironing?
- A. Towels
 - B. Aprons
 - C. Gloves
 - D. Socks
21. Which of the following kills microorganisms in water before disinfecting clothing?
- A. Adding soap
 - B. Boiling water
 - C. Storing in a bucket
 - D. Filtering
22. A farmer uses sunlight to disinfect dustcoats. What should they do to ensure full disinfection?
- A. Place the dustcoat in shade
 - B. Spread it wide under direct sun
 - C. Fold it on a chair
 - D. Sprinkle it with perfume



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23. Why is it important to disinfect aprons used in food preparation?
- A. To make them white
 - B. To prevent spread of foodborne diseases
 - C. To remove oil stains
 - D. To look smart
24. Which one of the following is not a household article?
- A. Towels
 - B. Gloves
 - C. Aprons
 - D. Wheelbarrow
25. Which method of disinfection uses osmotic pressure to kill germs?
- A. Sunlight
 - B. Salting
 - C. Boiling
 - D. Ironing
26. When using disinfectants, which precaution should be taken?
- A. Always mix many disinfectants together
 - B. Use the correct concentration
 - C. Avoid using gloves
 - D. Never rinse after soaking
27. Which article should NOT be boiled as a method of disinfection?
- A. Socks
 - B. Aprons
 - C. Leather gloves
 - D. Towels
28. The main role of ironing clothes after washing is:
- A. Killing germs with heat
 - B. Adding shine to fabric
 - C. Removing dirt
 - D. Reducing weight



KITCHEN AND BACKYARD GARDENING



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29. Which disinfecting method also helps to dry clothes faster?
- A. Boiling
 - B. Sunlight
 - C. Salting
 - D. Soaking
30. Which one of the following articles can best be disinfected by sunlight?
- A. Handkerchiefs
 - B. Pots
 - C. Spoons
 - D. Cups
31. Which method of disinfecting clothing is most suitable during rainy weather?
- A. Sunlight
 - B. Ironing
 - C. Salting
 - D. Boiling
32. Why should disinfected articles be properly rinsed after soaking in disinfectants?
- A. To make them shiny
 - B. To avoid skin irritation from chemicals
 - C. To add fragrance
 - D. To make drying faster
33. Which article is often disinfected by boiling in hospitals?
- A. Uniforms
 - B. Towels
 - C. Bed sheets
 - D. Socks
34. Which of these is the quickest disinfection method?
- A. Sunlight
 - B. Boiling
 - C. Salting
 - D. Ironing



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COOKING

35. A learner places their wet handkerchief on a hot iron box surface for a few seconds. What is achieved?
- A. Whitening
 - B. Disinfection through heat
 - C. Drying only
 - D. Coloring
36. Which disinfection method also prevents bad smell in socks?
- A. Salting
 - B. Painting
 - C. Ironing
 - D. Perfuming
37. Which article should be disinfected regularly because it comes into contact with the nose and mouth?
- A. Gloves
 - B. Handkerchiefs
 - C. Towels
 - D. Dustcoats
38. Why is it advisable to disinfect clothes before storing them?
- A. To save space
 - B. To prevent spread of germs in storage
 - C. To add perfume
 - D. To make clothes neat
39. Which is a limitation of disinfecting clothes using sunlight?
- A. It is expensive
 - B. It is not effective on rainy days
 - C. It damages fabric
 - D. It uses electricity
40. Which disinfecting method is suitable for metallic spoons and forks?
- A. Boiling
 - B. Sunlight
 - C. Ironing
 - D. Salting



41. What is the main reason for disinfecting farm dustcoats?
- A. To prevent spread of pests and diseases
 - B. To change the fabric color
 - C. To make them fashionable
 - D. To reduce weight
42. Which method is best suited for disinfecting baby clothes?
- A. Ironing
 - B. Boiling
 - C. Salting
 - D. Sunlight
43. Which disinfecting method does not kill germs effectively when clothes are thick?
- A. Sunlight
 - B. Boiling
 - C. Ironing
 - D. Using disinfectants
44. Why is it unsafe to mix many disinfectants together?
- A. It is cheap
 - B. It may produce harmful fumes
 - C. It makes disinfection stronger
 - D. It reduces germs faster
45. Which method also helps to remove wrinkles from clothes?
- A. Salting
 - B. Ironing
 - C. Boiling
 - D. Sunlight
46. Which of the following cannot be disinfected by salting?
- A. Socks
 - B. Handkerchiefs
 - C. Aprons
 - D. Towels



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COOKING

47. Which method of disinfection saves both time and electricity?
- Sunlight
 - Boiling
 - Ironing
 - Using disinfectants
48. Why is it important to disinfect gloves used in milking cows?
- To keep them soft
 - To avoid spreading mastitis
 - To make them white
 - To reduce weight
49. Which article is most suitable for disinfection by boiling in salty water?
- Socks
 - Aprons
 - Gloves
 - Dustcoats
50. What is the general importance of disinfecting household articles?
- To promote personal hygiene and prevent diseases
 - To decorate the house
 - To make clothes attractive
 - To save money

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Answer ALL questions. Each carries 20 marks.

Question 1 (20 marks)

a) Define disinfection of clothing and household articles. (2 marks)

.....

.....



b) List five methods of disinfecting clothing and household articles. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

c) Explain the steps of disinfecting an apron using boiling method. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

d) State three articles that can be disinfected by ironing. (3 marks)

- i).....
- ii).....
- iii)

e) Give five reasons why disinfection of household articles is important. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

Question 2 (20 marks)

a) Explain how sunlight disinfects clothing. (3 marks)

- i).....
- ii).....
- iii)

b) Mention three household articles that can be disinfected using sunlight. (3 marks)

- i).....
- ii).....
- iii)
- iv)



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c) Describe how salting is done on socks to disinfect them. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

d) State two limitations of salting as a disinfection method. (2 marks)

- i).....
- ii).....

e) Give four advantages of using sunlight as a disinfection method. (7 marks)

- i).....
- ii).....
- iii)
- iv)

Question 3 (20 marks)

a) Outline the procedure of disinfecting towels using disinfectants. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

b) Mention three disinfectants commonly used at home. (3 marks)

- i).....
- ii).....
- iii)

c) Explain why boiling is more effective than salting in disinfecting articles. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)



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d) State two safety precautions to take when boiling articles. (2 marks)

i).....

ii).....

e) Give five personal hygiene benefits of disinfecting clothing. (5 marks)

i).....

ii).....

iii)

iv)

v)

Question 4 (20 marks)

a) State four articles that should be disinfected regularly at home. (4 marks)

i).....

ii).....

iii)

iv)

b) Describe the process of disinfecting gloves using ironing. (5 marks)

i).....

ii).....

iii)

iv)

v)

c) Explain three reasons why aprons should be disinfected. (6 marks)

i).....

ii).....

iii)

d) State two challenges farmers face when disinfecting clothes using boiling. (2 marks)

i).....

ii).....



e) Suggest three modern ways of improving household disinfection. (3 marks)

- i).....
- ii).....
- iii)

Question 5 (20 marks)

a) Explain the importance of responsibility in disinfecting personal items. (4 marks)

- i).....
- ii).....
- iii)
- iv)

b) Describe the procedure of disinfecting a dustcoat using boiling. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

c) Outline four differences between sunlight disinfection and ironing disinfection. (4 marks)

- i).....
- ii).....
- iii)
- iv)

d) State two dangers of using too much disinfectant when cleaning. (2 marks)

- i).....
- ii).....

e) Suggest five ways learners can promote personal hygiene through disinfection. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)



SET 2

SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all questions in the spaces provided. Show your working clearly where needed.

Question 1 (10 Marks)

Mrs. Wanjiku's family has been experiencing frequent minor infections. The doctor advised her to disinfect clothing and household items to improve hygiene.

a) State **three reasons** why disinfecting clothing and household articles is important. (3 Marks)

- i).....
- ii).....
- iii)

b) Describe **four methods** of disinfecting clothing and household articles. For each method, give **one example** of an item it can disinfect. (4 Marks)

- i).....
- ii).....
- iii)
- iv)
- v)

c) Mrs. Wanjiku chooses to use direct sunlight for disinfection. Explain **how sunlight disinfects** and state **two conditions** needed for it to work well. (3 Marks)

- i).....
- ii).....
- iii)

Question 2 (10 Marks)



a) Name the method of disinfection shown in the picture. (1 Mark)

- i).....



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b) Explain **how this method works** to disinfect clothing. (2 Marks)

- i).....
- ii).....

c) State **three advantages** of using sunlight to disinfect clothes. (3 Marks)

- i).....
- ii).....
- iii)

d) Mention **two other methods** that could be used to disinfect socks. (4 Marks)

- i).....
- ii).....

Question 3 (10 Marks)

A Grade 9 learner is demonstrating how to disinfect a plastic cutting board and a kitchen towel.

a) (i) Describe the steps for disinfecting a **plastic cutting board** used for raw meat with a chemical disinfectant. (4 Marks)

- i).....
- ii).....
- iii)
- iv)

(ii) Give **two safety precautions** to observe when using chemical disinfectants. (2 Marks)

- i).....
- ii).....

b) Explain how to disinfect a **used kitchen towel** by boiling. (3 Marks)

- i).....
- ii).....
- iii)

c) Suggest **one other method** (not boiling) that can be used to disinfect a kitchen towel. (1 Mark)

- i).....
- ii).....
- iii)
- iv)
- v)



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Question 4 (10 Marks)

Mr. John works on a farm and wears a dustcoat and gloves. He wants to disinfect them properly after use.

a) Give **three reasons** why his farm clothing and gloves should be disinfected regularly. (3 Marks)

- i).....
- ii).....
- iii)

b) Explain how **ironing** helps in disinfecting clothing such as a dustcoat. (2 Marks)

- i).....
- ii).....

c) Mr. John considers using **salting** to disinfect some items. Describe the process of salting and name **one item** that can be disinfected this way. (3 Marks)

- i).....
- ii).....

d) Suggest **two ways** Mr. John can maintain hygiene of his work gloves apart from disinfecting. (2 Marks)

- i).....
- ii).....

Question 5 (10 Marks)

A health campaign in a village teaches people to disinfect household articles to reduce infections.

a) List **four household articles** (other than clothing) that should be disinfected regularly. (4 Marks)

- i).....
- ii).....
- iii)
- iv)

b) For **two of the articles** listed in (a), suggest a suitable method of disinfection and explain why it is appropriate. (4 Marks)

- i).....
- ii).....

c) Explain **two ways** in which regular disinfection of clothing and household items leads to a healthy home. (2 Marks)

- i).....
- ii).....



STRAND 4.0: PRODUCTION TECHNIQUES

SUB-STRAND 4.1: GRAFTING IN PLANTS

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

- Grafting is best described as:
 - Growing plants from seed only
 - Joining two plant parts to grow as one
 - Using water to propagate plants
 - Cutting stems and planting them in soil
- Which part of the plant provides the root system in grafting?
 - Scion
 - Rootstock
 - Bud
 - Branch
- The scion in grafting should come from:
 - A diseased plant
 - A healthy plant with desired traits
 - Any part of the rootstock
 - A dried plant stem
- What is the main reason for grafting plants?
 - To increase seed germination
 - To combine qualities of two plants
 - To produce more weeds
 - To prevent photosynthesis
- Which tool is commonly used during grafting?
 - Hoe
 - Grafting knife
 - Wheelbarrow
 - Water can
- The union in grafting is successful when:
 - The scion and rootstock cambium layers meet
 - Only the roots touch the soil
 - The scion is larger than the rootstock
 - The tape is never removed
- Which plant is commonly propagated by grafting in Kenya?
 - Maize
 - Mango
 - Beans
 - Kale
- The main precaution when using a grafting knife is:
 - Using it without sharpening
 - Keeping it dirty
 - Handling it carefully to avoid injury
 - Using it on stones

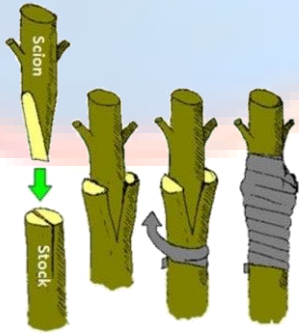


KITCHEN AND BACKYARD GARDENING



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9. What type of graft is shown in the diagram below?



- A. Cleft graft
 - B. Whip graft
 - C. Side graft
 - D. Bud graft
10. Removal of grafting tape is done:
- A. Immediately after grafting
 - B. Before watering
 - C. After the graft union is successful
 - D. During planting
11. Which type of graft is used to repair damaged trees?
- A. Bridge graft
 - B. Side graft
 - C. Whip graft
 - D. Bud graft
12. The whip graft is mostly used when:
- A. The scion is much thicker than the rootstock
 - B. The scion and rootstock are of equal thickness
 - C. The plant is mature
 - D. The plant is dying
13. The diagram shows a bud being inserted under the bark. This method is called:



- A. Side graft
- B. Bud graft
- C. Top graft
- D. Bridge graft

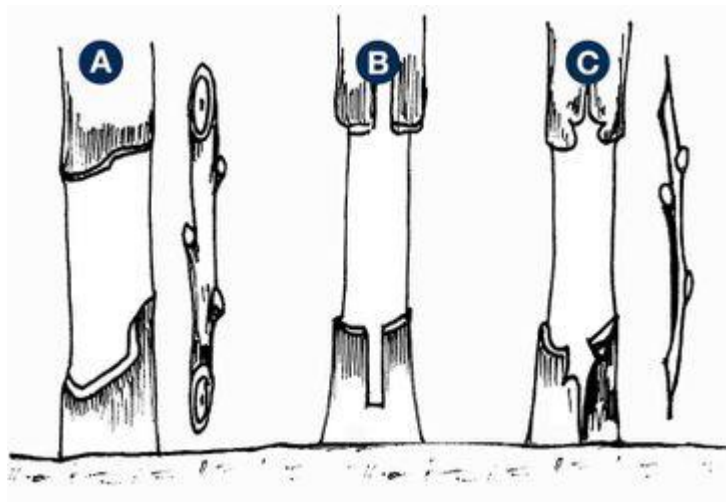


KITCHEN AND BACKYARD GARDENING



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14. Aesthetic grafting is mainly done:
 - A. To improve the taste of fruit
 - B. For beautification of plants
 - C. To control weeds
 - D. To make plants shorter
15. Grafting for rejuvenation is important because:
 - A. It makes old plants young and productive
 - B. It reduces sunlight
 - C. It increases weeds
 - D. It produces more seeds
16. Which grafting method is used in fruit trees with thick bark?
 - A. Budding
 - B. Cleft grafting
 - C. Whip grafting
 - D. Side grafting
17. Which grafting method is suitable when rootstock is larger than scion?
 - A. Whip graft
 - B. Cleft graft
 - C. Bud graft
 - D. None of the above
18. In side grafting, the scion is inserted:
 - A. On the root
 - B. On the side of the rootstock stem
 - C. On the leaves
 - D. On the fruit
19. The diagram shows a graft used for damaged plants. This is:



- A. Whip graft
- B. Side graft
- C. Bridge graft
- D. Bud graft



KITCHEN AND BACKYARD GARDENING



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20. Which of the following is NOT a purpose of grafting?
 - A. Repair
 - B. Rejuvenation
 - C. Aesthetic
 - D. Soil fertility
21. After grafting, watering helps to:
 - A. Kill the plant
 - B. Keep the plant healthy and growing
 - C. Dry the soil
 - D. Stop root formation
22. Why should extra buds on the rootstock be removed?
 - A. They compete with the scion for nutrients
 - B. They beautify the plant
 - C. They improve graft success
 - D. They produce more fruits
23. What should be done if the scion dries up after grafting?
 - A. Leave it alone
 - B. Replace with a fresh scion
 - C. Stop watering
 - D. Harvest immediately
24. Protecting the graft union can be done using:
 - A. Plastic tape
 - B. Stones
 - C. Fertilizer
 - D. Water can
25. Which care practice prevents infection at the graft union?
 - A. Using dirty tools
 - B. Using sharp and clean tools
 - C. Leaving wounds exposed
 - D. Cutting roots
26. The main reason for removing graft tape is:
 - A. To allow the graft to breathe
 - B. To reduce watering
 - C. To stop plant growth
 - D. To kill the plant
27. Too much sunlight on the graft union may cause:
 - A. Faster healing
 - B. Drying and death of scion
 - C. Better growth
 - D. More fruit
28. In handling grafting knives, learners should:
 - A. Share without care
 - B. Use them carelessly
 - C. Follow safety measures to avoid accidents
 - D. Hide them in soil



KITCHEN AND BACKYARD GARDENING



COOKING

29. Why should the graft union be tied tightly?
- To prevent cambium contact
 - To secure the scion and rootstock together
 - To stop plant growth
 - To damage the union
30. The diagram shows a successful graft. Which sign proves success?



- Drying of scion
- New leaf growth
- Broken scion
- Falling bark

SET 2

STRAND 4.0: PRODUCTION TECHNIQUES

SUB-STRAND 4.1: GRAFTING IN PLANTS

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

- A farmer, Narok County, has an old, unproductive mango tree that produces small, sour fruits. They want to start getting large, sweet mangoes without planting a new tree from scratch. Which method of plant propagation would be most suitable to achieve this improvement quickly?
 - Planting seeds from a sweet mango.
 - Air layering.
 - Grafting.
 - Cuttings.
- A learner is learning about grafting. They are told that it involves joining two plant parts: the scion and the rootstock. What is the primary function of the scion in grafting?
 - To provide a strong root system.
 - To provide the desired aerial characteristics
 - To anchor the plant to the soil.
 - To absorb water and nutrients from the soil.
- A farmer has a valuable citrus tree whose trunk was accidentally damaged by a grazing animal. To save the tree, they decide to perform a bridge graft. This grafting purpose is primarily for:
 - Aesthetic improvement.
 - Rejuvenation of an old plant.
 - Repairing a damaged plant.
 - Changing fruit type.



KITCHEN AND BACKYARD GARDENING



COOKING

4. In a school agriculture club, learners are carrying out grafting on young avocado seedlings. After joining the scion and rootstock, what crucial step must they take to ensure a successful union?
- A) Expose the union to direct harsh sunlight.
 - B) Securely bind the union with grafting tape or wax.
 - C) Keep the plant completely dry for a week.
 - D) Remove all leaves from the scion.
5. A gardener wants to grow a rose bush that produces two different colored flowers on the same plant. Which grafting purpose would achieve this desired mixed-color appearance?
- A) Repair.
 - B) Rejuvenation.
 - C) Aesthetic.
 - D) Pest resistance.
6. Why is personal safety in the use of sharp grafting tools an important consideration when performing grafting?
- A) It slows down the grafting process.
 - B) It prevents injuries such as cuts to the hands.
 - C) It makes the tools last longer.
 - D) It is only necessary for professional grafters.
7. A learner has grafted a scion onto a rootstock. After a few weeks, they notice small shoots emerging from the rootstock below the graft union. What action should they take regarding these shoots?
- A) Allow them to grow freely.
 - B) Remove them immediately to prevent competition with the scion.
 - C) Cover them with soil.
 - D) Water them more frequently.
8. A farmer wants to make their fruit trees more tolerant to specific soil conditions or diseases. They choose a rootstock known for these desirable traits. This use of grafting is for:
- A) Aesthetic purposes.
 - B) Rejuvenation.
 - C) Plant improvement.
 - D) Repair.
9. After a successful graft union has formed and healed, what should be done with the grafting tape or material used to bind the union?
- A) Leave it indefinitely.
 - B) Remove it carefully to prevent girdling and allow the stem to grow.
 - C) Cover it with more tape.
 - D) Water it frequently.
10. Which of the following is a key characteristic of a healthy scion selected for grafting?
- A) Old, woody, and diseased.
 - B) Young, vigorous, disease-free, and dormant or semi-dormant.
 - C) Very thin and dried out.
 - D) From a wild, unknown plant.
11. During practical grafting, why is it recommended to use a sharp, sterilized knife?
- A) To reduce time used in grafting.
 - B) To make clean cuts that promote quick healing and prevent infections.
 - C) To make the scion longer.
 - D) To cut off the rootstock completely.



KITCHEN AND BACKYARD GARDENING



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12. A farmer has an old, declining fruit tree that produces very little fruit. To restore its vigor and fruiting capacity, they decide to graft young, healthy scions onto its existing branches. This purpose of grafting is known as:
 - A) Repair.
 - B) Aesthetic.
 - C) Rejuvenation.
 - D) Rootstock development.
13. What is the primary purpose of grafting wax or grafting tape after joining the scion and rootstock?
 - A) To provide nutrients to the scion.
 - B) To hold the parts firmly together and prevent desiccation (drying out) and infection.
 - C) To make the union look appealing.
 - D) To repel pests.
14. A learner is performing a cleft graft. This method involves making a split in the rootstock. What kind of tool is essential for making precise cuts in grafting?
 - A) A blunt knife.
 - B) A specialized grafting knife or a very sharp, clean razor.
 - C) A pair of scissors.
 - D) A saw.
15. Why is it important to ensure that the cambium layers of the scion and rootstock are aligned during grafting?
 - A) To make the graft visually appealing.
 - B) The cambium layers contain the cells responsible for growth and forming the union.
 - C) To prevent the scion from falling off immediately.
 - D) To make the plant produce more leaves.
16. Which of the following plants is commonly propagated by grafting in Kenya to improve fruit quality or disease resistance?
 - A) Maize.
 - B) Mangoes, citrus, and avocados.
 - C) Sukuma wiki.
 - D) Potatoes.
17. A farmer, Narok County, wants to grow a specific variety of avocado that is resistant to a common soil-borne disease in their area. They would achieve this by grafting the desired avocado variety onto a rootstock known for:
 - A) Producing larger fruits
 - B) Disease resistance.
 - C) Faster growth.
 - D) Aesthetic appeal.
18. When carrying out side grafting, why is the scion cut into a wedge shape before insertion?
 - A) To make it fit tightly into the cut on the rootstock.
 - B) To increase the number of leaves.
 - C) To make the scion look longer.
 - D) To stop the scion from drying quickly.
19. A learner takes care of a newly grafted plant by regularly watering it. Why is adequate watering crucial for a grafted plant?
 - A) To wash away the grafting tape.
 - B) To provide moisture essential for the healing process and the scion's initial growth.
 - C) To make the scion fall off.
 - D) To prevent the rootstock from growing.



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20. A branch of a valuable fruit tree accidentally snapped, leaving a clean break. A farmer uses grafting to re-attach the broken part. This is an example of grafting for:
 - A) Aesthetic purposes.
 - B) Rejuvenation.
 - C) Repair.
 - D) New variety creation.
21. What is the characteristic of a successfully united graft?
 - A) The scion has withered and died.
 - B) New growth (buds sprouting, leaves forming) appears on the scion, and the graft union shows callus formation.
 - C) The grafting tape has fallen off without new growth.
 - D) The rootstock has grown vigorously, overshadowing the scion.
22. A farmer grafts a specific type of bougainvillea onto a different rootstock to create a standard or tree form bougainvillea for ornamental purposes. This is grafting for:
 - A) Repair.
 - B) Plant improvement.
 - C) Rejuvenation.
 - D) Disease resistance.
23. Why is it important to keep the graft union protected from extreme weather conditions (e.g., strong wind, direct sun) immediately after grafting?
 - A) To prevent it from rotting.
 - B) To facilitate healing and prevent desiccation or physical damage.
 - C) To make it grow slower.
 - D) To change the fruit color.
24. A learner is practicing grafting. They choose a rootstock that is genetically compatible with the scion. Why is genetic compatibility between scion and rootstock essential for successful grafting?
 - A) It makes the plant look similar.
 - B) It ensures that the tissues can join and grow together successfully.
 - C) It makes the grafting process faster.
 - D) It prevents the plant from flowering.
25. Which method of plant propagation is grafting considered to be?
 - A) Sexual propagation.
 - B) Asexual or vegetative propagation.
 - C) Seed propagation.
 - D) Spore propagation.
26. A young farmer is advised to graft a productive but disease-susceptible local avocado variety onto a resistant rootstock. This specific use of grafting is for:
 - A) Increasing water absorption.
 - B) Improving disease resistance.
 - C) Repairing damaged roots.
 - D) Aesthetic purposes.
27. What is the correct post-grafting care for a young grafted plant?
 - A) Ignoring it until maturity.
 - B) Regular watering, shading, and monitoring for pests.
 - C) Removing the scion after one month.
 - D) Leaving the graft union open to air.



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28. A farmer has an old, leggy coffee plant that produces few berries at the top. They decide to cut it back severely and graft new, vigorous scions onto the old stem. This aims at:
- Increasing its height.
 - Rejuvenation.
 - Making it susceptible to disease.
 - Changing its species.
29. Which of the following is a common challenge faced when learning to graft, often requiring patience and practice?
- Grafting is always successful on the first try.
 - Achieving precise cuts and proper alignment of cambium layers.
 - Finding grafting tools.
 - Watering the plant.

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Answer ALL questions.

Question 1 (20 marks)

- a) Define grafting as used in agriculture. (2 marks)

.....

.....

.....

- b) Explain why grafting is important in fruit tree production. (5 marks)

i).....

ii).....

iii)

iv)

v)

- c) Describe four methods of grafting used in plants. (8 marks)

i).....

ii).....

iii)

iv)

- d) State three precautions to observe when grafting. (3 marks)

i).....

ii).....

iii)



e) Give two reasons why a graft may fail. (2 marks)

i).....

ii).....

Question 2 (20 marks)

a) State two reasons for grafting plants. (2 marks)

i).....

ii).....

b) With the aid of a diagram, describe the steps of whip grafting. (8 marks)

c) Explain three reasons for caring for grafted plants. (6 marks)

i).....

ii).....

iii)

d) Identify two tools used during grafting. (2 marks)

i).....

ii).....

e) Suggest two crops commonly propagated by grafting in Kenya. (2 marks)

i).....

ii).....



Question 3 (20 marks)

a) Differentiate between a scion and a rootstock. (4 marks)

.....

.....

.....

b) Explain how grafting is used for plant rejuvenation. (4 marks)

i).....

ii).....

iii)

iv)

c) List four practices done to care for grafted plants. (8 marks)

i).....

ii).....

iii)

iv)

d) Explain why grafting knives must be handled carefully. (2 marks)

i).....

ii).....

e) Name two advantages of using grafted plants over seedlings. (2 marks)

i).....

ii).....

Question 4 (20 marks)

a) Define aesthetic grafting. (2 marks)

.....

.....

b) Explain three purposes of aesthetic grafting. (6 marks)

i).....

ii).....

iii)



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c) Describe four steps followed when carrying out a cleft graft. (8 marks)

- i).....
- ii).....
- iii)
- iv)

d) State two signs of a successful graft. (2 marks)

- i).....
- ii).....

e) Identify two reasons why grafted plants should be watered. (2 marks)

- i).....
- ii).....

Question 5 (20 marks)

a) Explain bridge grafting and state its importance. (4 marks)

- i).....
- ii).....
- iii)
- iv)

b) Using diagrams, describe how bud grafting is carried out. (6 marks)

c) State four safety measures to observe when carrying out grafting. (4 marks)

- i).....
- ii).....
- iii)
- iv)



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d) Mention three problems that may affect grafted plants. (3 marks)

- i).....
- ii).....
- iii)

e) Give three benefits of grafting in crop improvement. (3 marks)

- i).....
- ii).....

SET 2

SECTION B: STRUCTURED QUESTIONS (100 Marks)

Instructions: Answer all questions in the spaces provided. Show your working clearly where necessary.

Question 1 (10 Marks)

Mr. Obuya, a farmer , wants to improve his mango trees using grafting.

a) What is grafting in plant propagation? (2 Marks)

.....

.....

.....

b) State two characteristics of a good scion. (2 Marks)

- i).....
- ii).....

c) State two characteristics of a good rootstock. (2 Marks)

- i).....
- ii).....
- iii)

d) Mention four steps Mr. Obuya should follow immediately after joining the scion and rootstock to ensure success. (4 Marks)

- i).....
- ii).....
- iii)
- iv)



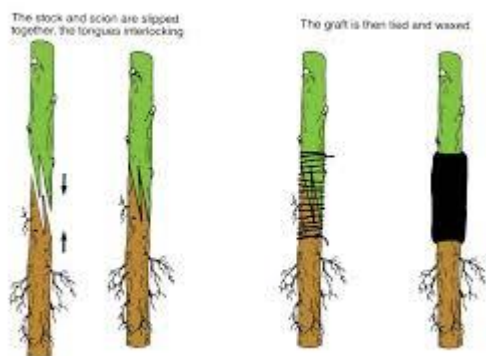
KITCHEN AND BACKYARD GARDENING



COOKING

Question 2 (10 Marks)

Study the grafting technique below.



a) Identify the grafting method shown in the diagram. (1 Mark)

i).....

b) Why must the cambium layers of the scion and rootstock be in contact? (2 Marks)

i).....

ii).....

c) Give three purposes of grafting and provide an example for each. (6 Marks)

i).....

ii).....

iii)

d) State one reason why safety is important when using sharp grafting tools. (1 Mark)

i).....

ii).....

Question 3 (10 Marks)

Grade 9 learners carried out grafting on citrus plants.

a) i) What should be done if shoots appear from the rootstock below the graft union? Give a reason. (2 Marks)

i).....

ii).....



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ii) Why is watering important after grafting? (2 Marks)

i).....

ii).....

b) Explain how grafting is used for rejuvenation, giving one example. (3 Marks)

i).....

ii).....

iii)

c) When should grafting tape be removed? What happens if it is left on for too long? (3 Marks)

i).....

ii).....

iii)

Question 4 (10 Marks)

An avocado tree was damaged at the lower trunk by a storm.

a) Explain how grafting can be used to repair the damaged tree. (4 Marks)

i).....

ii).....

b) State three benefits of grafting for plant improvement. (3 Marks)

i).....

ii).....

iii)

c) Explain how grafting helps in modern crop production techniques. (3 Marks)

i).....

ii).....

iii)



KITCHEN AND BACKYARD GARDENING



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Question 5 (10 Marks)

Learners are preparing a presentation on the uses of grafting.

a) Mention two plants (other than fruit trees) that are commonly grafted. For each plant, state one reason for grafting. (4 Marks)

- i).....
- ii).....

b) Explain two ways grafting is used to increase crop production. (3 Marks)

- i).....
- ii).....

c) State three examples of grafting used for beauty or ornamental purposes. (3 Marks)

- i).....
- ii).....
- iii)



STRAND 4.0: PRODUCTION TECHNIQUES
SUB-STRAND 4.2: HOMEMADE SUN DRYER
SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Instructions: Choose the best answer for each question.

1. A farmer harvests a large amount of traditional vegetables that risk spoiling within days. To preserve them cheaply for later use, which method is most suitable?
A) Canning in tins.
B) Freezing in an industrial freezer.
C) Using a homemade sun dryer.
D) Preserving with chemicals.
2. During a lesson, a Grade 9 learner explains that the transparent cover of a sun dryer is very important. What is its main role?
A) To make the dryer attractive.
B) To trap heat by allowing sunlight in, raising the drying temperature.
C) To block air completely.
D) To keep away birds.
3. A farmer prepares sukuma wiki for drying. Which preparation step is most important before placing the vegetables in the dryer?
A) Soaking them in water.
B) Chopping and sometimes blanching, then spreading thinly.
C) Stacking them in thick piles.
D) Covering them with soil.
4. Learners in Narok are building a sun dryer using wooden planks. To ensure proper airflow, what feature should they include?
A) Airtight sealing of all sides.
B) Ventilation openings at the top and bottom.
C) Bright paint on the inside.
D) Making the dryer very small.
5. A family wants to preserve their garden vegetables using a sun dryer. What principle makes this method effective?
A) Cooking the vegetables.
B) Removing moisture to stop microbial growth and enzyme activity.
C) Freezing the vegetables.
D) Adding preservatives.
6. A Grade 9 learner designs a dryer with a mesh tray. What is the advantage of this tray over a solid surface?
A) Vegetables stick to it.
B) It allows good airflow around the vegetables for even drying.
C) It keeps vegetables cool.
D) It stores water.
7. A farmer slices tomatoes before placing them in a homemade sun dryer. Why is slicing necessary?
A) To make tomatoes heavier.
B) To increase surface area for faster drying.
C) To lower nutritional value.
D) To make them spoil faster.



KITCHEN AND BACKYARD GARDENING



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8. A women's group in Narok is comparing drying methods. Why is a homemade sun dryer better for them than buying an industrial dryer?
 - A) It needs special training to use.
 - B) It is cheap, energy-efficient, and uses locally available materials.
 - C) It dries vegetables instantly.
 - D) It leaves vegetables with high moisture.
9. A learner notices that their vegetables are drying too slowly in the sun dryer. What could be the likely cause?
 - A) Hot weather outside.
 - B) Too many vegetables stacked thickly or poor airflow.
 - C) Too much sunlight.
 - D) Using fresh vegetables.
10. Farmers in Western Kenya often dry vegetables for later sale. Which vegetable is most commonly dried using this method?
 - A) Watermelon.
 - B) Sukuma wiki, kale, and African leafy vegetables.
 - C) Lettuce.
 - D) Cucumber.
11. A learner paints the inside of their sun dryer black. Why is this done?
 - A) To make it dirty.
 - B) Black absorbs more heat from sunlight, raising internal temperature.
 - C) To hide the vegetables.
 - D) To block sunlight.
12. A farmer wants to reduce their electricity bills. How does using a sun dryer help the environment compared to an electric dryer?
 - A) By consuming more fossil fuels.
 - B) By saving energy and reducing the carbon footprint.
 - C) By polluting the air.
 - D) By causing deforestation.
13. A family chops vegetables and places them inside their sun dryer. How should they arrange the vegetables for efficient drying?
 - A) In one thick heap.
 - B) In a single thin layer.
 - C) Under a heavy cloth.
 - D) Mixed with raw vegetables.
14. After a bumper harvest, a farmer builds a sun dryer. What is the main benefit of this technology?
 - A) To make vegetables rot faster.
 - B) To extend shelf life and reduce post-harvest losses.
 - C) To increase vegetable prices.
 - D) To make vegetables grow quickly.
15. A farmer wants to store dried vegetables for six months. What should be their final moisture content after drying?
 - A) Over 50%.
 - B) 20–30%
 - C) Below 10–15%
 - D) Still wet.



KITCHEN AND BACKYARD GARDENING



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16. While drying vegetables, which external factors does a sun dryer protect them from?
 - A) Too much sun.
 - B) Rain, dust, insects, and direct ground contact.
 - C) Air circulation.
 - D) Wind.
17. A learner observes that vegetables at the top tray are drying faster than those at the bottom. What should they do?
 - A) Remove the cover.
 - B) Turn or redistribute vegetables regularly.
 - C) Add more fresh vegetables.
 - D) Pile them thicker.
18. A farmer uses a sun dryer during rainy season shortages. How does this improve household food security?
 - A) By reducing available vegetables.
 - B) By ensuring off-season supply of vegetables.
 - C) By spoiling vegetables.
 - D) By increasing imports.
19. A youth group is building dryers from timber, plastic sheets, and wire mesh. These materials are described as:
 - A) Expensive imports.
 - B) Locally available and cheap.
 - C) Non-recyclable.
 - D) House-building materials.
20. What role does airflow play in a sun dryer?
 - A) Making vegetables wet.
 - B) Carrying away evaporated moisture.
 - C) Cooling the dryer.
 - D) Attracting insects.
21. A family lives in a rainy area. Which design feature must their sun dryer include?
 - A) An open roof.
 - B) A waterproof transparent cover.
 - C) A heavy stone base.
 - D) No ventilation.
22. During a nutrition lesson, learners discuss dried vegetables. Why are they still healthy to eat months later?
 - A) Because all vitamins are destroyed.
 - B) Because many vitamins and minerals are retained after drying.
 - C) Because they taste bland.
 - D) Because chemicals are added.
23. A farmer fears insects may attack vegetables inside the dryer. Which part of the design helps prevent this?
 - A) Solid wooden walls.
 - B) Fine mesh or netting.
 - C) A metal sheet.
 - D) Leaving it uncovered.
24. A Grade 9 learner wants to maximize efficiency of their dryer. Where should they place it?
 - A) Away from the sun.
 - B) In a shady area.
 - C) In full sunlight throughout the day.
 - D) Inside a house.



25. A farmer checks dried amaranthus leaves. Which sign shows they are ready for storage?
 - A) Soft and bendy.
 - B) Brittle and snap easily.
 - C) Still wet.
 - D) Turned black.
26. A household builds a sun dryer and uses it regularly. What long-term financial benefit do they gain?
 - A) Higher electricity bills.
 - B) Reduced food waste and money saved from off-season purchases.
 - C) Increased transport costs.
 - D) Higher demand for fresh vegetables.
27. A group of learners visit the library to find information about homemade sun dryers. What skill are they developing?
 - A) Avoiding learning.
 - B) Research and information literacy skills.
 - C) Learning only outdated methods.
 - D) Instantly becoming experts.
28. Apart from vegetables, which fruit is commonly dried in Kenya using homemade sun dryers?
 - A) Mangoes.
 - B) Avocados.
 - C) Pawpaw.
 - D) Oranges.
29. A farmer wants their dryer to last for many years. What simple maintenance practice should they carry out?
 - A) Leaving it exposed to rain.
 - B) Cleaning trays and covers regularly.
 - C) Ignoring broken parts.
 - D) Storing wet vegetables inside.
30. A teacher asks learners: *How can innovative technology help preserve vegetables?* What is the best answer in the case of a sun dryer?
 - A) By depending only on weather changes.
 - B) By designing simple solar-powered devices for drying.
 - C) By using expensive electric machinery.
 - D) By applying chemical preservatives.

SET 2

Strand 4.0: Production Techniques

Sub-strand 4.2: Homemade Sun Dryer

SECTION A: MULTIPLE CHOICE QUESTIONS (50 Marks)

(Each question carries 1 mark)

Q1.

James harvested a lot of sukuma wiki (kales) during the rainy season. He wants to store some for the dry season. Which method should he use to reduce moisture content without electricity?

- | | |
|-------------|---------------|
| A. Freezing | B. Sun drying |
| C. Smoking | D. Boiling |



KITCHEN AND BACKYARD GARDENING



COOKING

Q2.

Sarah constructed a small wooden frame covered with clear polythene in her garden. She uses it to dry vegetables. What is this structure called?

- A. Shade house
- B. Greenhouse
- C. Homemade sun dryer
- D. Storage shed

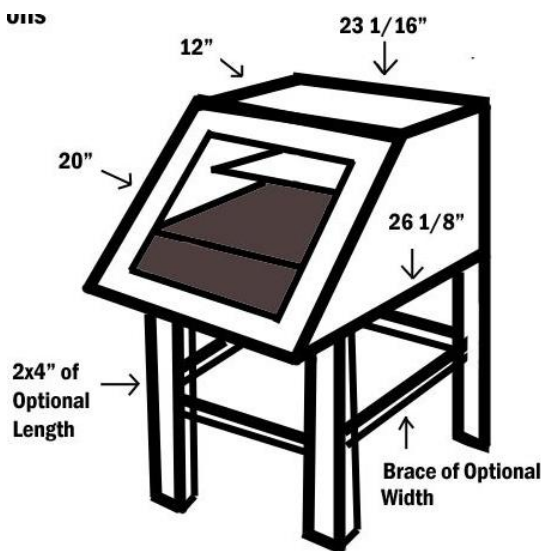
Q3.

A group of learners preserved amaranth leaves by spreading them inside a sun dryer. What was the **main purpose** of using the dryer?

- A. To cook food
- B. To dry vegetables and fruits
- C. To provide shade
- D. To heat water

Q4.

The sketch below shows a simple structure made with timber and polythene.



What is its function?

- A. Germinating seeds
- B. Preserving vegetables
- C. Raising seedlings
- D. Growing mushrooms

Q5.

During construction, Peter was asked to cover the dryer with a material that allows sunlight to pass through. Which material is most suitable?

- A. Glass
- B. Clear polythene sheet
- C. Sisal sack
- D. Canvas



KITCHEN AND BACKYARD GARDENING



COOKING

Q6.

A teacher advised learners to use strong timber for the frame of the dryer. Which timber should they choose?

- A. Freshly cut green wood
- B. Rotten timber
- C. Strong dry timber
- D. Wet bamboo

Q7.

Linnet compared vegetables dried in open sun with those dried inside a sun dryer. Which was the main benefit of using the sun dryer?

- A. Prevents dust, insects, and contamination
- B. Increases moisture
- C. Exposes to rainwater
- D. Makes them wet again

Q8.

A farmer in Eldoret constructed a dryer in his compound. To capture maximum sunlight, in which direction should he place it?

- A. East–West
- B. North–South
- C. Under a tree
- D. Inside the house

Q9.

Before vegetables are put in the sun dryer, Amina prepares them first. What should she do?

- A. Mix with soil
- B. Wash and slice
- C. Boil for one hour
- D. Store in a sack

Q10.

After using a sun dryer, a farmer noticed his vegetables kept their nutrients. What is this an advantage of?

- A. Sun drying
- B. Refrigeration
- C. Canning
- D. Freezing

Q11.

Kibet stored vegetables that were not completely dry. What is most likely to happen?

- A. They become sweeter
- B. They rot and grow mould
- C. They increase in size
- D. They lose colour only

Q12.

In a school demonstration, the teacher preserved sukuma wiki using a sun dryer. Which other vegetable could also be dried in the same way?

- A. Tomatoes
- B. Sukuma wiki
- C. Potatoes
- D. Onions



KITCHEN AND BACKYARD GARDENING

COOKING

Q13.
The learners drilled holes in the dryer. Why were these holes important?
A. To allow circulation of air
B. To let in animals
C. To make the dryer heavy
D. To reduce light

Q14.
The agriculture teacher asked why clear polythene is used instead of black polythene. Which answer was correct?
A. It is transparent and allows sunlight in
B. It absorbs water
C. It is very heavy
D. It blocks sunlight

Q15.
During drying, Jane noticed dust settling on vegetables. What was missing in her dryer?
A. A door
B. A protective cover
C. A window
D. A roof

Q16.
The school's agriculture club raised the dryer 1 metre above the ground. Why was this done?
A. To allow circulation of air
B. To reduce cost of timber
C. To increase vegetable size
D. To let animals eat vegetables

Q17.
Musa compared electricity-powered dryers with homemade ones. Which energy source is used in the homemade dryer?
A. Electricity B. Gas C. Sunlight D. Firewood

Q18.
Vegetables in the dryer were placed on trays instead of the floor. Why were trays used?
A. To increase drying surface
B. To add flavour
C. To reduce colour loss
D. To remove seeds

Q19.
Apart from vegetables, Wanjiku used her dryer to preserve another food. Which one?
A. Fish
B. Milk
C. Meat broth
D. Soup



KITCHEN AND BACKYARD GARDENING



COOKING

Q20.

A learner listed the following steps:

- i. Wash vegetables
- ii. Slice vegetables
- iii. Dry vegetables

What does this order show?

- A. Correct preparation for drying
- B. Wrong preparation
- C. Storing process
- D. Cooking process

Q21.

In Kericho, a farmer dried spinach during the dry season. Later, he found mould on the leaves. What was the likely cause?

- A. Drying was not complete
- B. Vegetables were too fresh
- C. The dryer was too clean
- D. Sunlight was too much

Q22.

A school built a sun dryer but placed it under a big tree. Why did the vegetables take long to dry?

- A. The shade blocked sunlight
- B. The tree added dust
- C. The tree produced insects
- D. The dryer had no trays

Q23.

Lucy made her dryer very small. What is one disadvantage of a small dryer?

- A. Dries few vegetables at a time
- B. Uses too much polythene
- C. Blocks too much air
- D. Becomes too heavy

Q24.

When using the dryer, learners were told to avoid putting wet soil inside. Why?

- A. It can cause mould on vegetables
- B. It adds nutrients
- C. It dries faster
- D. It increases weight

Q25.

Vegetables dried in a homemade sun dryer are usually:

- A. Soft and watery
- B. Crisp and light
- C. Fresh and heavy
- D. Bitter and raw



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COOKING

Q26.

At the school farm, learners placed vegetables on the ground inside the sun dryer instead of trays. What problem might this cause?

- A. Vegetables dry evenly
- B. Vegetables may get dirty
- C. Vegetables become more nutritious
- D. Vegetables grow again

Q27.

A farmer harvested spinach and wanted to dry them quickly. He opened both sides of the dryer to allow air circulation. What was the benefit of this?

- A. It increased mould growth
- B. It allowed faster drying
- C. It reduced sunlight
- D. It made vegetables wet

Q28.

John used a black polythene sheet to cover his dryer. Why did his vegetables take longer to dry?

- A. Black polythene blocked sunlight
- B. Black polythene let in too much light
- C. Black polythene added nutrients
- D. Black polythene was too light

Q29.

Amina wanted to use her dryer to dry tomatoes. Which form of tomatoes should she prepare first?

- A. Whole raw tomatoes
- B. Sliced tomatoes
- C. Rotten tomatoes
- D. Unwashed tomatoes

Q30.

The agriculture club placed their dryer on bare ground. After some time, termites destroyed the timber. What should they have done?

- A. Painted the timber with oil
- B. Added more termites
- C. Used wet timber
- D. Covered with soil

Q31.

Moses dried his vegetables in an open field without a cover. Why did they spoil quickly?

- A. They absorbed too much dust and insects
- B. They dried faster than expected
- C. They gained more vitamins
- D. They changed colour only



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Q32.

Jane dried vegetables in the sun dryer and stored them immediately in a plastic bag while still warm. What is the risk?

- A. Moisture will condense and cause mould
- B. Vegetables will remain fresh
- C. Vegetables will increase in size
- D. Vegetables will cook inside the bag

Q33.

During a school project, learners were told to clean the dryer before use. Why is cleaning important?

- A. To remove dust and pests
- B. To add nutrients
- C. To reduce sunlight
- D. To keep timber soft

Q34.

A farmer in Machakos used the dryer to preserve vegetables for the market. What is one economic benefit of this?

- A. Reduced market prices
- B. Prolonged shelf life
- C. Reduced quality
- D. Increased wastage

Q35.

The learners forgot to slice vegetables before drying. What was the effect?

- A. Vegetables dried slower
- B. Vegetables dried faster
- C. Vegetables changed colour
- D. Vegetables lost nutrients

Q36.

Paul dried his vegetables during a rainy week. Why was this a challenge?

- A. Too much sunlight
- B. Low sunlight reduced drying speed
- C. Excessive dust in the air
- D. Vegetables became too crispy

Q37.

A group of learners added wire mesh at the base of the dryer. Why was this useful?

- A. To prevent animals and insects from entering
- B. To make vegetables sweeter
- C. To reduce weight of vegetables
- D. To block sunlight



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Q38.

Mary wanted to preserve vegetables for her boarding school. Which advantage of sun drying made it possible?

- A. Vegetables become lighter and easy to transport
- B. Vegetables remain heavy
- C. Vegetables produce water
- D. Vegetables rot quickly

Q39.

During harvesting, Peter collected rotten vegetables and placed them in the dryer. What will happen?

- A. They will dry normally
- B. They will spoil and spread rot to others
- C. They will gain nutrients
- D. They will become crisp

Q40.

Learners noticed that vegetables dried in the dryer had better colour compared to open sun drying. Why?

- A. Sun dryer reduces nutrient loss
- B. Open sun adds more colour
- C. Sun dryer removes all vitamins
- D. Open sun prevents drying

Q41.

During storage, the teacher advised learners to use airtight containers for dried vegetables. Why?

- A. To prevent moisture and pests
- B. To increase air circulation
- C. To make vegetables heavier
- D. To change vegetable colour

Q42.

In a community project, women preserved vegetables using sun dryers. How did this help their families?

- A. Reduced food wastage
- B. Increased dependence on markets
- C. Increased hunger
- D. Reduced variety of food

Q43.

A farmer placed vegetables in thick layers inside the dryer. What problem could this cause?

- A. Uneven drying
- B. Faster drying
- C. No effect at all
- D. Sweeter vegetables

Q44.

During an agriculture lesson, the teacher asked why the dryer should be raised. What was the correct answer?

- A. To allow air circulation and prevent contamination
- B. To keep vegetables on the ground
- C. To block sunlight
- D. To reduce space



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Q45.

A farmer harvested cabbages and dried them in the dryer. After drying, what should he do before storage?

- A. Cool the vegetables
- B. Mix with water
- C. Pack while still hot
- D. Place in the sun again

Q46.

Learners built a dryer but left no ventilation holes. What will happen to the vegetables?

- A. They will dry faster
- B. They may rot due to poor air flow
- C. They will taste better
- D. They will last longer

Q47.

Jane used a dryer to dry cowpeas leaves. Which other leafy vegetable could also be preserved this way?

- A. Spinach
- B. Pumpkin
- C. Sweet potato tubers
- D. Groundnuts

Q48.

The agriculture teacher advised learners to cut vegetables into thin slices. Why was this necessary?

- A. Thin slices dry faster
- B. Thin slices rot easily
- C. Thin slices gain water
- D. Thin slices prevent light

Q49.

A farmer wants to sell dried vegetables in Nairobi. Why is packaging important?

- A. To keep them clean and safe
- B. To make them heavier
- C. To add water to vegetables
- D. To reduce quality

Q50.

During a school competition, one team made a sun dryer facing North-South. Why did their vegetables dry slowly?

- A. Dryer did not capture enough sunlight
- B. Dryer was too clean
- C. Dryer was too small
- D. Dryer had too much timber



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SECTION B: STRUCTURED QUESTIONS (100 Marks)

Question 1 (20 marks)

The Grade 9 agriculture teacher asked learners to explain what a homemade sun dryer is.

(a) Define a homemade sun dryer. (2 marks)

.....

.....

.....

(b) State four materials used in constructing it. (4 marks)

- i).....
- ii).....
- iii)
- iv)

(c) Explain three reasons why vegetables should be dried in a sun dryer instead of open sun. (6 marks)

- i).....
- ii).....
- iii)

(d) Draw a well-labelled diagram of a homemade sun dryer. (8 marks)



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Question 2 (20 marks)

Amina harvested a lot of amaranth leaves. She wants to dry them for use during the dry season.

(a) Outline the steps she should follow before placing vegetables in the dryer. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

(b) Explain how she can construct a homemade sun dryer using locally available materials. (7 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

(c) State four advantages of using a homemade sun dryer. (8 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

Question 3 (20 marks)

In a village, a farmer dried vegetables but later found mould growing on them.

(a) State three possible causes of mould. (6 marks)

- i).....
- ii).....
- iii)



(b) Suggest four ways to avoid mould during drying. (8 marks)

- i).....
- ii).....
- iii)
- iv)

(c) Give three characteristics of well-dried vegetables. (6 marks)

- i).....
- ii).....
- iii)

Question 4 (20 marks)

The learners of Greenfield School compared open sun drying and using a homemade sun dryer.

(a) State three differences between the two methods. (6 marks)

- i).....
- ii).....
- iii)

(b) List four safety precautions to observe when using a homemade sun dryer. (8 marks)

- i).....
- ii).....
- iii)
- iv)

(c) Explain how the orientation of a sun dryer affects its efficiency. (6 marks)

- i).....
- ii).....
- iii)



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Question 5 (20 marks)

During practical lessons, learners were asked to construct a sun dryer in the school garden.

(a) Write down the procedure they should follow. (6 marks)

- i).....
- ii).....
- iii)

(b) List five challenges they may face when constructing it. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)

(c) Suggest four solutions to these challenges. (4 marks)

- i).....
- ii).....
- iii)
- iv)

(d) Explain the importance of preserving vegetables with a sun dryer to the community. (5 marks)

- i).....
- ii).....
- iii)
- iv)
- v)



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SET 2

SECTION B: STRUCTURED QUESTIONS (50 Marks)

Instructions: Answer all the questions in the spaces provided. Show your working where necessary.

Question 1 (10 Marks)

Farmers, Narok County, lose many vegetables after harvest because they spoil quickly. To reduce these losses, they decide to make **homemade sun dryers**.

a) State four features of a good homemade sun dryer. (4 Marks)

- i).....
- ii).....
- iii)
- iv)

b) Write the steps for making a simple homemade sun dryer using local materials. (4 Marks)

- i).....
- ii).....
- iii)
- iv)

c) Explain one way the dryer helps reduce vegetable spoilage. (2 Marks)

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Question 2 (10 Marks)

The diagram below shows a simple **cabinet-type homemade sun dryer** with a transparent cover, mesh trays, and ventilation holes.



a) What type of energy is mainly used in this dryer? (1 Mark)

i).....

b) Explain three ways the features shown in the diagram help vegetables dry faster. (3 Marks)

i).....

ii).....

iii)

c) List three preparation steps done on vegetables like sukuma wiki before drying. (3 Marks)

i).....

ii).....

iii)

d) Explain how using homemade sun dryers improves food supply in the family. (3 Marks)

i).....

ii).....

iii)

Question 3 (10 Marks)

A Grade 9 Agriculture class is making a homemade sun dryer as a school project.

a) Name five local materials they can use to build the dryer. (5 Marks)

i).....

ii).....

iii)

iv)

v)

b) State two factors that affect how long vegetables take to dry in the sun dryer. (2 Marks)

i).....

ii).....

iii)

iv)



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c) After drying vegetables, explain three ways to store them properly so that they remain good for use. (3 Marks)

- i).....
- ii).....
- iii)

Question 4 (10 Marks)

A farmer lives in an area where the weather changes between sunny and cloudy.

a) Describe two ways the farmer can improve the dryer to absorb more sunlight and retain heat. (2 Marks)

- i).....
- ii).....

b) Give three ways a sun dryer protects vegetables compared to drying them on the ground. (3 Marks)

- i).....
- ii).....
- iii)

c) Suggest two ways the farmer can stop insects from entering the dryer. (3 Marks)

- i).....
- ii).....

d) Explain one skill learners gain when they design and build a sun dryer. (2 Marks)

- i).....

Question 5 (10 Marks)

A youth group wants to teach villagers about using homemade sun dryers to preserve food.

a) State two economic benefits of using sun dryers in the home. (2 Marks)

- i).....
- ii).....

b) Explain how sun dryers help reduce food waste in the community. (3 Marks)

- i).....
- ii).....
- iii)



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c) State two differences between drying vegetables in the open sun and drying them in a sun dryer in terms of hygiene and quality. (2 Marks)

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.....

.....

d) Explain three community benefits of using homemade sun dryers. (3 Marks)

- i).....
- ii).....
- iii)

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